

CargoExpress PH: A Progressive Web-Based Cargo Delivery Booking and Tracking System

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Bea	A.	Sarong
Quin Ann	P.	Amigo
Joven	M.	Tibuc

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APPROVAL SHEET

This thesis entitled **CargoExpress PH: A Progressive Web-Based Cargo Delivery Booking and Tracking System** prepared and submitted by Bea A. Sarong, Quin Ann P. Amigo, and Joven M. Tibuc, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology has been examined and recommended for acceptance and approval for oral defense.

THE THESIS COMMITTEE

SHELLA C. OLAGUIR, PhD
Dean

ANGELICA T. SIT, LPT
Adviser

JULIE O. BITASOLO, PhD
English Critic

THE EXAMINING TRIBUNAL

Approved by the Committee on Oral Examination with a grade of _____.

MARIETTA C. MACALOLOT, PhD
Chairperson

SHELLA C. OLAGUIR, PhD
Member

JULIE O. BITASOLO, PhD
Member

BENJAMIN N. OMAMALIN, PhD
Member

Accepted and approved as partial fulfillment of the requirement for the degree of Bachelor of Science in Information Technology.

September 21, 2026
Date of Oral Defense

SHELLA C. OLAGUIR, PhD
Dean, College of Computing and Information Sciences

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ABSTRACT

CargoExpress PH is a comprehensive **Progressive Web Application (PWA)** (Na define or na explain bani sa study ninyu nga PWA) (yes mam gihilight ko po ung section na naaexplain ung PWA tapos ina-add ko rin sa definition of terms) and web-based logistics platform with an integrated digital booking **interface** and a real-time tracking dashboard designed to manage, dispatch, and monitor cargo shipments across domestic and international routes. It aimed to enhance the scheduling, tracking, and fulfillment of freight deliveries by addressing the limitations, communication gaps, and manual inefficiencies of existing traditional cargo forwarding and paper-based receipt systems. The system integrates modules for shipment lifecycle management, including a 5-step digital booking **process** with vessel capacity validation, secure digital payments via PayMongo, real-time shipment status tracking, automated administrative dispatch dashboards, and an integrated feedback and support chat module for structured customer issue resolution. The study was conducted within the context of Philippine cargo forwarding operations (such as Manila-to-Bohol sea freight routes), with participants including shipping customers, logistics administrators, dispatch personnel, and delivery riders. The system was developed using a developmental research design and an Agile-based System Development Life Cycle (SDLC), with data gathered through operational workflow analyses, stakeholder interviews, and document reviews involving logistics managers, frontline staff, and customers. By leveraging technology, CargoExpress PH enables logistics administrators to improve pricing and weight-tracking accuracy, ensure secure digital ledger management, and gain real-time visibility into fleet capacity and shipment statuses. Overall, CargoExpress PH enhances operational transparency, improves customer trust through live tracking, and promotes a more efficient, organized, and digitally transformed cargo forwarding ecosystem.

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Chapter 1

THE PROBLEM AND ITS SCOPE

INTRODUCTION

Rationale

The rapid advancement of information and communication technology has transformed how organizations manage information, deliver services, and communicate with customers. In the logistics industry, digital technologies have improved operational efficiency by automating business processes, enhancing data accuracy, facilitating faster communication, and increasing access to information. The adoption of web-based systems, online payment platforms, and digital communication technologies has enabled logistics service providers to improve service quality, streamline operations, and respond more effectively to the growing demand for reliable and efficient cargo delivery services.

Despite these technological advancements, several limitations remain evident in existing logistics and cargo management systems. Many available solutions are designed for large-scale logistics operations and do not fully address the operational needs of small cargo delivery businesses. Common limitations include limited shipment visibility, fragmented booking and payment processes, insufficient customer communication, and the absence of integrated cargo management within a single platform. Some tracking solutions also rely on continuous GPS monitoring and specialized technologies, making them costly and impractical for smaller logistics providers. These limitations emphasize the need

for a practical, affordable, and integrated web-based cargo management system that supports the daily operations of small cargo delivery businesses while improving administrative efficiency and customer service.

To address these identified limitations, this study proposes CargoExpress PH, a Progressive Web Application (PWA) designed for small cargo delivery businesses operating between Bohol and Manila. A Progressive Web Application functions like a traditional website but offers the experience of a native mobile app. Through PWA technology, users can easily access the platform using any web browser and install it directly to their mobile devices or desktop computers without the need to download it from an app store. This approach provides a lightweight, accessible, and cost-effective solution. The proposed system provides online cargo booking, shipment tracking through predefined status updates, centralized cargo management, dynamic pricing based on cargo weight, payment monitoring with GCash and PayMongo QR integration, and communication through chatbot and administrator chat support. By integrating these features into a single platform, CargoExpress PH aimed to improve operational efficiency, enhance shipment visibility, simplify transaction management, and provide a more convenient and reliable cargo delivery experience for both administrators and customers.

Literature Background

The rapid advancement of information technology and web-based applications has fundamentally transformed modern industry frameworks by introducing automated processes, centralized data management, and enhanced

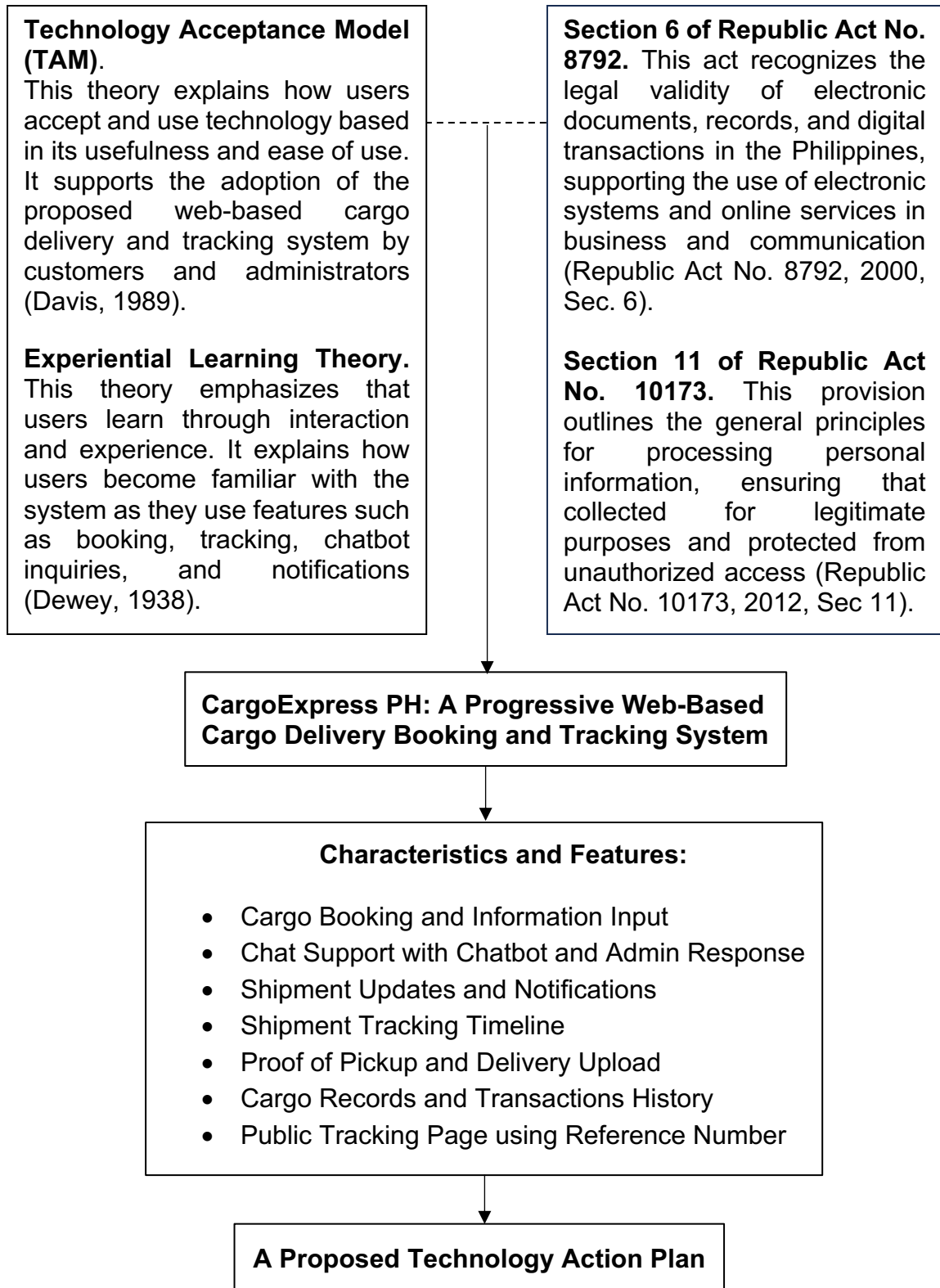


Figure 1. Theoretical and Conceptual Framework

accessibility. In today's digital landscape, organizations increasingly leverage web systems to phase out manual, error-prone methodologies in favor of structured digital environments that foster operational efficiency, seamless communication, and data integrity. These web platforms serve as vital mechanisms for streamlining enterprise workflows, providing secure information storage, and enabling data transparency between service providers and end-users across various consumer-facing industries.

The Technology Acceptance Model (TAM), proposed by Fred Davis (1989), explains how users come to accept and use a new technology based on two primary constructs: perceived usefulness and perceived ease of use. Perceived usefulness pertains to how a user believes that a system can improve performance or efficiency, while perceived ease of use relates to how easy the system is to understand and operate. TAM has been widely applied in information systems research to predict and explain user adoption across various domains, including e-commerce, digital education, and logistics platforms. The model suggests that when users find a system both useful and easy to use, they are more likely to adopt it willingly and incorporate it into their daily tasks.

In the context of the present study, TAM provided a relevant framework for understanding how customers and administrators would likely adopt the proposed web-based cargo delivery booking and tracking system. If users perceived that the system was useful in simplifying cargo booking and tracking processes, and if they found the interface easy to navigate, they would be more inclined to use it effectively. This principle guided the design of the system's user interface and the

selection of core features, including the booking module, tracking timeline, and notification system, all of which were developed to enhance user experience and promote system acceptance.

The Experiential Learning Theory, introduced by John Dewey (1938), emphasizes that learning occurs through direct experience and active interaction with one's environment. Rather than relying solely on passive instruction, individuals develop understanding and competence by engaging with a process, reflecting on their actions, and refining their approach over time. This theory has been applied in various contexts involving technology adoption, where users gradually become familiar with a system by exploring its features and learning through trial and repeated use. The theory explains that users develop better understanding and confidence through actual experience and repeated interaction.

This theory was relevant to the proposed system because users could learn and become familiar with the platform through continuous interaction and actual use. As customers and administrators engaged with features such as booking cargo, checking tracking updates, and communicating through the system, they gained a better understanding of its functionality and developed confidence in using the application. Rather than requiring formal training, the system was designed to allow users to learn progressively through hands-on interaction, reinforcing the experiential learning cycle of action, observation, and refinement.

Challenges Faced by the Indian Transportation Sector in Implementing New Age Technologies tried to articulate the various types of challenges faced by the

Indian transportation sector in implementing new age technologies for better control and visibility throughout the entire supply chain (Tripathy, 2021). The study discussed the challenges affecting digital transformation in transportation through literature review and survey responses from logistics professionals. It emphasized that minimizing manual intervention, reducing the risk of manipulated data, and incorporating features such as real-time tracking could improve transportation management and customer satisfaction. However, the study primarily identified existing challenges and recommended future improvements without developing an operational digital transportation management system.

CargoExpress PH addressed these challenges by providing a centralized web-based cargo delivery booking and tracking system that digitized shipment management and reduced dependence on manual record keeping. The proposed system enabled customers to monitor shipment progress through a tracking timeline and a public tracking page using a reference number, improving shipment visibility throughout the delivery process. It also automated cargo booking, trip scheduling, tracking number generation, status updates, and payment monitoring, providing a practical solution for small-scale inter-island cargo delivery operations.

Cloud-Based Booking Platforms in Warehouse Operations evaluates the implementation costs of cloud-based booking platforms (CBBPs) to manage vehicle queues in warehouses and investigates how they support inter-organizational logistics processes (Giuffrida et al., 2021). To examine the impact of CBBPs, the researchers interviewed logistics managers regarding platform processes, key features, and their effects on inter-organizational logistics. They

also conducted analytical modeling simulations to evaluate the economic impact of different investment scenarios. The findings revealed that the benefits and financial viability of cloud-based booking platforms varied depending on company size and whether the primary objective was cost reduction or time optimization. Furthermore, the study emphasized that adopting digital technologies in inter-organizational logistics environments presents implementation challenges. However, the platform was designed specifically for warehouse vehicle scheduling and did not address inter-island cargo booking and tracking operations.

In response to these limitations, CargoExpress PH extended the booking platform concept to a different operational context by integrating a trip scheduling module where administrators scheduled vehicle trips with defined capacity limits and assigned orders to available slots. The proposed system also incorporated tracking timeline, a public tracking page where receivers could look up shipments, and flexible payment monitoring options including Cash, GCash, and Pay Later. These features addressed the cargo delivery operations between Manila and Bohol rather than the warehouse environment. Accessibility through a web-based platform ensured that both customers and administrators could use the system from any device without requiring specialized infrastructure.

Online Courier Management System with Text Blast and QR Code Technology aimed to develop a courier service management system to manage and monitor the package; employ a QR Code technology to secure customers' information, and design a system that integrates a text blast technology for customers' notification about updates to their package (Regla & Dadiz, 2022). The

study found that all system modules functioned as expected during unit testing, and a Technology Acceptance Model evaluation yielded a grand mean score of 4.24, indicating very high user acceptance. However, the study focused primarily on package management, QR code security, and SMS notification features, without discussing broader operational functions such as trip scheduling, integrated payment management, and comprehensive cargo monitoring.

The study demonstrated the effectiveness of integrating digital technologies to improve courier service operations through package management, customer information security, and SMS notifications. However, its scope was limited to these functions and did not include operational features such as trip scheduling, payment monitoring, and comprehensive cargo tracking. To address this limitation, the proposed CargoExpress PH integrates online cargo booking, shipment tracking, trip scheduling, and payment monitoring through Cash, GCash, and Pay Later within a centralized web-based platform. These features provide administrators and customers with a more efficient system for managing cargo delivery operations for small businesses.

Design of a Web-Based Information System to Manage Delivery Activities aimed to design and developed a web-based information system that facilitated the management of delivery activities by integrating delivery records, customer information, shipping documents, invoices, and reporting into a centralized platform (Madyanti et al., 2023). The researchers developed the system to improve the efficiency of managing delivery transactions and administrative processes through a digital platform. The findings indicated that the system improved data

organization, simplified report generation, and enhanced operational efficiency in managing delivery activities. However, the study primarily concentrated on delivery administration and documentation without incorporating customer cargo booking, shipment tracking timelines, or integrated payment monitoring.

Unlike the previous study, CargoExpress PH expanded these functionalities by integrating customer and administrator modules into a single web-based platform. The proposed system allowed customers to submit cargo bookings online, monitor shipment progress through a tracking timeline, and view shipment information using a public tracking page. Administrators were also provided with trip scheduling, cargo assignment, payment monitoring, customer management, and centralized cargo operation features, offering a more comprehensive solution for cargo delivery management.

Logistics 4.0 – Digital Transformation with Smart Connected Tracking and Tracing Devices analyzed current tracking and tracing-focused applications that can provide value for logistics operations through a case study approach (Helo & Thai, 2024). The study highlighted that smart connected tracking and tracing devices supported digital transformation by providing real-time information across the supply chain, improving visibility and coordination in logistics operations. However, the study focused on tracking and tracing technologies and their role in logistics operations rather than developing a complete cargo booking and tracking management system.

These concepts were applied in the proposed system by providing a cargo

management system that improved shipment visibility for both customers and administrators. Instead of relying on specialized tracking devices, the proposed system implemented a tracking timeline that updated shipment progress through predefined logistics stages, allowing customers to monitor deliveries using a reference number. The system also integrated online booking, trip scheduling, payment monitoring, and cargo management features, providing a more comprehensive platform for small-scale inter-island cargo delivery operations.

Parcel Pulse: A Web-Based Courier Management System for Enhanced Operational Efficiency and Real-Time Tracking aimed to developed a web-based parcel management and tracking system that streamlined parcel handling processes for small and medium-sized enterprises (SMEs) through real-time tracking, efficient record management, and optimized delivery operations (Chaudhary et al., 2025). The study developed a system with separate interfaces for administrators, employees, and customers to improve parcel management and delivery monitoring. The findings showed that the system enhanced operational efficiency by providing organized parcel records, real-time shipment updates, and better coordination among users. However, the study focused mainly on parcel delivery operations and did not include cargo trip scheduling, flexible payment monitoring, or shipment management for inter-island cargo transport.

CargoExpress PH addressed these limitations by providing a proposed system specifically designed for transportation between Manila and Bohol. Aside from allowing customers to book shipments online and monitor their shipment status through a tracking timeline, the system also enabled administrators to

schedule cargo trips, assign bookings to available trips, monitor payments through Cash, GCash, and Pay Later options, and manage operations using a centralized dashboard. These features supported the operational requirements of small-scale cargo delivery businesses.

Design and Implementation of a Customer-Interactive Cargo Delivery System: A Case Study from Türkiye implemented an Interactive Cargo Delivery System, enabling recipients to manage their deliveries through a web-based platform accessible via short messaging system (SMS) (Kabil et al., 2025). The study developed a customer-interactive platform that allowed recipients to confirm delivery addresses, reschedule deliveries, select alternative drop-off locations, and monitor shipments through real-time courier tracking with geo-fenced delivery verification. The implementation of the system improved same-day delivery success, reduced branch return rates, and achieved a high level of customer satisfaction based on millions of completed shipments. However, the study focused primarily on enhancing last-mile delivery services and customer interaction within a national logistics provider, without addressing broader cargo management functions such as trip scheduling, payment monitoring, and public shipment tracking for inter-island cargo operations.

CargoExpress PH addressed these additional operational requirements by integrating cargo booking, trip scheduling, shipment tracking through a tracking timeline, payment monitoring using Cash, GCash, and Pay Later options, and a public tracking page into a centralized web-based platform. Unlike the earlier study, the proposed system supported both customer and administrator

operations, allowing administrators to manage cargo trips, update shipment statuses, and monitor deliveries while enabling customers to conveniently book, pay for, and track cargo shipments between Manila and Bohol.

Intercity Goods Transportation System aimed to developed a centralized transportation management system that streamlined shipment booking, vehicle allocation, billing, and shipment monitoring to improve the efficiency of intercity goods transportation (Tijare et al., 2026). The study demonstrated that integrating booking management, transport scheduling, billing, and shipment monitoring into a single platform improved operational coordination and supported more efficient transportation services. However, the study mainly emphasized transportation management processes and did not include customer-accessible shipment tracking, flexible payment monitoring, or web-based public shipment tracking.

The proposed system extended these concepts by providing a web-based cargo delivery booking and tracking platform that supported both customers and administrators. The system enabled customers to book shipments online, monitor shipment progress through a tracking timeline, and track cargo using a public tracking page. It also allowed administrators to manage trip schedules, assign shipments, monitor payments, update cargo status, and oversee cargo operations through a centralized dashboard, making the system more suitable for small-scale inter-island cargo delivery businesses.

The reviewed studies demonstrated that web-based systems and

information technology played a significant role in improving logistics, cargo delivery, and transportation operations by enhancing efficiency, shipment visibility, customer interaction, and data management. The studies consistently emphasized the importance of centralized platforms in streamlining cargo booking, delivery management, shipment tracking, and administrative processes while reducing manual errors and improving operational performance. However, most of the reviewed systems were developed for specific applications such as warehouse operations, enterprise logistics, last-mile delivery, courier services, or general cargo management, making them less suitable for the operational requirements of small-scale inter-island cargo businesses. Guided by the strengths and limitations identified in these studies, the proposed CargoExpress PH was designed as a centralized web-based cargo delivery booking and tracking system that integrates online booking, trip scheduling, tracking timeline, payment monitoring, public shipment tracking, and communication features into a single platform. By combining these essential logistics functions, the proposed system aimed to provide a practical, efficient, and accessible solution for managing cargo delivery operations between Manila and Bohol.

The development of the proposed system is supported by existing laws in the Philippines that promote the use of information technology and ensure the protection of digital transactions and personal information.

Moreover, Republic Act No. 8792, Section 6 stated that:

“Information shall not be denied legal effect, validity or enforceability solely on the grounds that it is in the data message purporting to give rise to such legal

effect, or that it is merely referred to in that electronic data message (Republic of the Philippines, 2000)".

This provision recognizes electronic data and digital transactions as legally valid. In relation to the proposed study, the web-based system allows customers to submit cargo booking information electronically and enables administrators to process shipment records through a digital platform. This supports the transition from traditional manual booking processes to electronic transactions within the logistics service.

Furthermore, Section 11 of Republic Act No. 10173

"The processing of personal information shall be allowed, subject to compliance with the requirements of this Act and other laws allowing disclosure of information to the public and adherence to the principles of transparency, legitimate purpose and proportionality. (Republic of the Philippines, 2012)".

In this regard, the state establishes the general data privacy principles governing the processing of personal information. The law emphasizes that personal data must be collected for legitimate purposes and protected from unauthorized access or misuse. In relation to the proposed study, the web-based system collects and stores customer information such as sender and receiver details, contact numbers, and cargo information. Therefore, the system must ensure that personal data is handled responsibly and securely in accordance with the principles of data privacy and protection.

THE PROBLEM

Statement of the Problem

This study aimed to design and develop a Web-Based Cargo Delivery Booking and Tracking System that improved and streamline the process of booking and monitoring cargo deliveries in selected areas such as Bohol and Manila.

Specifically, this study sought to address the following questions:

1. What is the current process of booking and tracking cargo deliveries in selected areas such as Bohol and Manila?
2. What challenges or constraints are encountered in the current cargo delivery booking and tracking system?
3. How should the proposed system be design in terms of:
 - A. System Rules;
 - B. Context Diagram;
 - C. Event List;
 - D. Process Specification;
 - E. Program Hierarchy;
 - F. Database Design;

- G. Data/Test Cases;
- H. Technical Requirements;
- I. Cost-Benefit Analysis; and
- J. Interface Design?

4. What technology action plan should be proposed to effectively implement the application?

Scope and Limitations

This study focuses on the development of CargoExpress PH, a web-based cargo delivery booking and tracking system for selected service areas including Bohol, Metro Manila, and nearby provinces such as Cavite, Batangas, Laguna, and Bulacan. The system allows customers to book deliveries by entering sender and receiver details, selecting routes, and tracking cargo through a step-by-step timeline.

The system is accessible through a web browser on different devices, making it convenient for both customers and administrators. Administrators can manage bookings, assign shipments to trip schedules, update delivery status, and upload proof of pickup and delivery. The system also includes chat support with a basic chatbot for inquiries.

The system supports cash and GCash payments through PayMongo, including a pay-later option where the administrator can record a full or partial payment amount and track the remaining balance. It also records whether the sender or receiver is responsible for the payment.

The system also includes a public tracking page where receivers can enter a reference number to view the cargo's current delivery status, shipment progress, and tracking timeline without requiring an account. This feature allows receivers to conveniently monitor updates such as booking confirmation, cargo pickup, trip assignment, and delivery progress through a centralized tracking interface accessible through a web browser.

However, the system had several limitations. It was limited to predefined routes between Bohol and selected areas near Metro Manila and did not support nationwide or international delivery. It required an active internet connection to function and could not operate offline. The system did not include real-time GPS tracking; instead, tracking updates depended on administrators manually updating the shipment status. The system did not support advanced AI-based route optimization or automatic complaint and dispute management features. These limitations defined the boundaries of the study and provided direction for future enhancements of the system.

Significance of the Study

The researchers believed that the success of this study would be beneficial to the following:

Customers. The system will provide customers with a more efficient and transparent way to book and track their shipments. It will allow them to monitor delivery status, view costs, and receive timely notifications.

Administrator. The system will help administrators manage shipments, payments, and update delivery statuses in a centralized platform. It will also allow accurate pricing based on weight and proper handling of promissory payments.

Future Researchers. This study will provide the researchers the opportunity to apply their knowledge and skills in developing a web-based logistics system. It will also serve as a reference for future researchers by providing insights into system design, database integration, and the implementation of booking and tracking features.

RESEARCH METHODOLOGY

Research Design

This study employed a developmental research design, which focuses on the design, development, and evaluation of a system intended to enhance existing operational processes. This approach is appropriate as the study aimed to design and develop CargoExpress PH, a web-based cargo delivery booking and tracking system.

The proposed system provides a structured and accessible platform where users can book shipments and monitor the status of their cargo through a timeline-based tracking feature. The system was designed to improve efficiency, organization, and accessibility of cargo delivery operations by utilizing a centralized web-based platform that supports booking management, trip scheduling, and status updates. The development of CargoExpress PH was

expected to support logistics operations by offering a more organized and user-friendly approach to handling cargo transactions, enabling both customers and administrators to access relevant information in a timely and convenient manner.

Research Environment and Participants

The study was conducted at Batuan, Bohol, where CargoExpress PH operates through a home-based setup that served as the central hub for managing cargo transactions. The operational environment involved daily booking coordination, shipment assignment, and delivery scheduling.

The participants consisted of fifteen (15) customers who used CargoExpress PH to send packages for personal or business purposes, and two (2) administrators or business owners who managed daily operations.

Data Gathering Instrument

The researchers were given permission to conduct interviews and discussions with the selected customers and administrators of the CargoExpress PH cargo delivery service to gather detailed information about the current cargo delivery workflow, booking and tracking processes, and communication methods within the operation. These methods were used to collect relevant data that served as the foundation for the development of the CargoExpress PH system.

Interview and Discussion

The researchers conducted interviews and discussions with the customers

and administrators of the CargoExpress PH cargo delivery service in Batuan, Bohol. These sessions focused on understanding the current cargo delivery process, including how booking requests were received, how shipment updates were communicated to customers, and how delivery schedules were coordinated.

The researchers also gathered input regarding the operational challenges encountered and the preferred features and requirements for the proposed web-based system. The data collected from these interviews and discussions served as the basis for the design and development of CargoExpress PH, ensuring that it aligned with the actual operational needs of the cargo delivery service.

Documentation Review

The researchers reviewed various documents, including cargo booking records, shipment tracking logs, payment records, and delivery schedules from the CargoExpress PH cargo delivery service. This process helped identify the essential data fields, as well as the input and output design required for the development of the CargoExpress PH system. Additionally, related studies on web-based logistics and cargo delivery systems were examined to determine potential features that could enhance system functionality, security, and efficiency.

Research Procedure

The development of the proposed system followed the System Development Life Cycle (SDLC) using the Agile Model, which emphasizes continuous development, flexibility, and user-centered improvements. The process

organized into five major phases to ensure a structured and systematic development of CargoExpress PH, a web-based cargo delivery booking and tracking system.

Phase I: Requirements Gathering

This phase involves gathering requirements through interviews and observations of the current cargo delivery process in Batuan, Bohol. The researchers identified issues such as booking procedures, lack of shipment tracking visibility, and fragmented communication between customers and administrators. These insights serve as the basis for designing a web-based system aimed at improving efficiency and service quality.

Phase II: Design the Requirements

This phase involves designing the requirements of the system based on the gathered data. The researchers define the system architecture and key components, including the cargo booking workflow, tracking timeline structure, trip management logic, and payment processing flow. Security features such as user authentication, role-based access control, and data protection are also integrated to ensure the system meets the operational and security needs of the cargo delivery service.

Phase III: Coding

This phase involves coding the system based on the designed architecture using React for the web interface and Supabase for the backend database and authentication. Core functionalities include cargo booking, tracking timeline display, trip and order management, payment monitoring with PayMongo/GCash integration and pay-later option, proof of pickup and delivery upload, admin chat support, public tracking page, notifications, and an administrator dashboard. A user-friendly interface is developed for both customers and administrators, and testing is conducted to ensure that all features function correctly and meet the identified requirements.

Phase IV: Testing

This phase involves testing the system to ensure all components function correctly. Each feature, including the booking process, tracking timeline, trip management, payment processing, chat support, and notification system, is checked for accuracy, functionality, and compatibility. Identified bugs and issues are resolved to ensure the system meets all requirements and is ready for deployment. Unit testing and system testing are conducted to verify that each module performs as expected, and the system is evaluated for performance, usability, and reliability under different conditions.

Phase V: Deployment

This phase involved the deployment of the CargoExpress PH system in the actual operational environment in Batuan, Bohol. The system was set up in the live environment, and user accounts for both customers and administrators were

configured to support cargo booking, tracking, and delivery operations. Necessary adjustments were implemented to ensure that the web-based system functioned properly within the operational workflow. System performance was monitored during initial use to identify possible issues and improvements, while additional support was provided to users to ensure effective utilization of the system.

Phase VI: Feedback

This phase involves gathering feedback from users after the system is deployed. Customers and administrators share their experiences, suggestions, and concerns regarding the system's performance and usability. The collected feedback is used to evaluate the effectiveness of the system and identify areas for improvement, ensuring continuous enhancement and better user satisfaction. The results are analyzed to guide future updates and enhancements of the system.

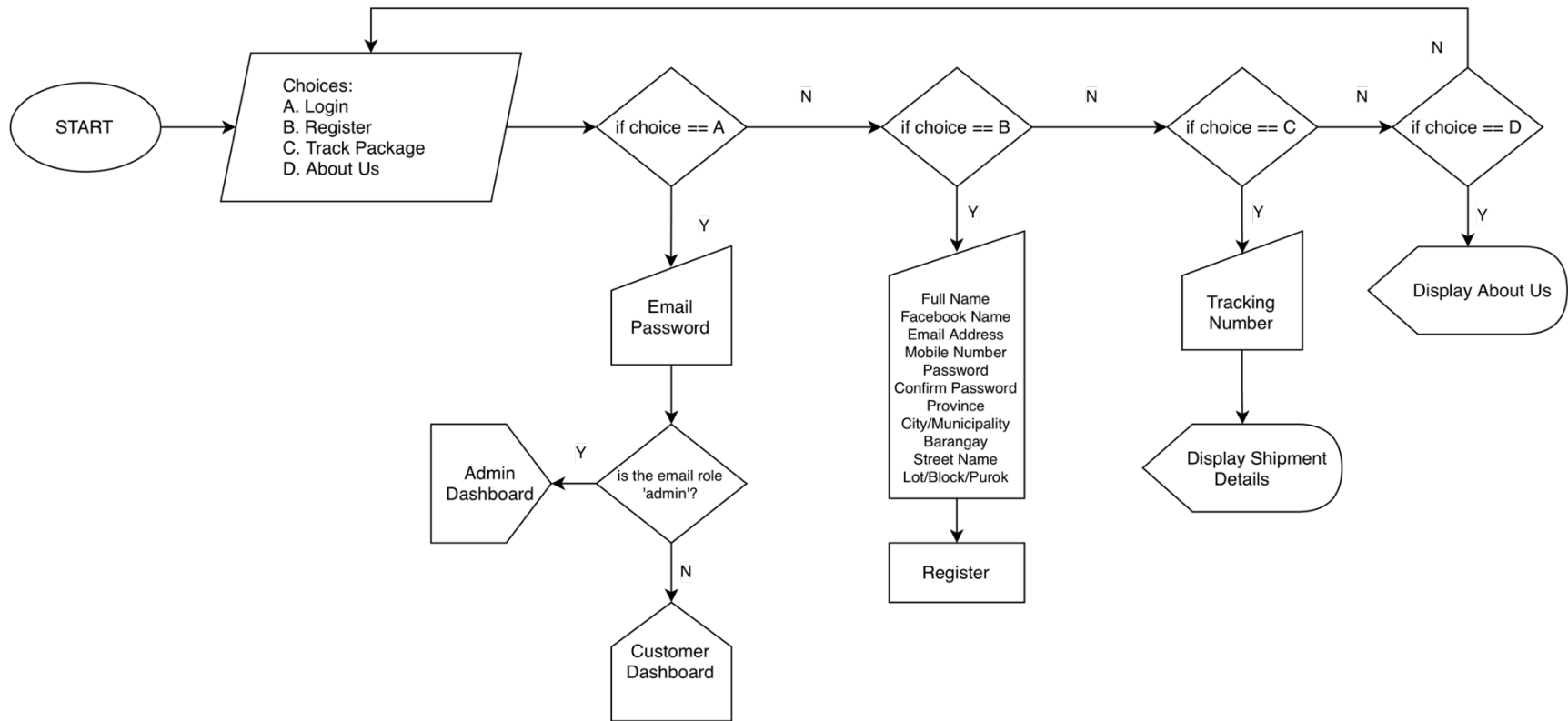


Figure 2. Landing Page Flowchart

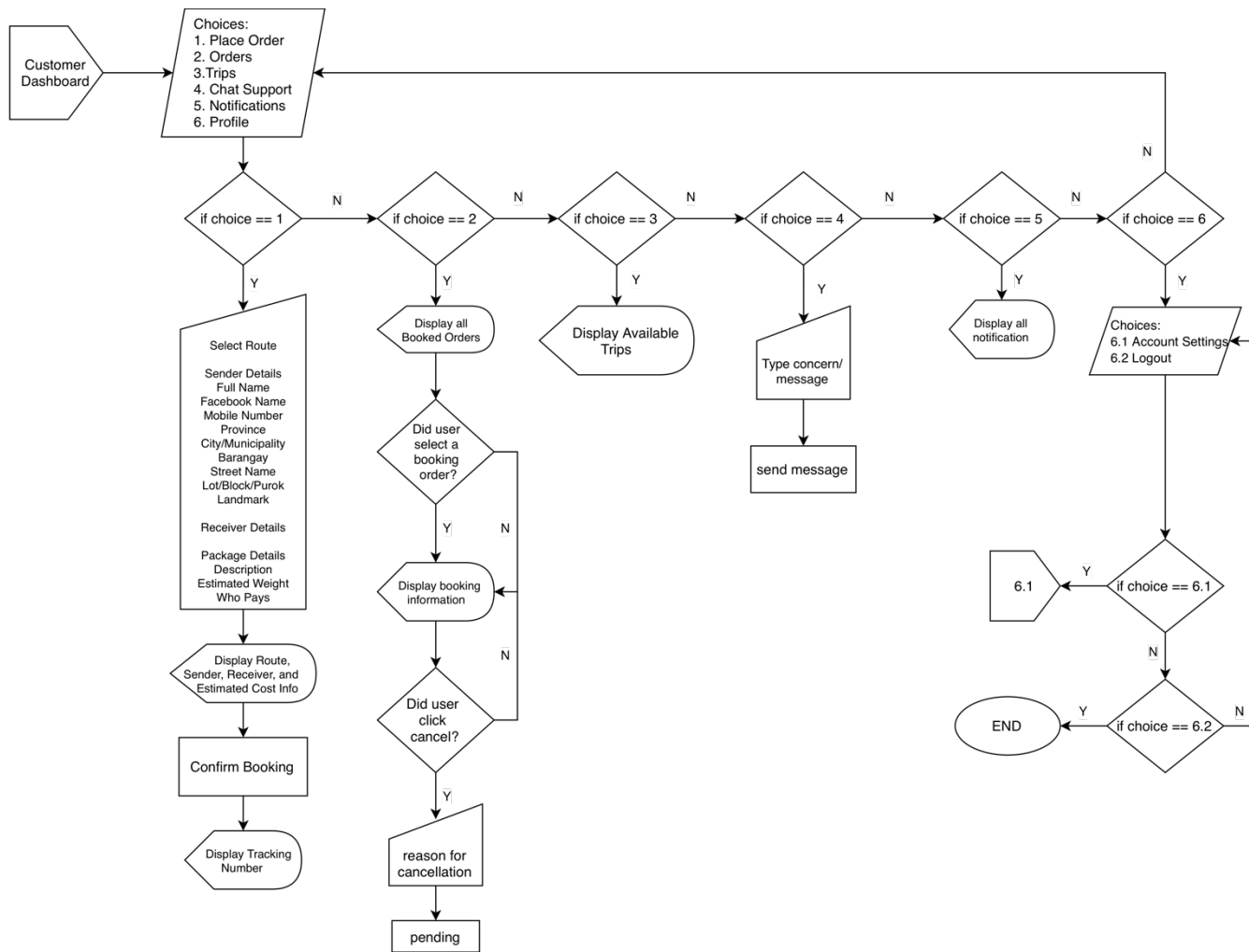


Figure 3. Customer Dashboard Flowchart

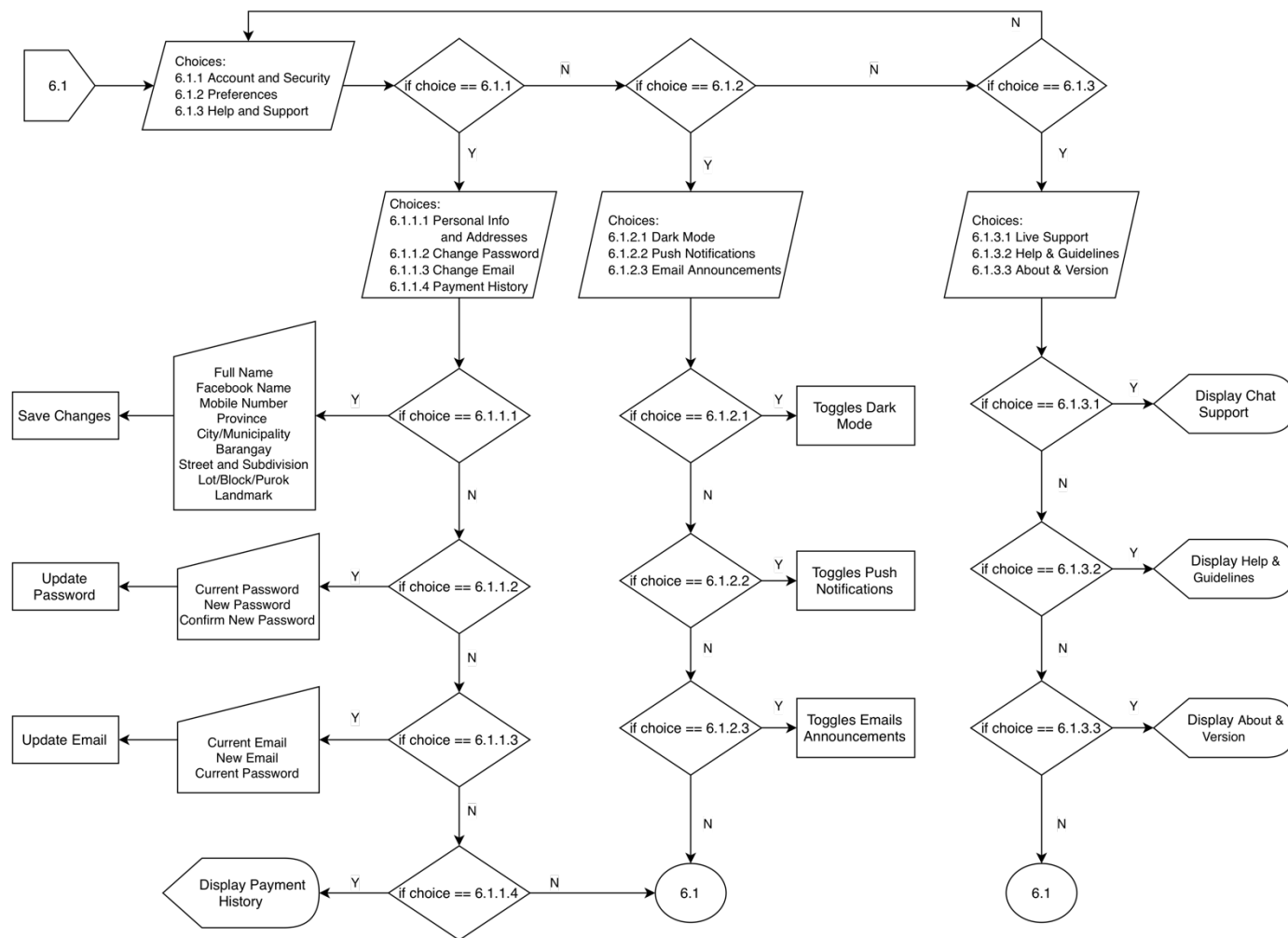


Figure 4. My Profiles Flowchart

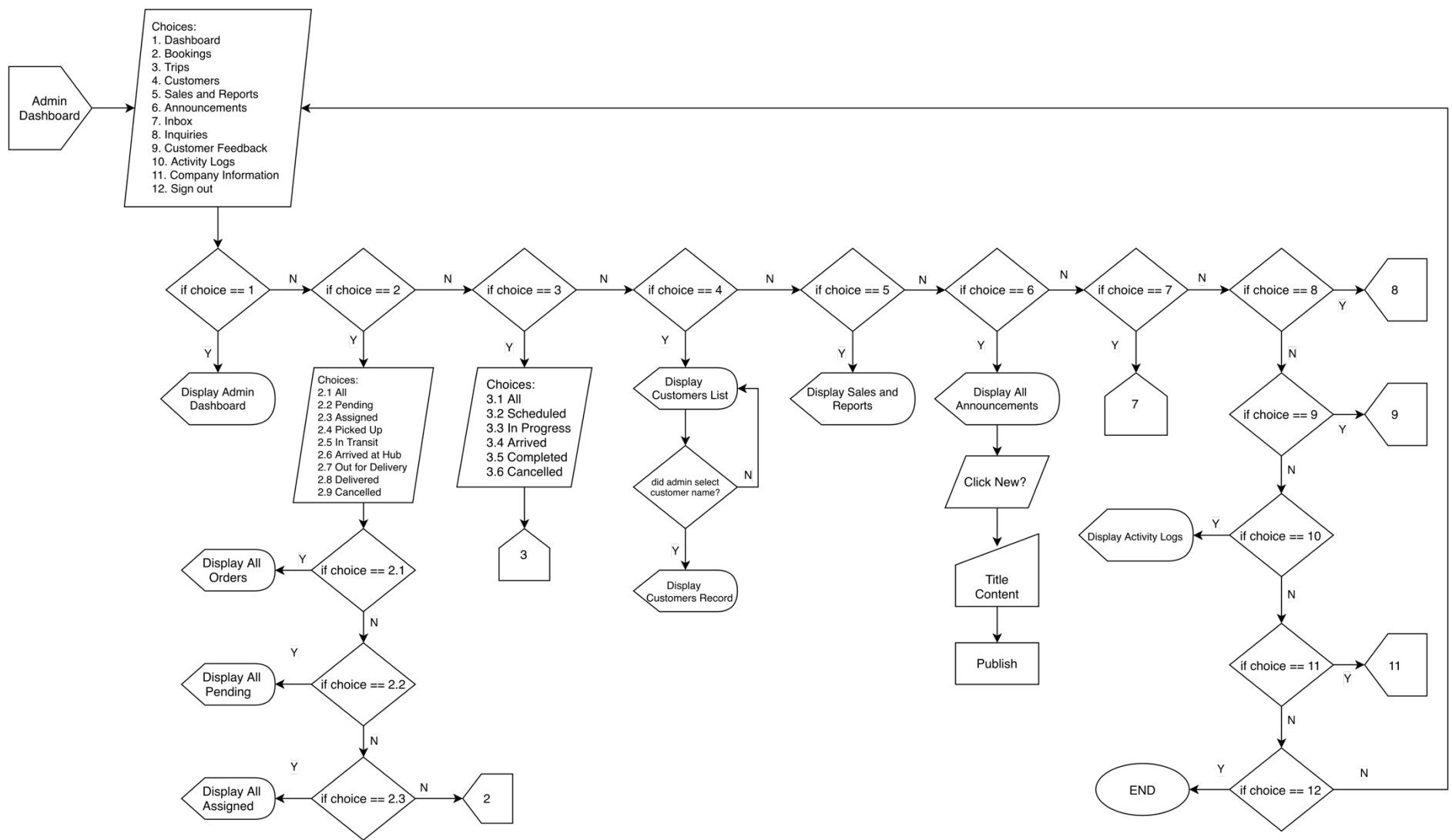


Figure 5. Admin Dashboard Flowchart

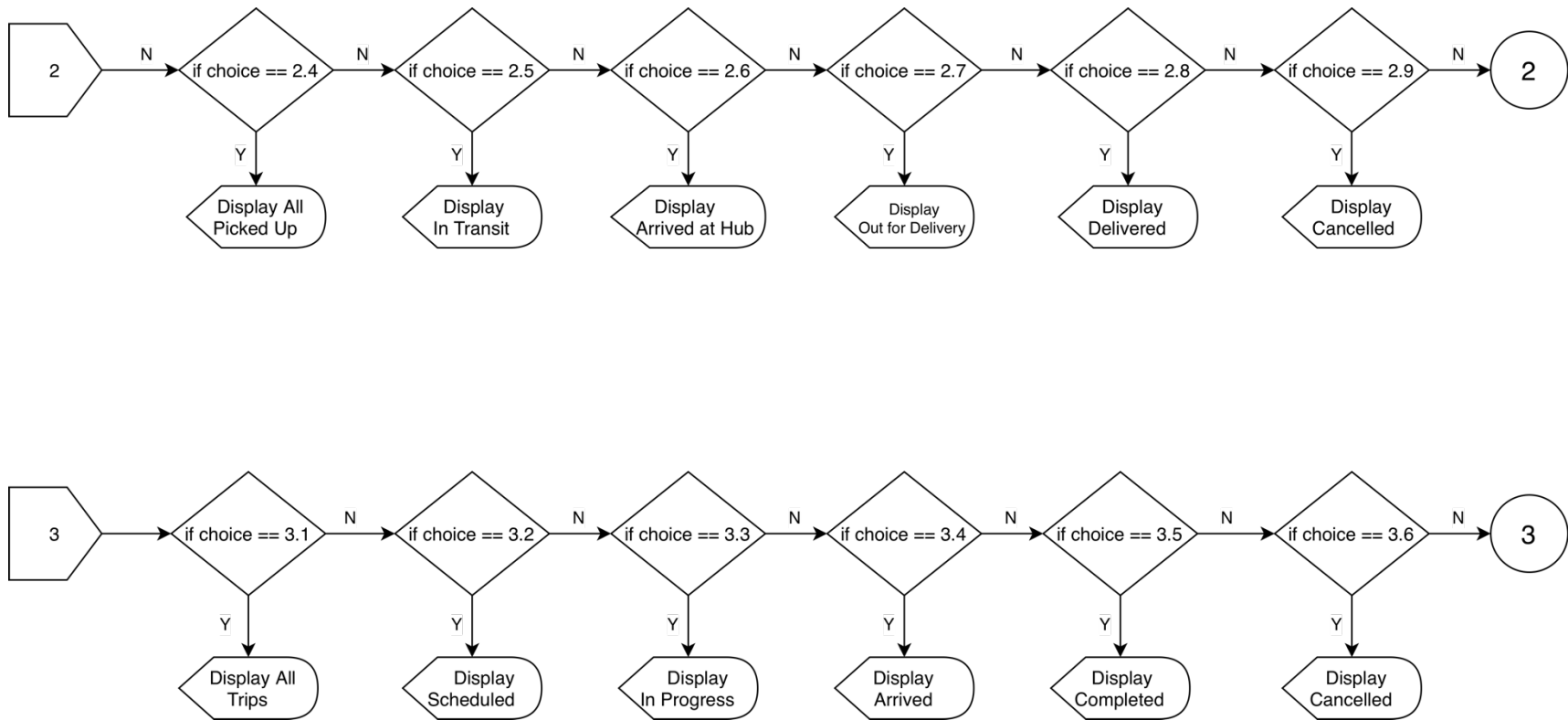


Figure 6. Orders and Trips Management Flowchart

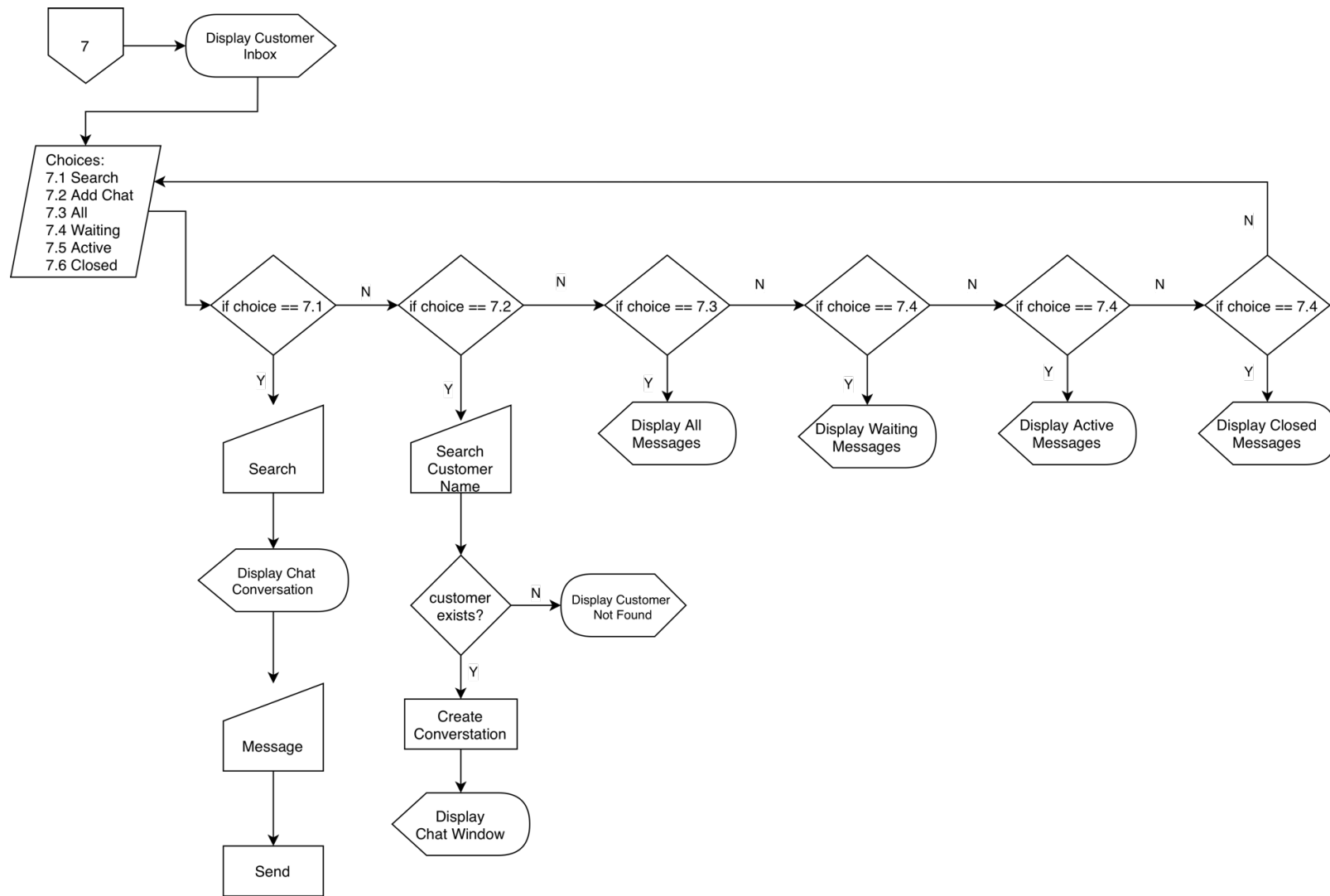


Figure 7. Inbox Admin Flowchart

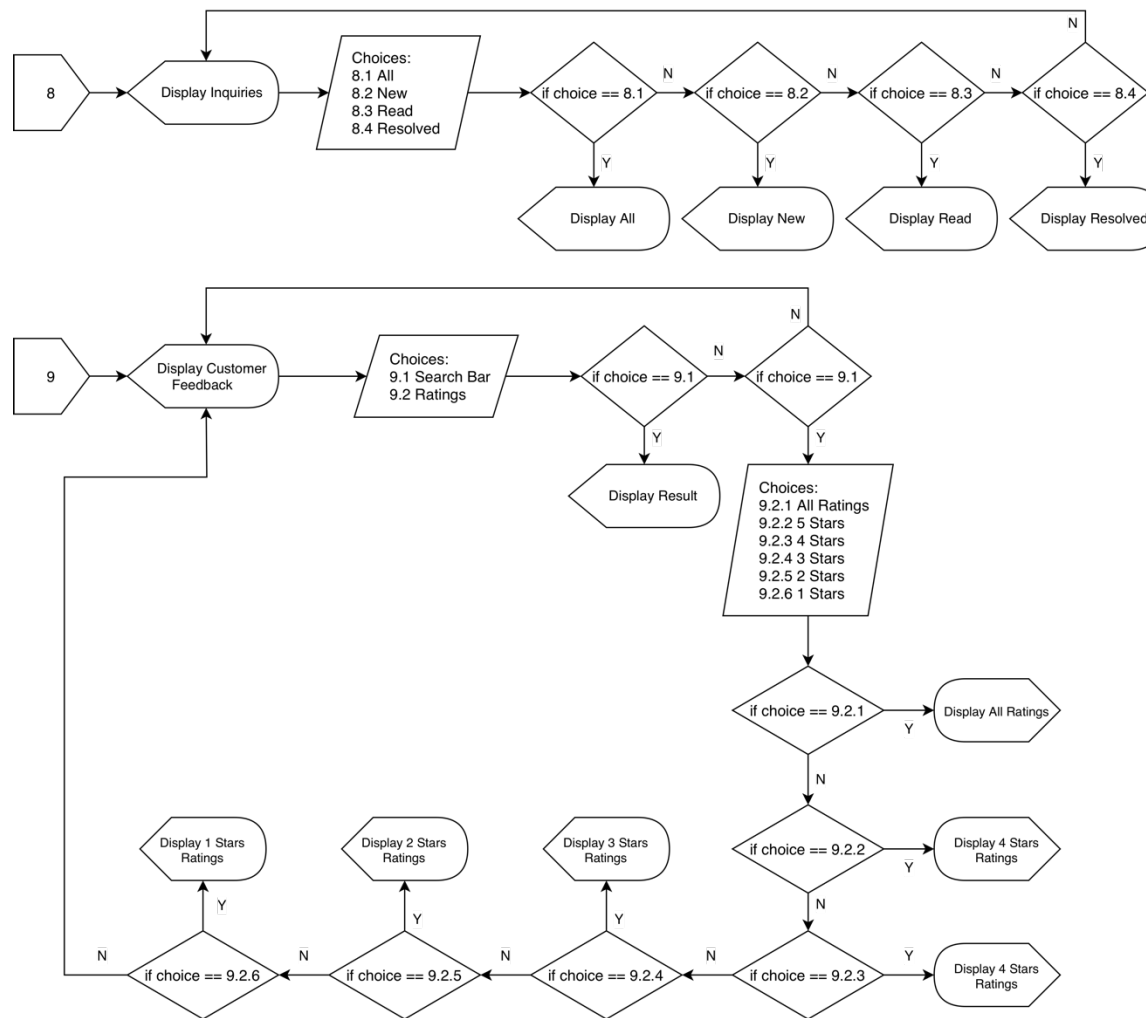


Figure 8. Inquiries and Customer Feedback Flowchart

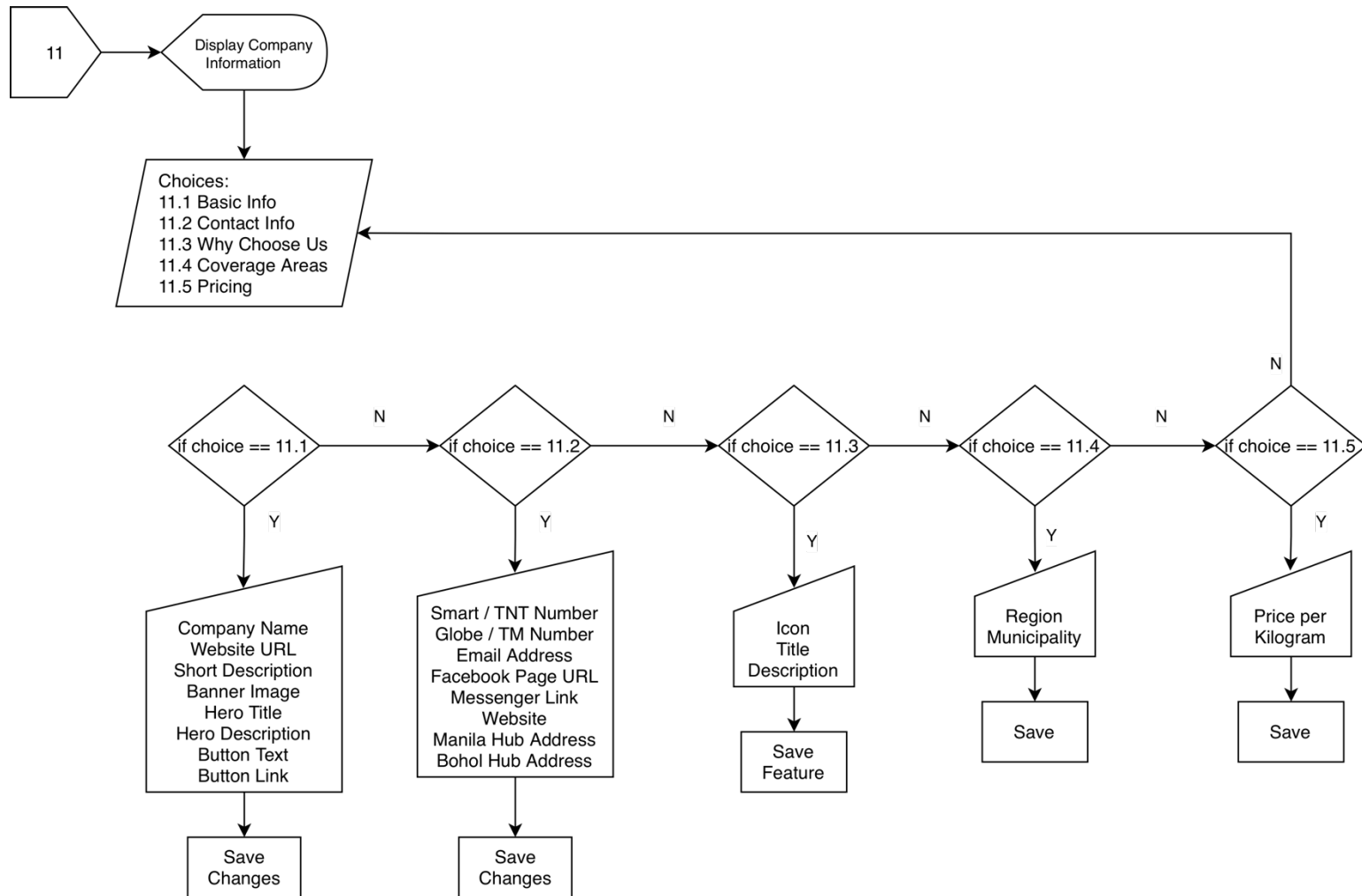


Figure 9. Company Information Flowchart

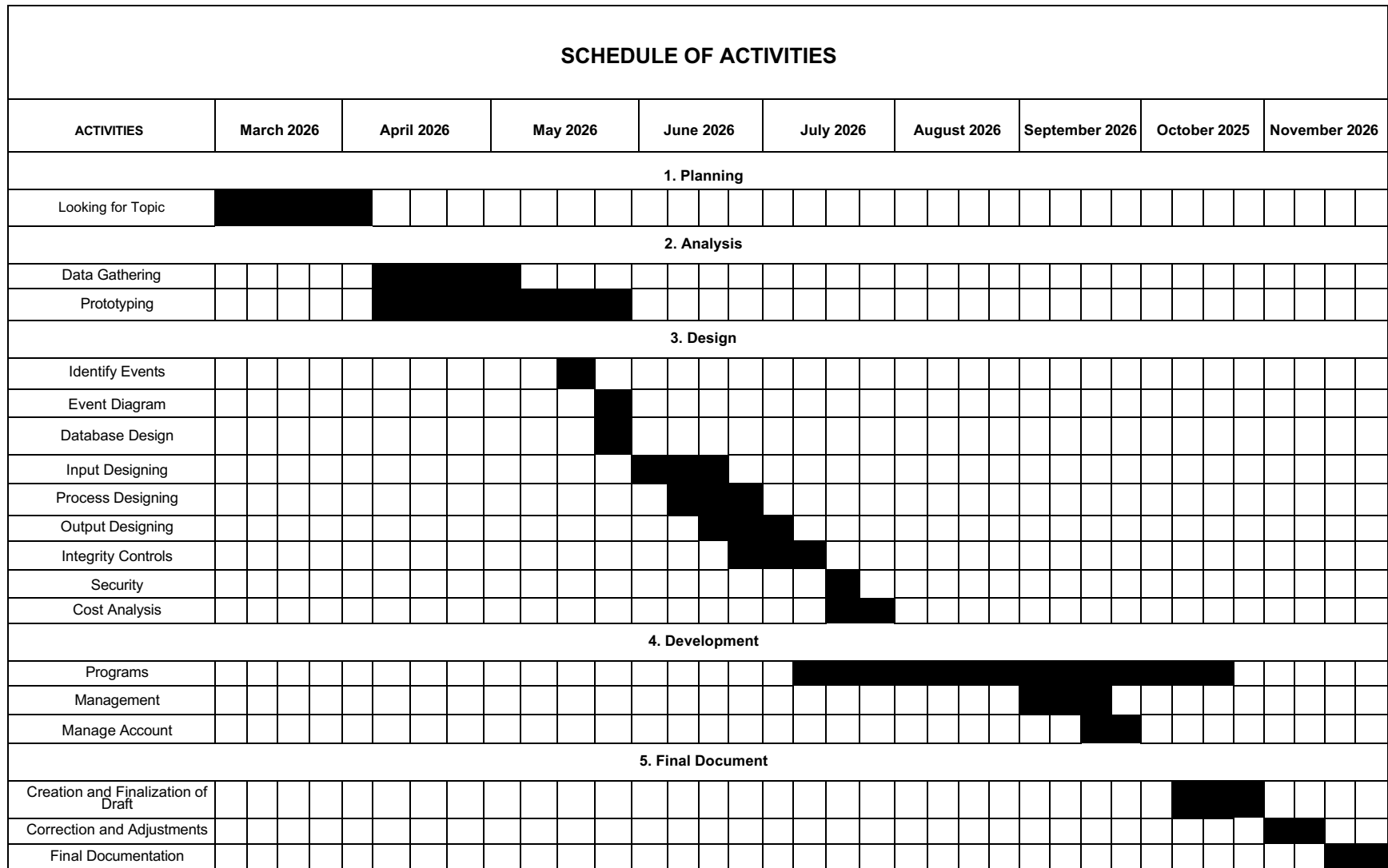


Figure 10: Gantt Chart

DEFINITION OF TERMS

For better understanding of the study, the following terms are operationally defined

CargoExpress PH – A web-based cargo delivery booking and tracking system developed in this study that allows customers to book shipments, monitor cargo status through a tracking timeline, and enables administrators to manage delivery operations.

Pay-Later Transaction – A system feature that allows partial or delayed payment while tracking the remaining balance.

Progressive Web Application (PWA) – A website that functions like a native application and is installable directly to a user's phone or computer.

Public Tracking Page – A system feature that allows receivers to view shipment status using a reference number without logging in.

Reference Number – A unique system-generated identifier used to track and access shipment details.

Tracking Timeline - Shows the progress of a cargo through predefined status updates

Trip Management – A system function that allows the administrator to create schedules, assign cargo to trips, and update delivery status.

Chapter 2

PRESENTATION OF THE PRESENT AND PROPOSED SYSTEM

A. PRESENT SYSTEM

System Narrative

CargoExpress PH currently manages its cargo delivery transactions through several personal and social communication channels. Before every scheduled trip, the client posts an announcement on the business Facebook page and personal Facebook account to inform customers about the upcoming travel schedule and the availability of cargo delivery services. Customers who are interested in sending cargo may respond by commenting on the Facebook post, sending a private message through Messenger, sending a text message to the client's personal mobile number, or calling directly to ask about the trip schedule, pickup arrangements, delivery coverage, and other shipment-related concerns.

The client personally entertains these inquiries and provides the necessary information regarding the available trip, expected schedule, pickup arrangement, and other details requested by the customer. Communication may continue through Facebook Messenger, text messaging, or phone calls depending on the customer's preferred means of contact. Some customers also use these communication channels to confirm that they will be sending baggage or cargo for the upcoming trip. Once the customer confirms the shipment, the client provides or arranges the schedule for when the cargo will be picked up from the sender.

On the agreed pickup schedule, the cargo is collected from the sender. The

final delivery charge is determined only after the shipment has been physically weighed. After obtaining the actual weight, the client informs the customer of the corresponding amount and confirms the payment arrangement. Payment may be made through cash or GCash and may be settled either in full or partially. Depending on the agreement, the sender may pay for the shipment, while in some transactions the receiver settles the payment after receiving the cargo. In cases where the customer is unable to complete the payment immediately, the customer may provide a promised payment date indicating when the remaining balance will be settled.

After the cargo has been collected and the payment arrangement has been confirmed, the shipment is brought to the storage area or warehouse where it is kept together with the other cargo scheduled for the same trip. The shipments are gathered and prepared until they are ready for loading.

While the cargo is being prepared for transportation, customers may continue to contact the client through Messenger, text messages, or phone calls to ask for updates regarding their shipment or the status of the trip. The client personally responds and provides the available information regarding the cargo, planned departure, and current progress of the trip.

Before departure, the cargo is loaded into the vehicle together with the other shipments assigned to the trip. Once loading has been completed, the client posts photographs on Facebook to inform customers that the cargo has already been loaded and that the trip is ready for departure. The vehicle then travels toward the

scheduled destination.

Upon arrival at the destination, the cargo is unloaded and arranged according to the planned delivery sequence. The client and delivery personnel determine which shipments will be delivered first and organize the delivery order based on the route they intend to take. Customers whose shipments are scheduled for delivery are contacted and informed accordingly.

The delivery continues from one customer to the next until the shipments have been handed over to their respective receivers. If a remaining payment is due, the sender or receiver may settle the balance during delivery. In some cases, the cargo may already be delivered even if the payment has not yet been completed. The customer then provides a promised payment date for settling the remaining balance through the agreed payment method. Once the cargo has been received, the payment arrangement has been confirmed, and any remaining balance is subsequently settled according to the agreed schedule, the transaction for that shipment is considered completed.

Constraints

The study identified several constraints associated with the present cargo delivery process of CargoExpress PH. The business currently relies on Facebook posts, Facebook Messenger, text messages, and phone calls for trip announcements, customer inquiries, shipment confirmations, pickup arrangements, and delivery updates. Since these transactions are communicated through different channels, shipment information and customer conversations are

handled separately and must be reviewed manually whenever the client needs to confirm cargo details, pickup schedules, payment arrangements, or delivery information.

Another constraint involves the manual recording and monitoring of shipment and payment information. The final shipping amount is determined only after the cargo has been physically picked up and weighed, after which the client communicates the corresponding amount to the sender or receiver. Payments may be made through cash or GCash and may be settled in full or partially. In some cases, the shipment may already be delivered while a remaining balance is still unpaid, and the customer provides a promised payment date for settling the balance. These different payment arrangements require the client to manually monitor the amount paid, remaining balance, responsible payer, and agreed payment date for each shipment.

Shipment monitoring also depends on direct communication between the client and the customers. Customers who want to know the status of their cargo usually send messages, text the client's personal number, or make phone calls to request updates. The client responds individually based on the available information regarding the cargo, trip departure, current location, and expected delivery. Facebook posts and photographs are also used to announce when cargo loading has been completed and when the vehicle is ready to depart.

Cargo preparation and delivery scheduling are likewise handled manually. After pickup, shipments are brought to the warehouse and grouped with the other

cargo scheduled for the trip. Upon arrival at the destination, the cargo is unloaded and arranged before the client and delivery personnel determine which shipments will be delivered first. The sequence of deliveries is planned based on their intended route, and customers are contacted individually as their cargo is prepared for delivery.

The present process therefore requires continuous manual coordination of customer inquiries, booking confirmations, cargo pickup, shipment information, payment balances, promised payment dates, trip updates, and delivery arrangements. As transactions increase, the client must manage several communication channels and shipment records at the same time while ensuring that the information provided to customers remains updated throughout the delivery process.

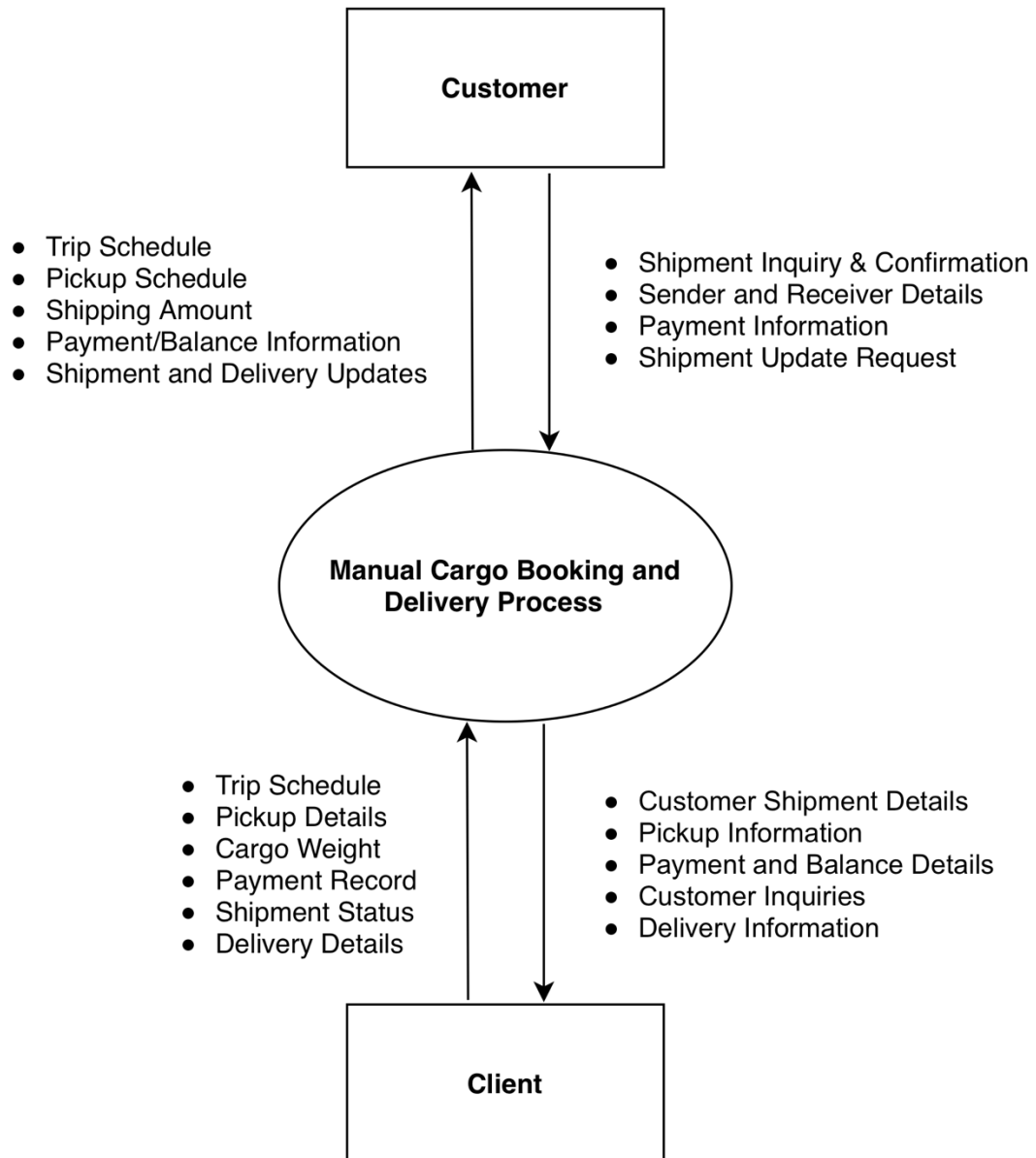


FIGURE 11. Context Diagram of Present System

Event List

1. Customer Inquiry and Shipment Confirmation
2. Cargo Pickup Scheduling
3. Cargo Pickup and Weighing
4. Payment Arrangement and Recording
5. Cargo Storage and Trip Preparation
6. Shipment Status Inquiry and Updates
7. Cargo Loading and Trip Departure
8. Cargo Arrival and Delivery Arrangement
9. Cargo Delivery and Balance Settlement

Event List Diagram

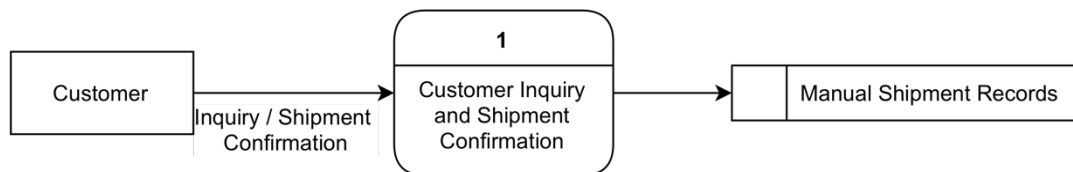


FIGURE 12. Customer Inquiry and Shipment Confirmation

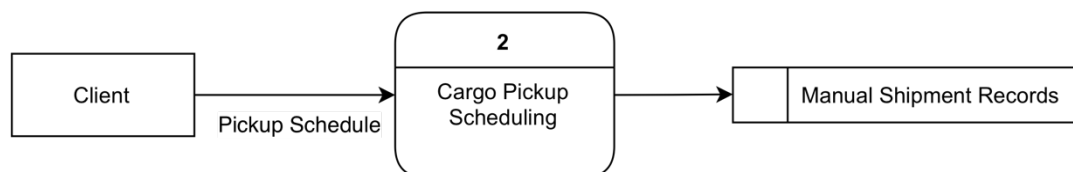


FIGURE 13. Cargo Pickup Scheduling



FIGURE 14. Cargo Pickup and Weighing

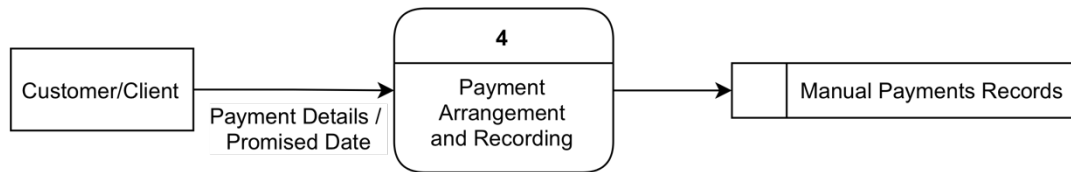


FIGURE 15. Payment Arrangement and Recording

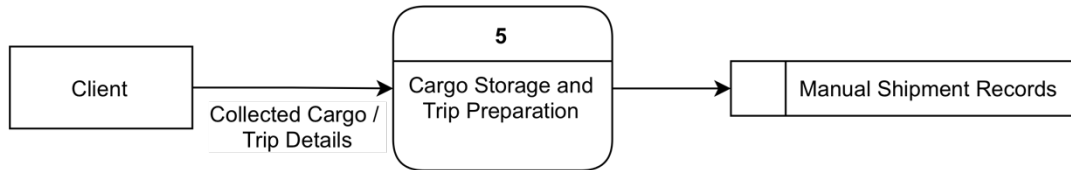


FIGURE 16. Cargo Storage and Trip Preparation

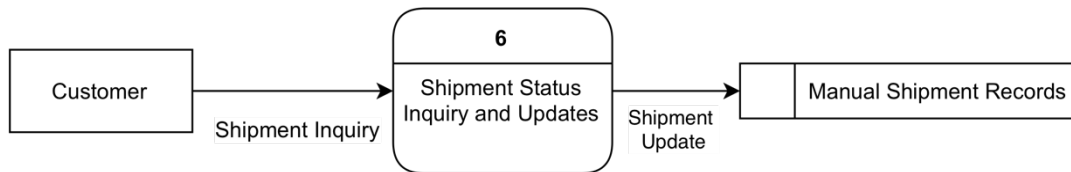


FIGURE 17. Shipment Status and Inquiry and Updates

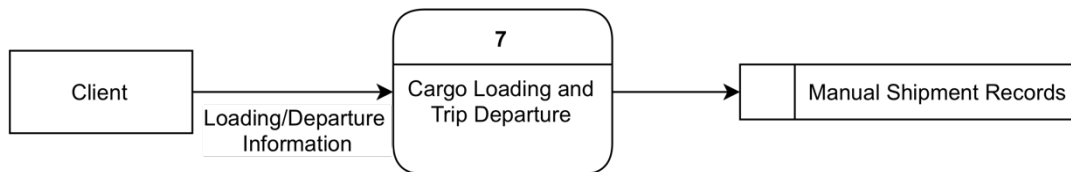


FIGURE 18. Cargo Loading and Trip Departure

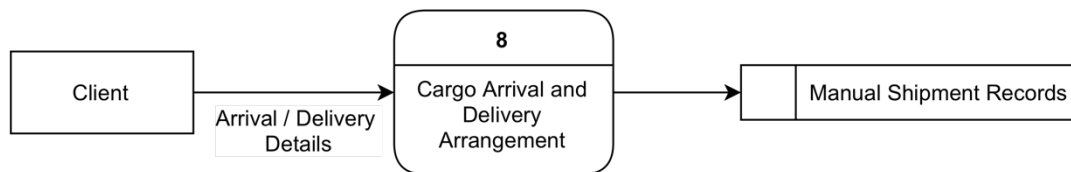


FIGURE 19. Cargo Arrival and Delivery Arrangement

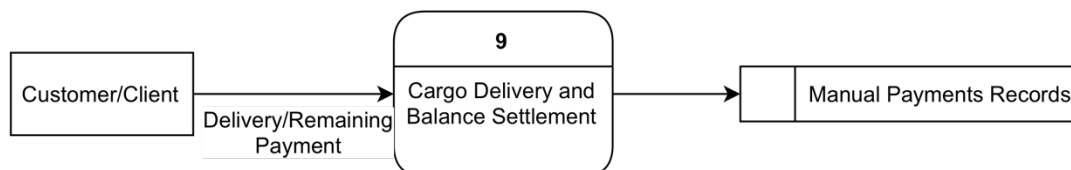


FIGURE 20. Cargo Delivery and Balance Settlement

B. PROPOSED SYSTEM

System Rules

The proposed CargoExpress PH system operates through a web-based application accessible to visitors, registered customers, and the administrator. Visitors may access the public pages to view company information, available trip schedules, frequently asked questions, and shipment tracking. A visitor may also send an inquiry through the public contact form without creating an account.

A customer who wishes to send cargo first creates an account by providing personal information, contact details, Facebook name, email address, password, and complete address. After registration, the customer may log in to the system and access the customer dashboard. From the dashboard, the customer can view available trip schedules, existing bookings, shipment information, notifications, payment records, and other customer functions.

To send cargo, the customer opens the booking page and selects the appropriate route. The customer may also select a currently scheduled trip when one is available. The system then collects the sender's information, including the sender's name, contact number, Facebook name, and complete pickup address. The customer also provides the corresponding information of the receiver and describes the cargo or items that will be sent. During booking, the customer indicates whether the sender or receiver will be responsible for payment and may select a preferred payment method. The customer then reviews all entered information before confirming the booking.

After confirmation, the system records the booking and generates a unique tracking number for the shipment. If a specific trip has already been selected, the shipment is associated with that trip. Otherwise, the booking remains available for later assignment by the administrator. Bookings outside the standard service coverage are flagged for administrator review. The confirmed tracking number may be used by the customer to monitor the shipment through the system.

The administrator receives the submitted bookings through the administrative interface. The administrator may review the booking information and assign unassigned cargo to an available trip. The administrator may also create a booking directly for a walk-in customer and may later associate that booking with a registered customer account. Trips are created by the administrator by specifying the route, departure date, and other necessary trip information. The default carrying capacity and shipping rate are obtained from Company Information. Matching unassigned bookings may also be automatically associated with a newly created trip.

When the cargo is physically collected from the sender, the administrator processes the pickup through the system. The shipment is weighed using the actual physical weight of the cargo, and the administrator records this weight in the booking. The total shipping cost is then calculated based on the recorded weight and the applicable price per kilogram. The administrator may apply a discount with a recorded reason to determine the final amount payable. The system does not require the customer to provide the final cargo weight or total shipping cost during

booking because these values are established during pickup. The administrator may also print a QR label for each box, allowing an authorized administrator to scan and open the booking details. All boxes share the booking's shipment status, and scanning does not automatically confirm delivery.

During pickup, the administrator also confirms whether the sender or receiver will be responsible for payment. If the sender is responsible, the payment may be collected through cash or GCash and may be paid either in full or partially. When only part of the amount is paid, the remaining balance is recorded together with a promised payment date. GCash payments may be processed through the integrated PayMongo service, while cash and manually recorded GCash transactions may also be entered by the administrator. If the receiver is responsible for payment, the collection may instead be deferred until delivery. The administrator also attaches pickup photographs before completing the pickup process. The administrator may apply a discount with a recorded reason. The discount is deducted from the shipping cost to determine the final amount payable. When a booking is cancelled, its previous shipping balance is no longer treated as an amount to collect. The administrator reviews the payments received and records the refund or any amount retained with the customer's agreement. Pending and completed refunds are displayed separately.

Once the pickup has been completed, the shipment is recorded as picked up. When the administrator starts the assigned trip, the shipments associated with that trip are updated to indicate that they are in transit. A trip may only start on its

scheduled departure date based on Philippine time. An overdue trip must be rescheduled with a reason before it can start. The system records the actual departure and hub arrival times separately from the planned schedule. When the vehicle reaches the destination, the administrator marks the trip as “Arrived at Hub,” which updates the related shipments to indicate that they have arrived at the destination hub. The system follows the shipment progression from pending or assigned status through pickup, transit, arrival, delivery, and completion.

After the cargo arrives at the destination hub, the administrator processes each shipment individually for delivery. An eligible shipment is updated to indicate that it is out for delivery. The actual ordering of delivery stops is determined operationally by the delivery personnel, since the current proposed system does not contain an automated route-sequencing or route-optimization function.

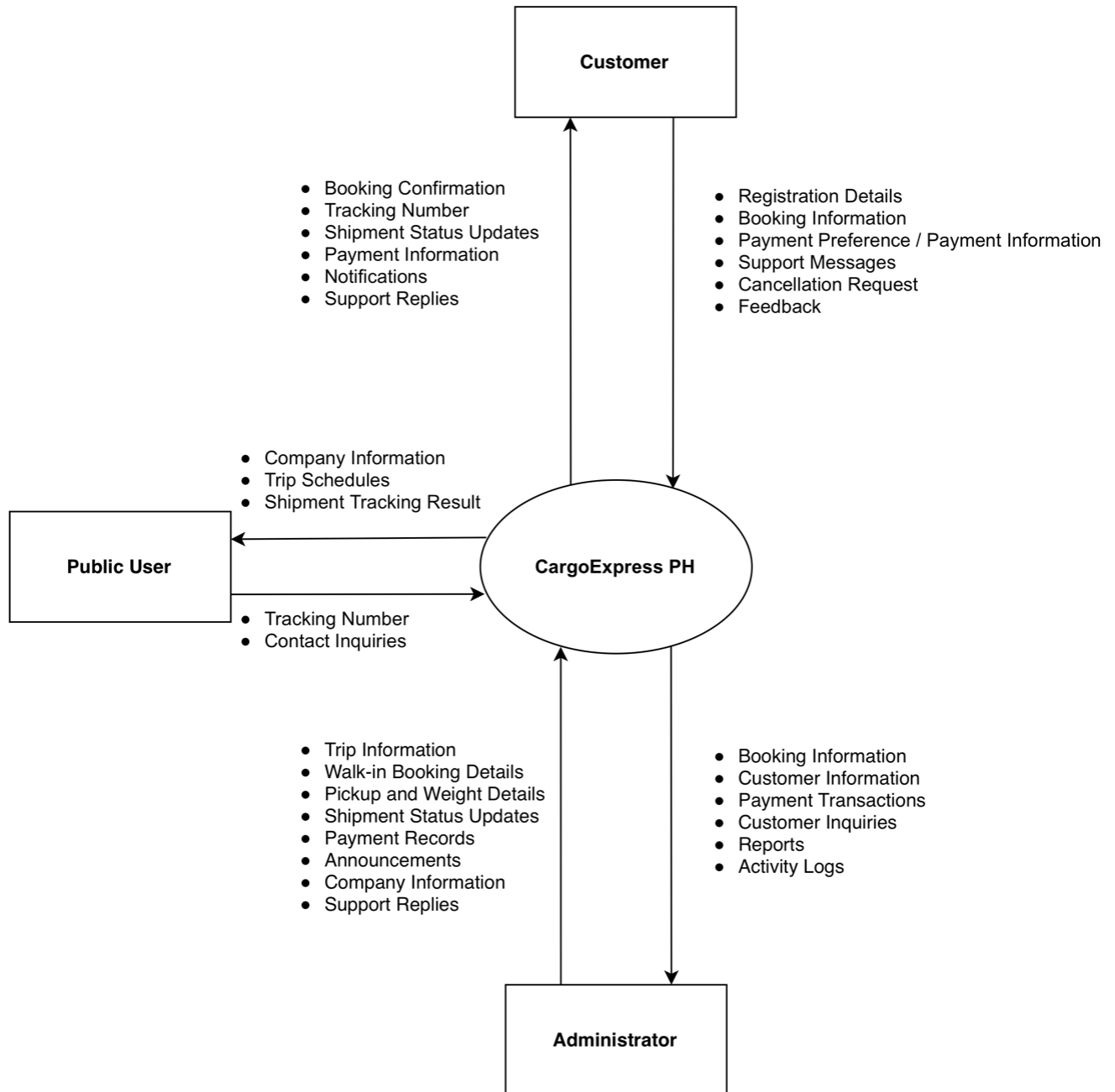
During delivery, the administrator records proof-of-delivery photographs. If an outstanding balance remains, the applicable payment is collected using the available payment options. Once the delivery information and payment details have been recorded, the shipment is marked as delivered. After receiving the shipment, the customer may submit a rating and written feedback regarding the completed transaction.

Throughout the shipment process, the customer may monitor the cargo using the tracking number through the public tracking page or through the customer's account. The tracking interface displays the current shipment status

and its progression through the delivery process. Registered customers may also receive system notifications when shipment information changes.

Customers may communicate with CargoExpress PH through the system's customer support function. Logged-in customers use a menu-driven support assistant to select topics and view information about their shipments, payments, bookings, and delivery concerns. Customers who need further assistance may select "Talk to an Admin" to communicate with the administrator through the system inbox. Visitors who are not logged in may separately submit inquiries through the public contact form.

The administrator may also publish announcements regarding company or trip information, manage registered customer records and their booking histories, review customer feedback, maintain company information, monitor outstanding balances, generate and print monthly reports organized by trip, showing payments received, refunds, and remaining balances, and review recorded activity logs associated with system transactions.



lets talk in person

FIGURE 21. Context Diagram of Proposed System

Event List

1. Customer Registration
2. Cargo Booking
3. Shipment Tracking
4. Trip Scheduling and Cargo Assignment
5. Cargo Pickup and Weighing
6. Payment Recording and Monitoring
7. Shipment Status Monitoring and Updates
8. Cargo Delivery and Completion
9. Cancellation Request and Review
10. Customer Feedback Submission
11. Viewing Customer Records
12. Generating Reports
13. Managing Company Information and Coverage Areas
14. Publishing Announcements
15. Customer Chat Support
16. Public Contact Inquiry Management

Event List Diagram



FIGURE 22. Customer Registration

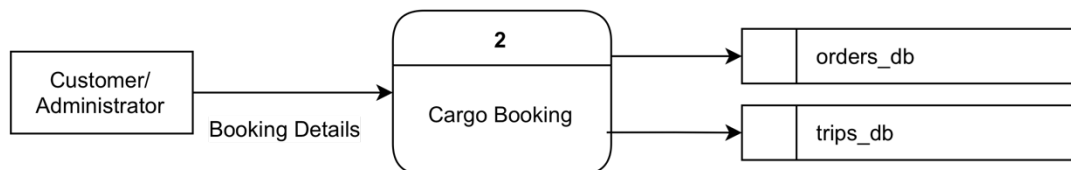


FIGURE 23. Cargo Booking

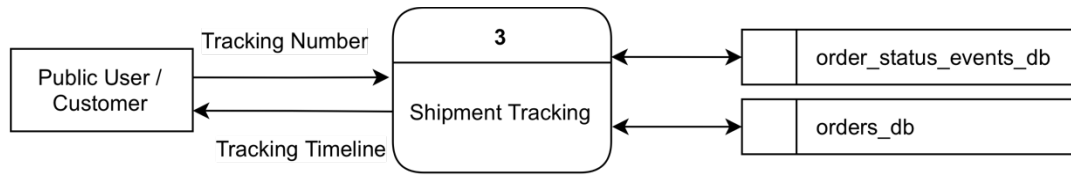


FIGURE 24. Shipment Tracking

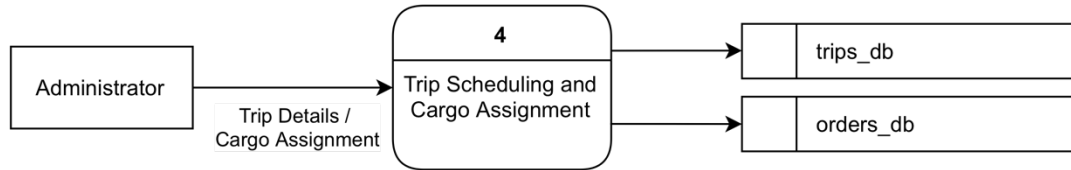


FIGURE 25. Trip Scheduling and Cargo Assignment

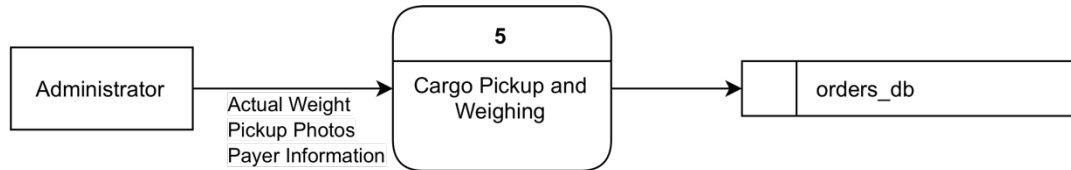


FIGURE 26. Cargo Pickup and Weighing

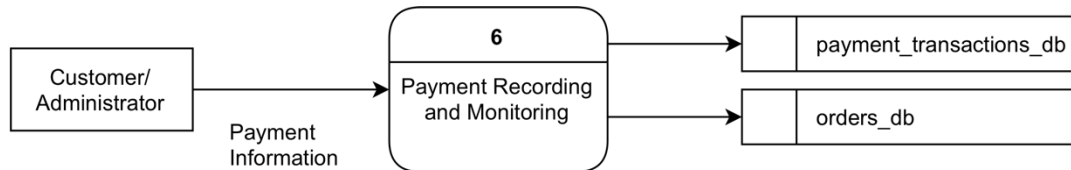


FIGURE 27. Payment Recording and Monitoring

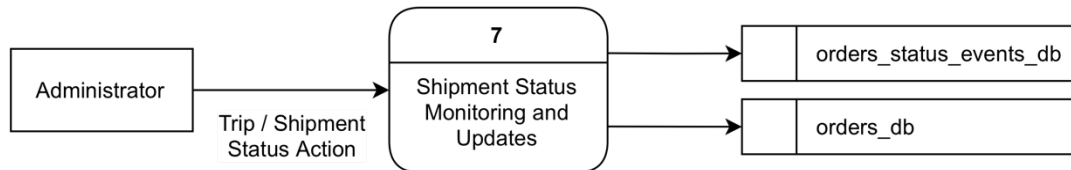


FIGURE 28. Shipment Status Monitoring and Updates

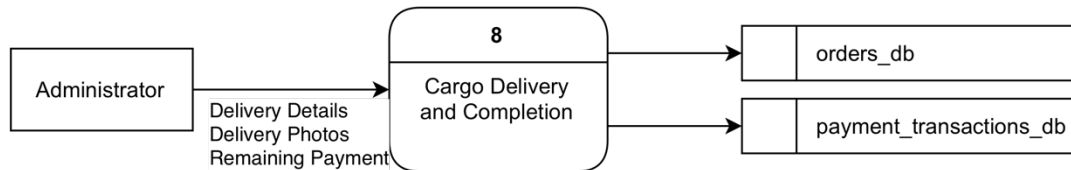


FIGURE 29. Cargo Delivery and Completion

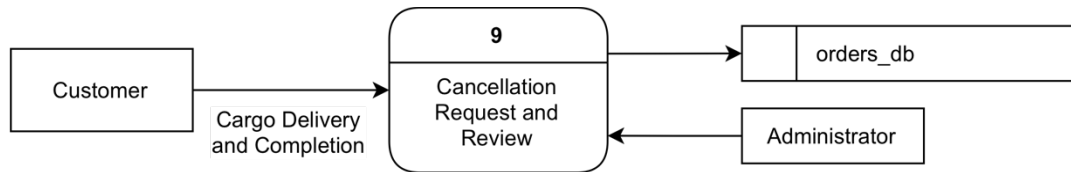


FIGURE 30. Cancellation Request and Review

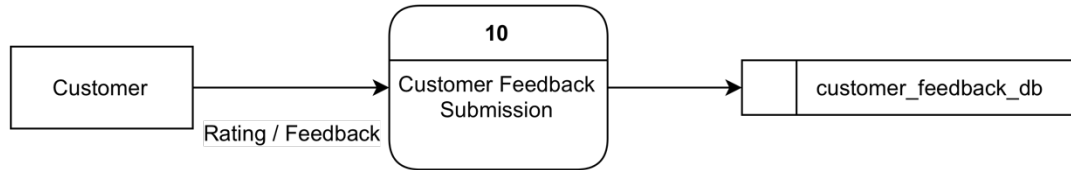


FIGURE 31. Customer Feedback Submission

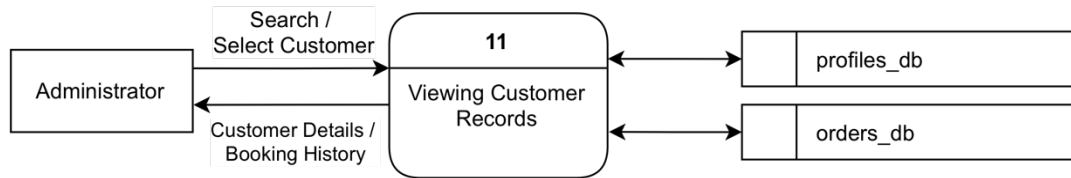


FIGURE 32. Viewing Customer Records

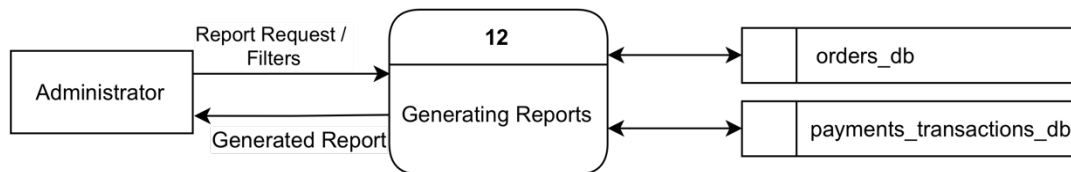


FIGURE 33. Generating Reports

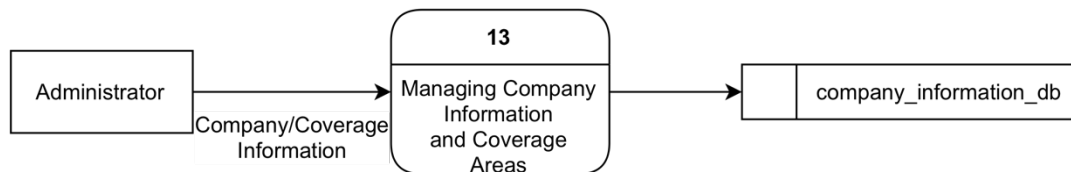


FIGURE 34. Managing Company Information and Coverage Areas

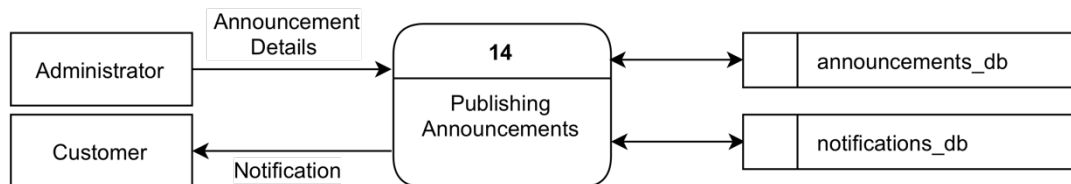


FIGURE 35. Publishing Announcements

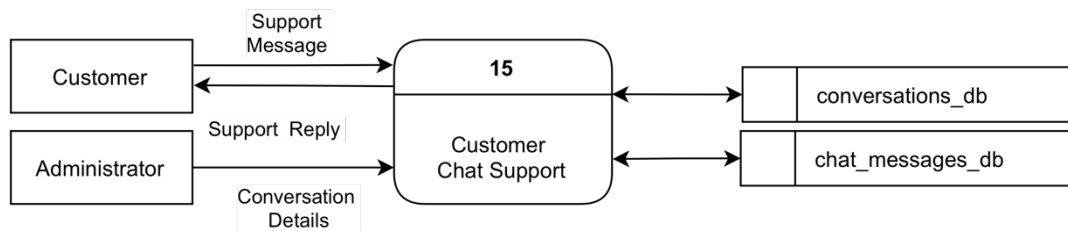


FIGURE 36. Customer Chat Support

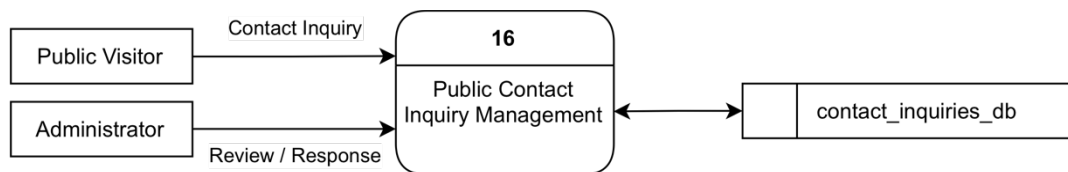


FIGURE 37. Public Contact Inquiry Management

Process Specification

1. Home and About Us

Begin

Display Company Overview

If Visitor selects "Sign In"

Redirect to Login Page

End If

If Visitor scrolls to Contact Us

Display Contact Form

Visitor enters Name, Mobile Number, Email Address, and Message

Visitor optionally checks "Receive announcement updates by email"

Click "Send Message"

If input is valid

System records the inquiry

Display "Message sent! We will contact you soon."

Else

Display field validation errors

End If

End If

End

2. Track Package Process

Begin

Display Tracking Search Box

User enters Tracking Number

Click "Track"

If tracking number is valid and found

 Display Current Status, Route, and Package Details

 Display Shipment Journey Timeline and progress bar

 System automatically refreshes the tracking status

Else

 Display "Shipment Not Found" message

End If

If repeated searches occur too quickly

 Display a temporary cooldown message

End If

End

3. Trip Schedules Process

Begin

 Display list of scheduled trips with Route, Departure Date, and Available Capacity

 If User is not logged in and selects a trip

 Display "Log in to book this schedule" message

 Redirect to Login Page

 Else If Customer selects a trip

 Redirect to Book Shipment with the route and trip pre-selected

End If

End

4. Login Process

Begin

Display Login Form

User enters Email Address and Password

Click "Sign In"

If credentials are valid

 If account role is Administrator

 Redirect to Admin Dashboard

 Else

 Redirect to Customer Home

 End If

Else

 Display "Incorrect password or email" message

End If

End

5. Customer Registration Process

Begin

Display Registration Form — Account Step

Customer enters Full Name, Facebook Name, Email Address, Mobile Number,
Password, and Confirm Password

Click "Continue to Address"

If account input is valid

Display Registration Form — Address Step

Customer selects Province and City/Municipality

Customer enters Barangay, Street, Lot/Block/Purok, and Landmark

Customer checks "I agree to the Terms of Service and Privacy Policy"

Customer optionally checks "Email me announcements"

Click "Create Account"

If registration is successful

 System saves the account details

 Display "Account Created!" confirmation

 Redirect to Customer Home

Else

 Display error message

End If

Else

 Display field validation errors

End If

End

6. Password Recovery Process

Begin

 Display Email Input Form

 User enters Email Address

 Click "Send Reset Link"

 If the reset request is accepted

Display "Check Your Inbox" confirmation

If the email belongs to a registered account

System sends a password reset link

End If

Else

Display reset request error message

End If

If User opens a password reset link

System verifies the link

If link is invalid or expired

Display "Link Expired or Invalid" message

User may request a new link

Else

Display New Password and Confirm Password fields

User enters and confirms the new password

Click "Update Password"

If password meets requirements and both entries match

System submits the password update

If update is successful

Display "Password Updated!" confirmation

Redirect to Login Page

Else

Display password update error message

End If

Else

Display validation error

End If

End If

End If

End

7. Customer Home Process

Begin

Display greeting and booking summary (Total, Active, Delivered)

System loads latest bookings, announcements, and the next available trip

If a next available trip exists

Display trip route, dates, and available capacity

If Customer clicks "Book Cargo for This Trip"

Redirect to Book Shipment with the route and trip pre-selected

End If

End If

If announcements exist

Display latest announcements with option to view and comment

End If

If active shipments exist

Display active shipment cards

If Customer selects a shipment

Redirect to Order Detail

End If

End If

If Customer enters a tracking number in the search box

Click "Track"

Redirect to Track Package

End If

End

8. Book Shipment Process

Begin

Display Booking Form

Customer selects Route and optionally selects an available Trip

Customer enters Sender and Receiver Details

Customer enters Package Description, Who Pays, and Payment Preference

Customer reviews the booking information

Click "Confirm Booking"

If required information is valid and the selected trip is available

System saves the booking without final weight or shipping fee

System generates a Tracking Number

If a trip was selected

System marks the booking as Assigned

Else

System marks the booking as Pending

End If

If the pickup location is outside standard coverage

System flags the booking for administrator review

End If

Display booking confirmation and Tracking Number

Else

Display validation error

End If

End

9. Bookings Process

Begin

Display Customer's bookings and status filters

Customer selects a booking

Display booking details, tracking timeline, and shipment photos

If booking is Cancelled

Display Payment and Refund Summary

Else If the shipment has not been weighed

Display that the shipping fee is not yet available

Else

Display Shipping Fee, Discount, Amount Paid, and Remaining Balance

End If

If Customer edits contact details

If editing is allowed for the current booking status

Validate and save the changes

Else

Display that the details cannot be edited

End If

End If

If online payment is available and a balance remains

Customer enters an amount

If amount is valid and does not exceed the remaining balance

System opens PayMongo checkout

If payment is confirmed by the payment provider

System updates the payment record and balance

Display payment confirmation

Else

Display the current payment status

End If

Else

Display validation error

End If

End If

If Customer requests cancellation

If the booking is eligible for cancellation

Customer enters a reason

System records the request for administrator review

Display confirmation

Else

Display that cancellation cannot be requested

End If

End If

If booking is Delivered and no feedback has been submitted

Customer enters a rating and review

Click "Submit Feedback"

If input is valid

System saves the feedback

Display confirmation

Else

Display validation error

End If

End If

End

10. Notifications Process

Begin

Display notifications grouped by date, each marked read or unread

If Customer selects a notification

If notification relates to a booking or trip

Redirect to the related Bookings or Trips screen

Else If notification is an announcement

Display the full announcement and its comments

End If

End If

If Customer clicks "Mark all read"

System marks all notifications as read

End If

If Customer clicks "Clear all"

Display confirmation prompt

If confirmed

System removes all notifications

End If

End If

End

11. Chat Support Process

Begin

Display chat conversation window

Customer enters a message

Click "Send"

System's automated assistant responds first

If the assistant's answer helps

Conversation continues with the assistant

Else If Customer selects "Talk to an Admin" (or an urgent request is detected)

System escalates the conversation to an administrator

Display "Connecting you to an admin..." message

End If

If an administrator resolves the conversation and the Customer sends a new message soon after

Conversation reopens with the same administrator

Else

A new message begins a fresh assistant session

End If

End

12. Payment History Process

Begin

Display payment summary and bookings with remaining balances

Display confirmed payment records

Display refunds and their current status separately

Only successful refunds reduce the amount paid

If Customer selects a month filter

Display records matching the selected month

End If

If Customer selects a payment

Display payment details and reference information

End If

If Customer selects a booking with a remaining balance

Redirect to the booking details

End If

End

13. Profile Management Process

Begin

Display account summary and profile completion status

If Customer selects "Personal Info & Addresses"

Display current Full Name, Facebook Name, Mobile Number, and Address

Customer edits fields

Click "Save Changes"

If input is valid

System updates the profile

Display "Profile updated successfully!" message

Else

Display validation error

End If

Else If Customer selects "Change Password"

Customer enters Current Password, New Password, and Confirm Password

Click "Update Password"

If current password is correct and new password meets requirements

Display success message

Else

Display error message

End If

Else If Customer selects "Change Email"

Customer enters New Email Address and Current Password

Click "Update Email"

If input is valid

Display "Check your new inbox" message

Else

Display error message

End If

Else If Customer selects "Help & Guidelines" or "About & Version"

Display the corresponding reference content

End If

If Customer clicks "Sign Out"

Display confirmation prompt

If confirmed

End session and redirect to Login Page

End If

End If

End

14. Admin Dashboard Process

Begin

Display Pending Bookings, Awaiting Departure, In Transit, and Active Trips counts

Display current trip capacity gauge

Display order status distribution chart

Display Recent Orders list

If Admin selects a recent order

 Redirect to Bookings Management

End If

End

15. Bookings Management Process

Begin

Display bookings with search and status filters

Admin selects a booking

Display booking details, tracking timeline, and payment summary

If booking is flagged for coverage review

 Admin reviews the request and records a decision

End If

If booking is unassigned

 Admin selects an available Trip

 If assignment passes validation

 System assigns the booking to the trip

 Else

 Display assignment error

 End If

End If

If Admin selects "Process Pickup"

Admin enters Actual Weight

System calculates Shipping Cost using the applicable rate

Admin may enter an eligible Discount and its reason

System calculates the final amount payable

Admin selects Who Pays and the applicable payment option

Admin enters the amount collected, if any

Admin records a Promised Payment Date when required

Admin attaches pickup photos

Click "Confirm Pickup"

If pickup requirements, capacity, discount, and payment are valid

 System saves the pickup information

 System records any confirmed payment

 System marks the booking as Picked Up

Else

 Display validation error

End If

End If

If Admin selects "Out for Delivery"

 If actual weight is recorded and dispatch requirements are met

 System marks the booking as Out for Delivery

 Else

 Display the reason dispatch is blocked

End If

End If

If Admin selects "Complete Delivery"

Admin records any remaining payment or a Promised Payment Date

Admin attaches 1 to 3 delivery photos

If photos are valid and the balance is settled or a promise date exists

System saves the delivery information

System marks the booking as Delivered

Else

Display validation error

End If

End If

If a cancellation request is pending

Admin reviews the reason

Admin approves or declines the request

System records the decision and notifies the Customer

End If

If Admin selects "Record Payment"

Admin enters the amount and required payment details

If payment is confirmed and the amount is valid

System records the payment and updates the balance

Else

Display validation error or current payment status

End If

```
End If
If Admin requests trip reassignment
    Admin selects another Trip and enters a reason
    If reassignment passes validation
        System updates the trip assignment
    Else
        Display reassignment error
    End If
End If
End If
End

16. Walk-in Booking Process
Begin
    Display Booking Form
    Admin selects Route
    Admin enters Sender Details and Receiver Details
    Admin enters Package Details and states a Payment Preference
    Click "Create Booking"
    If input is valid
        System saves the booking, marks it as a walk-in booking, and generates a
Tracking Number
        Display Success Message with Tracking Number
    Else
        Display validation error
```

End If

End

17. Trip Management Process

Begin

Display trips with schedules, status, and capacity

If Admin selects "Create Trip"

Admin selects Route and enters Departure and Planned Arrival dates

System obtains default Capacity and Rate from Company Information

Admin selects whether to email subscribed customers

Click "Create Trip"

If input is valid and no trip exists for the same route and departure day

System saves the trip

System assigns matching pending bookings within available capacity

Display confirmation

Else

Display validation error

End If

End If

If Admin selects a trip

Display assigned bookings and capacity usage

If Admin selects "Start Trip"

System checks the current Philippine date and trip readiness

If the scheduled departure date has not arrived

```
        Block the start and display the scheduled date
    Else If the scheduled departure date has passed
        Require rescheduling
    Else If pickup, capacity, or cancellation checks fail
        Display the reason the trip cannot start
    Else
        Admin may enter an Estimated Hub Arrival
        If the start request is valid
            System records the Actual Departure
            System starts the trip
            System marks the applicable bookings as In Transit
        Else
            Display validation error
        End If
    End If
End If

If Admin selects "Mark Arrived"
    If the trip is eligible for arrival confirmation
        System records the Actual Hub Arrival
        System marks the applicable bookings as Arrived at Hub
    Else
        Display validation error
    End If
End If
```

End If

If Admin selects "Complete"

 If all bookings are Delivered or Cancelled

 If no non-cancelled booking has a remaining balance

 System marks the trip as Completed

 Else

 Display unsettled balance message

 End If

 Else

 Display unfinished booking message

 End If

End If

If Admin selects "Reschedule"

 Admin enters new dates and a reason

 If the request is valid

 System saves the schedule change and its history

 System notifies affected customers

 Else

 Display validation error

 End If

End If

If Admin selects "Cancel"

 Display confirmation prompt

If confirmed and cancellation is allowed

System processes the trip cancellation

Else

Keep the trip unchanged and display any validation error

End If

End If

End If

End

18. Customer Management Process

Begin

Display list of registered customers

Admin searches by name, email, phone, or province

If Admin selects a customer

Display customer profile and booking history

End If

End

19. Sales and Reports Process

Begin

Display Sales and Reports Page

Admin selects a month

Click "Generate Report"

System retrieves trips for the selected month

If trips are available

System retrieves the bookings currently assigned to each trip

System calculates Shipping Fees and Payments Received

System calculates Successful Refunds and Payments after Refunds

System calculates the Amount Still to Collect from active bookings

System identifies cancelled bookings and their settlement information

Display per-trip details and combined totals

If Admin selects "Print Report"

 Open the report print view

End If

Else

 Display no records message

End If

End

20. Announcements Process

Begin

 Display announcement list

 If Admin selects "New"

 Admin enters Title, Content, and Category

 Admin may select "Send Email"

 Click "Publish"

 If required information is valid

 System publishes the announcement

 System creates the applicable notifications

If "Send Email" is selected

System queues emails for subscribed recipients

End If

Display confirmation

Else

Display validation error

End If

End If

If Admin selects "Delete"

Display confirmation prompt

If confirmed

System removes the selected announcement

End If

End If

End

21. Inbox Process

Begin

Display customer conversations and their status

Admin selects a conversation

Display conversation history

Admin enters a reply

Click "Send"

If the message is saved successfully

System makes the reply available to the Customer

System marks the conversation as Waiting on Customer

Else

Display message sending error

End If

If Customer replies

System marks the conversation as Waiting

End If

If Admin selects "Resolved"

System marks the conversation as Resolved

End If

If a Waiting on Customer conversation has no customer reply for 7 days

System automatically marks it as Resolved

System keeps the conversation messages

End If

End

22. Contact Inquiries Process

Begin

Display inquiries with New, Read, and Resolved status filters

Admin selects an inquiry

Display the visitor's contact details and message

Admin may assign the inquiry to an administrator

Admin updates the inquiry status

```
If the update is successful
    System saves the changes
    Display confirmation
Else
    Display update error
End If
End

23. Customer Feedback
Begin
    Display list of customer reviews with rating and message
    Admin filters by rating or searches
    If Admin clicks "Hide" on a review
        System hides the review from public display
    Else If Admin clicks "Restore"
        System makes the review visible again
    End If
End

24. Activity Logs Process
Begin
    Display activity records retained within the last 7 days
    Display Date and Time, Admin, Module, Action, and Details
    Admin may search, filter by Module, or hide sign-in entries
    Display matching records
```

If Admin selects "Clear Filters"

Reset the filters

End If

If Admin selects "Export CSV"

Generate a CSV file containing the filtered records

End If

End

25. Company Information Process

Begin

Display Company Information settings

Admin selects a settings tab

Admin edits company details, contact information, coverage, or pricing

Click "Save Changes"

If input is valid

System saves the changes

Display confirmation

Else

Display validation error

End If

If a new trip is created

System applies the current default Capacity and Rate

End If

End

26. Storage Monitoring Process

Begin

Display storage and email service status

Display booking photo folders

If Admin selects a folder

Display its available photos

End If

If Admin selects a photo

Display the photo preview

End If

If Admin requests deletion

Display confirmation prompt

If confirmed and deletion is allowed

System permanently deletes the selected photos

System records the deletion result

Refresh the displayed photos

Else

Keep the photos and display any validation error

End If

End If

End

27. Admin Profile Process

Begin

Display Admin account information

If Admin selects "Change Email"

Admin enters New Email Address and Current Password

Click "Update Email"

If the request is accepted

Display instructions to check the new email inbox

Else

Display validation or update error

End If

Else If Admin selects "Change Password"

Admin enters Current Password, New Password, and Confirm Password

Click "Update Password"

If the password update succeeds

Display confirmation

Else

Display validation or update error

End If

End If

End

28. Package QR Labels Process

Begin

Admin opens a booking

Admin enters the number of boxes

If the quantity is between 1 and 50 and the update is allowed

System saves the quantity

Admin selects Print Labels

System generates one QR label for each box

Each label shows the Tracking Number and Box Number

Open the label print view

Else

Display validation error

End If

If a QR label is scanned

If the user is an authorized administrator

Open the booking details and identify the box number

Else

Require administrator authentication

End If

End If

Scanning does not change shipment status or confirm delivery

End

29. Cancellation Settlement Process

Begin

Admin opens a Cancelled booking with recorded payments

Display Payments Received, Successful Refunds, and Refunds in Progress

Display the current settlement decision

If no decision exists

- Display that the remaining money requires review

End If

Admin selects Full Refund or enters an Amount to Retain

If an amount will be retained

- Admin confirms the Customer's agreement

End If

Click "Save"

System validates the decision against payments and existing refunds

If the decision is valid

- System saves the settlement and its history
- If a refund remains due
 - Admin processes a PayMongo refund or records a manual refund
 - System displays the refund status
 - Only a successful refund is counted as money returned

End If

Display the updated settlement summary

Else

- Display validation error

End If

The cancelled booking's previous shipping balance is not collected

End

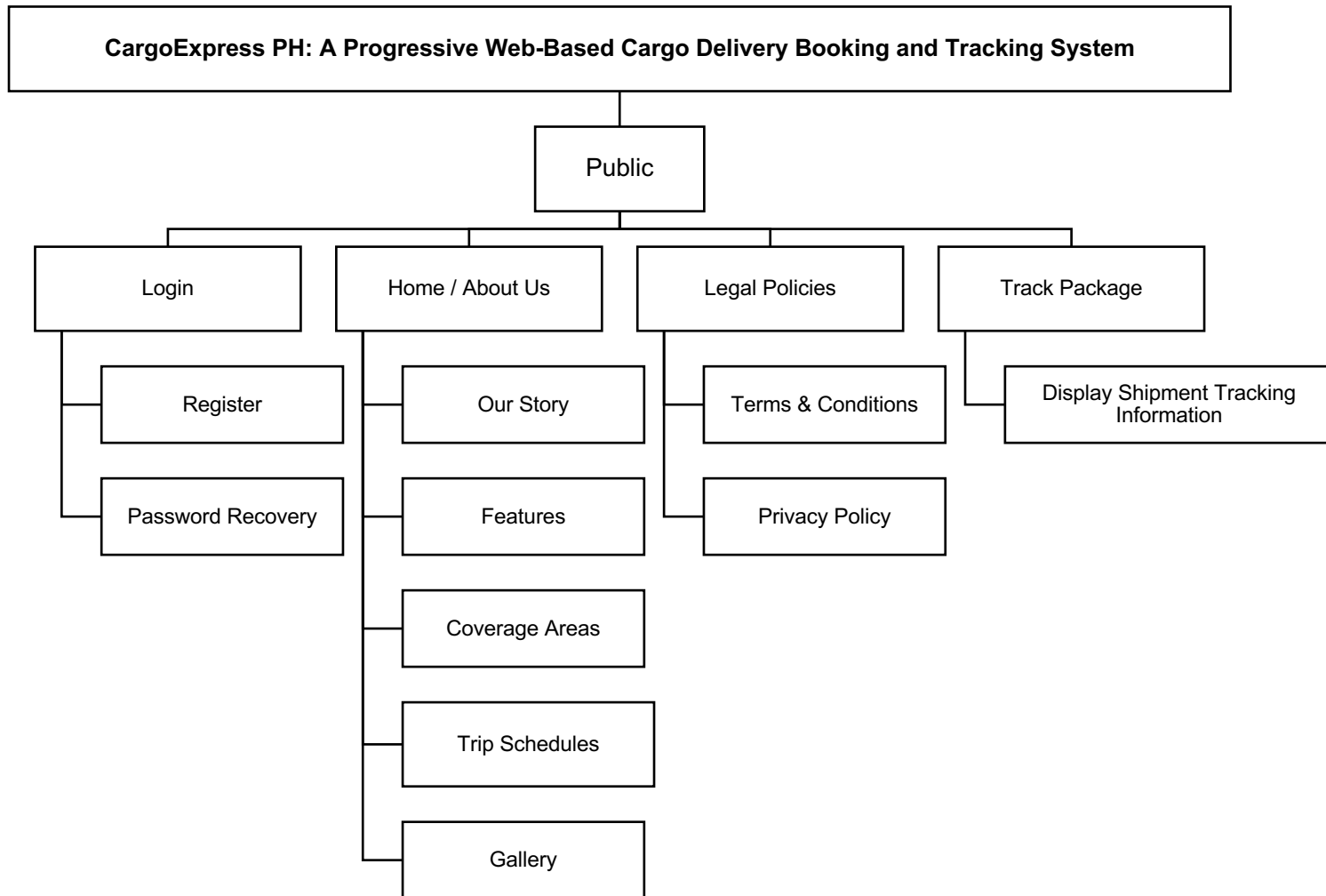


FIGURE 38. PROGRAM HIERARCHY (PUBLIC)

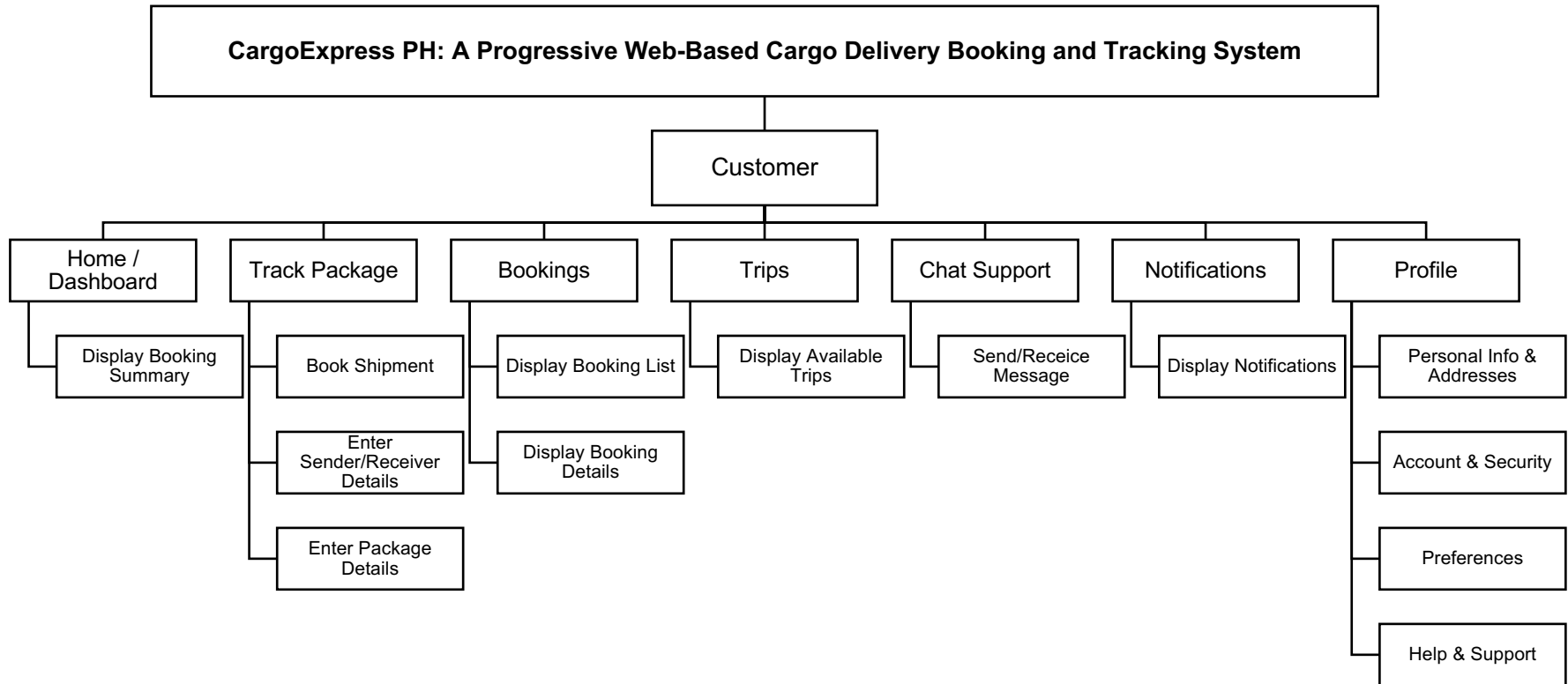


FIGURE 39. PROGRAM HIERARCHY (CUSTOMER)

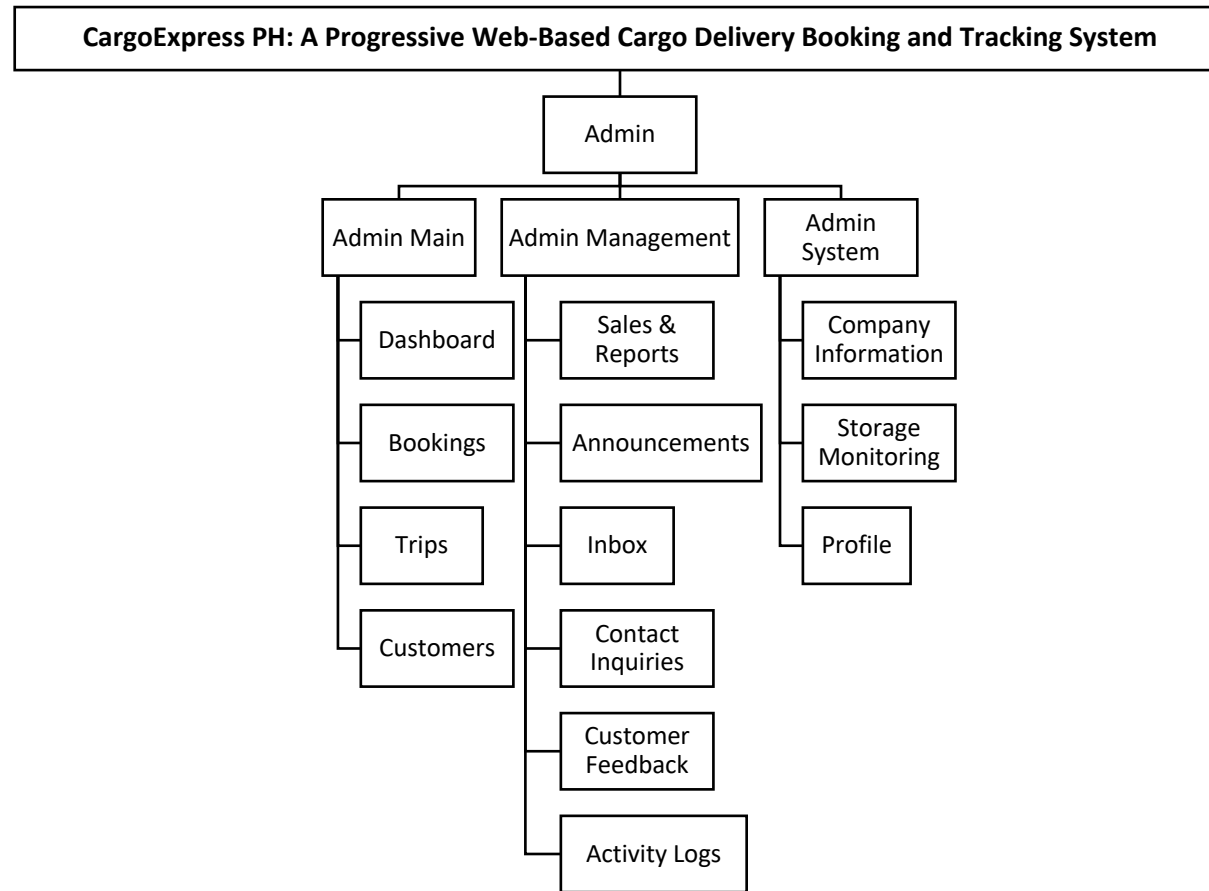


FIGURE 40. PROGRAM HIERARCHY (ADMIN)

Database Design

The following tables present the database structure of the CargoExpress PH system and describe the information stored and managed by each table.

Table 1. **Profiles:** This table contains information about the users, including their personal details, addresses, and `system roles.

Field name	Data type	Description
id	uuid	Unique identifier for each user linked to Supabase authentication.
name	varchar	Stores the full name of the user.
role	varchar	Identifies whether the user is an administrator or a customer.
email	varchar	Stores the unique email address used for logging in.
phone	varchar	Stores the primary contact number of the user.
address_lot_block	varchar	Stores the specific house, lot, or block details of the user's address.
address_street	varchar	Stores the street name of the user's address.
address_barangay	varchar	Stores the barangay of the user's address.
address_city	varchar	Stores the city or municipality of the user's address.
address_province	varchar	Stores the province of the user's address.
facebook_name	text	Stores the Facebook profile name of the user for social contact.
address_landmark	text	Stores a notable landmark near the user's address.

Table 2. **Trips:** This table contains records of scheduled cargo trips between origins and destinations.

Field name	Data type	Description
id	uuid	Unique identifier for each cargo trip record.
trip_number	varchar	Stores the unique alphanumeric code assigned to the trip.
origin	varchar	Indicates the starting location or port of the trip.
destination	varchar	Indicates the ending location or port of the trip.
departure_date	timestampz	Records the scheduled date and time of departure.
arrival_date	timestampz	Records the planned arrival date of the trip.
status	varchar	Indicates the current state of the trip.
departure_at	timestampz	Records the actual timestamp when the trip was started.

Table 3. **Orders:** This table contains information about individual cargo booking records and their tracking details.

Field name	Data type	Description
id	uuid	Unique identifier for each booking order.
user_id	uuid	Stores the identifier of the customer who placed the order.
trip_id	uuid	Stores the identifier of the assigned cargo trip for this order.
origin	varchar	Indicates the starting location of the cargo for this order.
destination	varchar	Indicates the destination location of the cargo for this order.
tracking_number	varchar	Stores the unique tracking code provided to the customer.
sender_first_name	varchar	Stores the first name of the sender.
sender_last_name	varchar	Stores the last name of the sender.
sender_phone	varchar	Stores the contact number of the sender.
receiver_first_name	varchar	Stores the first name of the receiver.
receiver_last_name	varchar	Store last name of the receiver.

receiver_phone	varchar	Stores the contact number of the receiver.
package_description	text	Stores the details and description of the items being shipped.
actual_weight	numeric	Records the verified weight of the package in kilograms.
shipping_cost	numeric	Records the shipping charge before discount.
payer_type	varchar	Indicates whether the sender or receiver is paying for the shipment.
payment_method	varchar	Indicates the chosen mode of payment such as cash or gcash.
payment_status	varchar	Indicates whether the order is unpaid, partially paid, or fully paid.
amount_paid	numeric	Records the amount paid after deducting successful refunds.
remaining_balance	numeric	Records the remaining shipping balance for an active booking.
promised_payment_date	date	Records the date when the customer promised to settle the balance.
status	varchar	Indicates the current tracking status of the order such as Pending or In Transit.
notes	text	Stores additional notes about the booking.
created_at	timestampz	Records the date and time when the order was placed.
updated_at	timestampz	Records the date and time when the order was last updated.
sender_facebook	text	Stores the Facebook profile name of the sender.
sender_city	text	Stores the city of the sender.
receiver_facebook	text	Stores the Facebook profile name of the receiver.
receiver_city	text	Stores the city of the receiver.
receiver_province	text	Stores the province of the receiver.
sender_province	text	Stores the province of the sender.
pickup_photos	jsonb	Stores a collection of photo URLs taken during pickup.
delivery_photos	jsonb	Stores a collection of photo URLs taken during delivery.
payment_reference	varchar	Stores the reference number for the payment transaction.
service_area_status	text	Indicates if the delivery address is within the standard service area.

service_area_remarks	text	Stores notes regarding the review of the service area.
featured_on_website	bool	Indicates if this order is showcased as a featured delivery on the website.
featured_title	text	Stores the title used when featuring this order.
featured_caption	text	Stores the caption used when featuring this order.
featured_image_type	text	Indicates the type of image used for featuring the order.
reassignment_history	jsonb	Stores a log of the different trips this order was assigned to.
cancellation_details	jsonb	Stores the complete details of the cancellation request, including the stated reason, previous order status, timestamps, and admin review notes.
payment_preference	text	Indicates the preferred payment method specified during booking.
package_quantity	integer	Total number of physical boxes or parcels in the shipment.
discount_amount	numeric	Total shipping discount amount applied (PHP).
discount_reason	text	Stores the reason for granting the discount.
discount_applied_by	uuid	Referencing the admin who applied the discount.
discount_applied_at	timestampz	Records the date and time the discount was applied.

Table 4. **Order_status_events**: This table provides a timeline log of all status changes for an order.

Field name	Data type	Description
id	uuid	Unique identifier for each status change event.
order_id	uuid	Stores the identifier of the order whose status was changed.
status	varchar	Records the new status that was applied to the order.

changed_at	timestampz	Records the date and time when the status change occurred.
changed_by	uuid	Stores the identifier of the user or admin who changed the status.
note	text	Stores additional remarks explaining why the status was changed.

Table 5. **Payment_transactions:** This table logs all finalized payment records and partial payments made for orders.

Field name	Data type	Description
id	uuid	Unique identifier for each transaction
order_id	uuid	Stores the identifier of the associated order being paid for.
amount	numeric	Records the exact monetary value paid in this transaction.
payment_method	text	Indicates the method used for this specific payment.
transaction_reference	text	Stores the reference number associated with the payment.
payment_status	text	Stores the payment status recorded for the transaction.
admin_id	uuid	Stores the identifier of the admin who processed or verified the payment.
admin_name	text	Stores the name of the processing admin for quick reference.
receipt_url	text	Stores the link to the uploaded proof of payment image.

Table 6. **Payment_refunds:** This table manages and tracks refund transactions processed through the system and payment gateways.

Field name	Data type	Description
id	uuid	Unique identifier for each payment refund record.
refund_id	text	Stores the unique identifier returned

		by the external payment gateway.
idempotency_key	uuid	Stable identifier to prevent duplicate refund processing.
payment_transaction_id	uuid	Links the refund to the original internal payment transaction record.
order_id	uuid	Stores the identifier of the associated order being refunded.
payment_id	text	Links the refund to the original external payment gateway transaction.
amount	numeric	Records the exact monetary value of the refund.
currency	text	Specifies the currency of the refund transaction.
status	text	Indicates the current processing state of the refund.
reason	text	Stores the justification or category for the refund request.
notes	text	Stores additional administrative notes regarding the refund.
livemode	bool	Indicates whether the refund belongs to the live or test environment.
initiated_by	uuid	Stores the identifier of the user or admin who requested the refund.
initiated_by_name	text	Stores the name of the user or admin who requested the refund.
last_error	text	Records technical diagnostics or the most recent error message if the refund failed.
last_event_id	text	Stores the identifier of the most recent gateway event processed for this refund.
provider_created_at	timestampz	Records the timestamp when the payment provider created the refund.
provider_updated_at	timestampz	Records the timestamp when the payment provider last updated the refund.
created_at	timestampz	Records the date and time when the refund record was created in the system.
updated_at	timestampz	Records the date and time when the refund record was last updated.
outcome_uncertain	bool	Indicates if the system cannot establish whether the provider accepted the request.
public_failure_reason	text	Stores sanitized, non-technical failure

		text safe for customer visibility.
succeeded_at	timestampz	Records when the refund was confirmed successful.
refund_channel	text	Whether the refund was processed through PayMongo or recorded manually.
return_method	text	Whether a manual refund was returned through cash or GCash
return_reference	text	Stores the GCash reference for a manual refund.
returned_at	timestampz	Records when the money was returned through a manual refund.

Table 7. **Announcements:** This table stores system announcements published by admins.

Field name	Data type	Description
id	uuid	Unique identifier for each announcement record.
title	varchar	Stores the headline or title of the announcement.
content	text	Stores the full body text of the announcement.
author_id	uuid	Stores the identifier of the admin who created the announcement.
is_active	bool	Indicates whether the announcement is currently visible to users.
created_at	timestampz	Records the date and time when the announcement was created.
updated_at	timestampz	Records the date and time when the announcement was last updated.
comments	jsonb	Stores user comments on the announcement.
send_email	bool	Indicates if the announcement should also be emailed to customers.
emailed_at	timestampz	Records when the announcement was emailed.

Table 8. **Notifications:** This table manages automated alerts and notifications sent to users.

Field name	Data type	Description
id	uuid	Unique identifier for each notification.
user_id	uuid	Stores the identifier of the user receiving the notification.
message	text	Stores the main content of the notification message.
type	varchar	Indicates the category of the notification such as order updates or general alerts.
reference_id	uuid	Stores the identifier of the related record like an order or trip ID.
is_read	bool	Indicates whether the user has already opened or read the notification.
created_at	timestampz	Records the date and time when the notification was created.

Table 9. **Conversations:** This table manages the support chat threads between customers and the system.

Field name	Data type	Description
id	uuid	Unique identifier for each conversation thread.
customer_id	uuid	Stores the identifier of the customer who owns the conversation.
created_at	timestampz	Records the date and time when the conversation started.
status	text	Indicates the current state of the chat determining whose turn it is to reply
first_response_at	timestampz	Records the date and time when the first admin response was sent.
last_customer_message_at	timestampz	Records the date and time of the last message sent by the customer.
resolved_at	timestampz	Records the date and time when the conversation was concluded.
bot_resolved	bool	Indicates whether the automated bot successfully resolved the issue.

Table 10. **Chat_messages:** This table stores the individual messages sent within a support conversation.

Field name	Data type	Description
id	uuid	Unique identifier for each individual chat message.
conversation_id	uuid	Stores the identifier of the conversation this message belongs to.
sender_id	uuid	Stores the identifier of the user who sent the message.
sender_role	varchar	Indicates the role of the sender such as admin or customer.
message	text	Stores the actual text content of the message.
created_at	timestampz	Records the date and time when the message was sent.

Table 11. **Contact_inquiries:** This table records messages sent by visitors through the public contact form.

Field name	Data type	Description
id	uuid	Unique identifier for each contact inquiry.
name	text	Stores the name of the person submitting the inquiry.
contact_phone	text	Stores the mobile number provided by the visitor.
message	text	Stores the actual question or concern of the visitor.
status	text	Indicates the current state of the inquiry such as new or resolved.
created_at	timestampz	Records the date and time the inquiry was received.
contact_email	text	Stores the email address provided by the visitor.
assigned_admin_id	uuid	Stores the identifier of the admin assigned to handle the inquiry.
resolved_at	timestampz	Records the date and time when the inquiry was marked as resolved.

Table 12. **Customer_feedback:** This table stores the ratings and reviews submitted by customers after a delivery.

Field name	Data type	Description
id	uuid	Unique identifier for each feedback entry.
order_id	uuid	Stores the identifier of the delivered order being reviewed.
customer_id	uuid	Stores the identifier of the customer providing the feedback.
rating	int4	Records the star rating given by the customer from 1 to 5.
message	text	Stores the written review or comment provided by the customer.

Table 13. **Company_information:** This table stores the configurable global details and settings of the company.

Field name	Data type	Description
id	uuid	Unique identifier for the single company information record.
name	text	Stores the display name of the company.
short_description	text	Stores a brief description or tagline of the company.
long_description	text	Stores the comprehensive background information about the company.
banner_image_url	text	Stores the link to the main banner image used on the website homepage.
banner_title	text	Stores the primary headline displayed on the homepage banner.
banner_description	text	Stores the subtext or supporting description on the homepage banner.
banner_button_text	text	Stores the label used for the primary call-to-action button.
banner_button_link	text	Stores the destination URL for the primary call-to-action button.
email	text	Stores the official contact email address of the company.
facebook	text	Stores the link to the company's official Facebook page.

smart_phone	text	Stores the company's primary Smart contact number.
globe_phone	text	Stores the company's primary Globe contact number.
manila_address	text	Stores the physical address of the Manila warehouse or office.
bohol_address	text	Stores the physical address of the Bohol warehouse or office.
updated_at	timestampz	Records the date and time when the company details were last updated.
default_price_per_kg	numeric	Stores the current rate per kilogram used when calculating the shipping charge at weighing.
default_capacity	int4	Stores the base cargo capacity in kilograms used for trip capacity checks.

Table 14. **Activity_logs**: This table tracks the actions performed by admins for system auditing.

Field Name	Data Type	Description
id	uuid	Unique identifier for each activity log entry.
admin_id	uuid	Stores the identifier of the admin who performed the action.
admin_name	text	Stores the name of the admin for quick reference.
module	text	Indicates the system module where the action occurred.
action	text	Stores a description of the specific action performed by the admin.
record_type	text	Indicates the type of data record that was modified.
record_id	uuid	Stores the identifier of the specific record that was modified.
record_ref	text	Stores a human-readable reference or title for the modified record.
new_value	jsonb	Records the state of the data after the modification in JSON format
details	text	Stores additional context or notes regarding the activity.

created_at	timestampz	Records the date and time when the action was logged.
client_event_id	uuid	Stores an idempotency key to prevent duplicate logs from the client.

Table 15. **Cancellation_settlements**: This table records the agreed financial settlement of a cancelled booking, including whether the payment will be fully refunded or an agreed amount will be retained.

Field Name	Data Type	Description
order_id	uuid	Identifies the cancelled booking and serves as the primary key.
decision_type	text	Indicates whether the decision is a full refund or retention of an agreed amount.
agreed_retained_amount	numeric	Records the agreed amount that the company will keep.
customer_agreement_confirmed	boolean	Indicates whether customer agreement to the retained amount has been recorded.
decided_by	uuid	Identifies the administrator who recorded the settlement decision.
decided_at	timestampz	Records the date and time of the settlement decision.

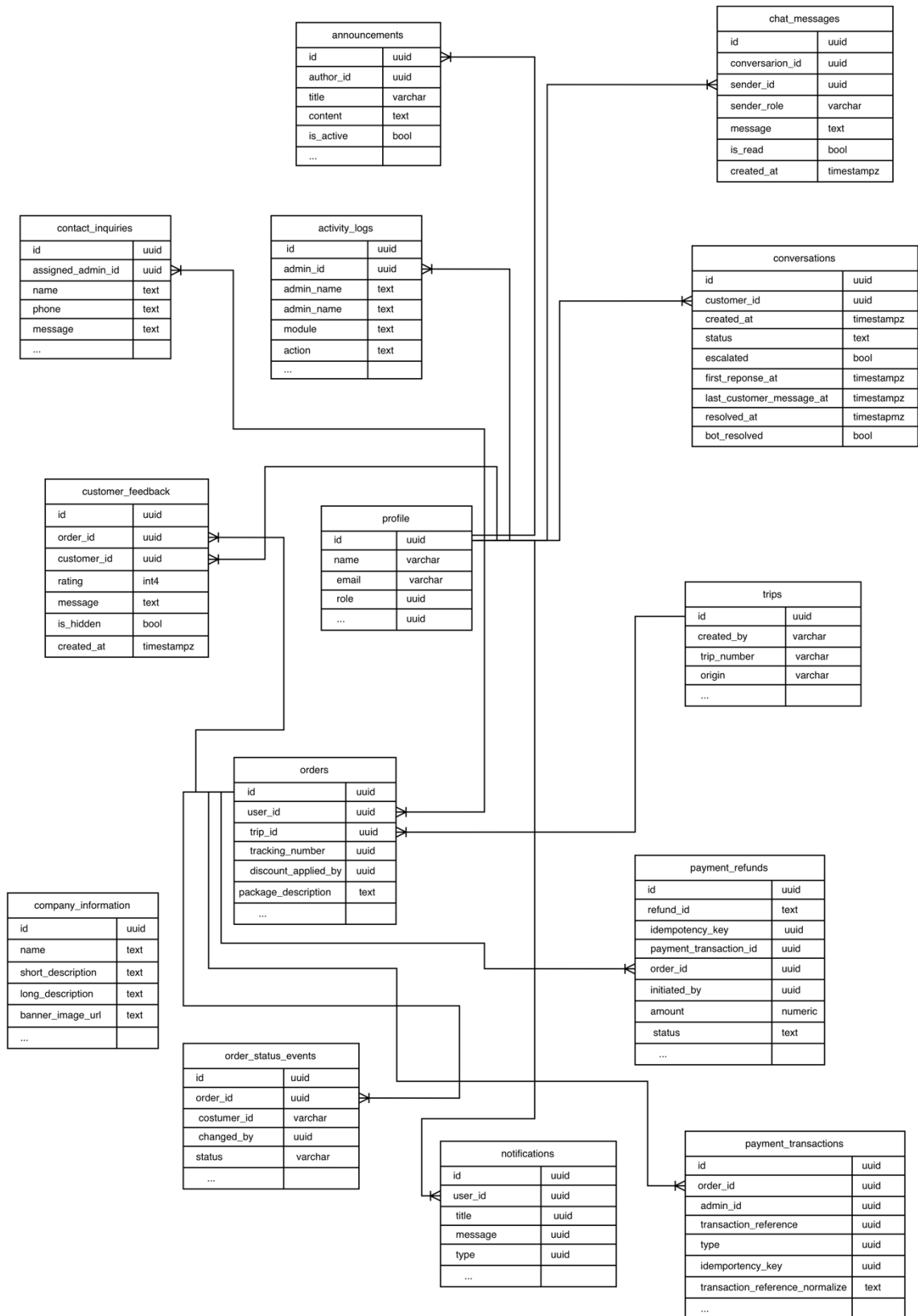
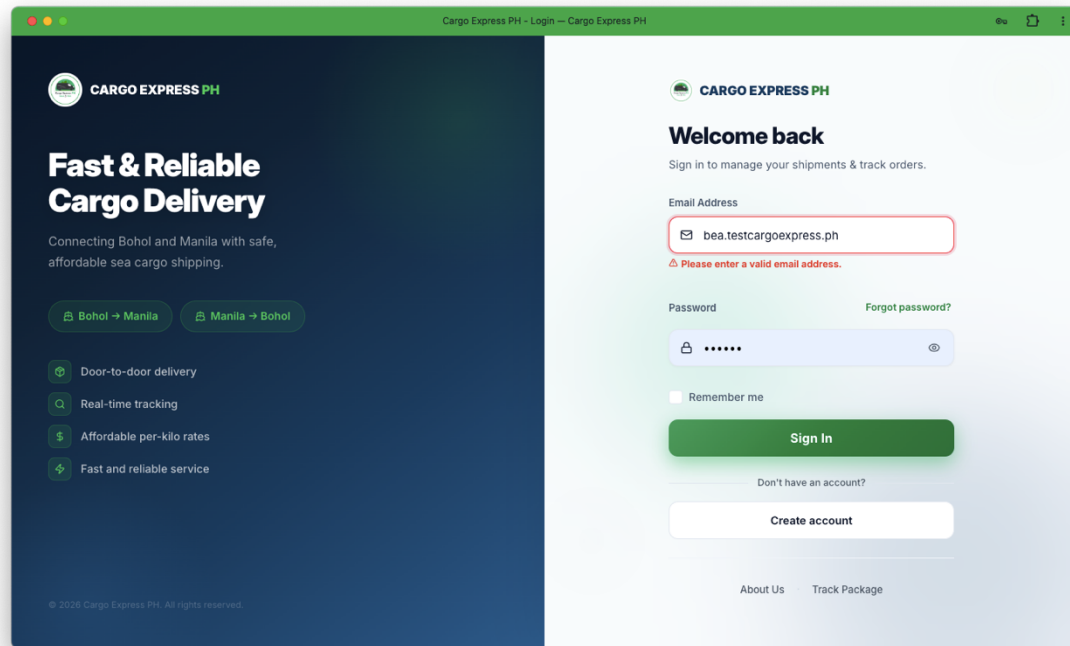


Figure 41. Database Schema

Test Data/Cases

Test Case 1: Client Log In



Preview 1: Invalid Login

INPUT: No email entered

Email missing @ symbol (bea.testcargoeexpress.ph)

Valid email (bea.test@cargoeexpress.ph)

PROCESS: System validates input. Only valid email format is allowed.

OUTPUT: Please enter a valid email address.

Test Case 2: Sender Booking Details

The screenshot displays the 'CargoExpress PH - Book Shipment' web interface. At the top, a navigation bar includes the logo, a '+ Book Shipment' button, and links for 'Bookings', 'Trips', 'Chat Support', and a user profile icon. Below this is a progress bar with five steps: 'Route' (completed), 'Sender' (active), 'Receiver', 'Package', and 'Review'. The 'Sender Details' form is the central focus. It contains several input fields: 'Full Name' (filled with 'Bea'), 'Mobile Number' (filled with '09032940343'), 'Province' (a dropdown menu showing 'Bohol'), 'Facebook Name' (filled with 'Bea Sarong'), 'City / Municipality' (a dropdown menu showing 'Select City' with a red border and a 'City is required.' error message), 'Barangay' (a dropdown menu showing 'Select a city first' with a red border and a 'Barangay is required.' error message), 'Street and Subdivision' (filled with 'NA'), 'Lot / Block / Purok' (empty with a red border and a 'Required field.' error message), and 'Landmark' (filled with 'Near what building/place?' with a red border and a 'Required field.' error message). A green 'Continue' button is at the bottom of the form.

Preview 2: Incomplete Sender Details

INPUT: No City/Municipality selected

No Barangay entered

No Lot/Block/Purok entered

No Landmark entered

PROCESS: The system verifies that all required sender details are complete before allowing the user to continue.

OUTPUT: Displays error messages under incomplete fields and blocks progress to the next step

Test Case 3: Receiver Booking Details

CargoExpress PH - Book Shipment - CargoExpress PH

CARGOEXPRESS PH + Book Shipment Bookings Trips Chat Support

Route Sender Receiver Package Review

Receiver Details

☐ Use my registered address for receiver details

Full Name ^{*} Recent Contacts

Bea Sarong

Mobile Number ^{*} Facebook Name ^{*}

XXXXXXXXXX Your name on Facebook

Mobile number is required. Required field.

Province ^{*} City / Municipality ^{*}

Metro Manila Makati

Barangay ^{*} Street and Subdivision (put NA if not applicable) ^{*}

Bangkal Rizal Street

Lot / Block / Purok ^{*} Landmark ^{*}

Lot 12, Block 5 Near Sari-Sari Store, Beside Church

Continue

Preview 3: Incomplete Receiver Details

INPUT: No mobile number entered

No Facebook name entered

PROCESS: System checks for the presence of all mandatory receiver details.

OUTPUT: Phone number is required, and Facebook Name is required

Test Case 4: Package Booking Details

Cargo Express PH - Book Shipment -- Cargo Express PH

CARGO EXPRESS PH + Place Order Orders Trips Chat Support

← BACK

Book Shipment

1 Route 2 Sender 3 Receiver 4 Package 5 Review

Package Details

What are you sending? *

e.g. Documents, 2 boxes of clothes, small appliance

⚠ Package description is required.
Describe your items. We weigh the parcel at pickup and the exact cost is confirmed then.

Who Pays?

Sender

Payment Preference (Optional)

I'll decide later

Letting us know how you plan to pay helps our team prepare for pickup or delivery.

Special Instructions / Notes (Optional)

e.g. Fragile item, deliver after 5 PM, etc.

Shipping Rate
₱80/kg
Your total is calculated when we weigh your parcel at pickup.

Review Booking

Preview 4: Missing Package Details

INPUT: No package description entered

PROCESS: System checks if required package details are provided before moving to the next step.

OUTPUT: Package description is required.

Test Case 5: Tracking Number

The screenshot shows a web browser window with the title "Cargo Express PH - Track Shipment - Cargo Express PH". The page has a green header bar. Below the header, the "CARGO EXPRESS PH" logo is displayed. The main heading is "Track Your Shipment" with the subtext "Live status updates — know exactly where your package is". A search bar contains the placeholder text "Enter tracking number (e.g. CE-20270101-0001)". To the right of the search bar is a green "Track" button. Below the search bar, a red error message reads "Enter a tracking number to search." Below this, a green cube icon is shown. The section "Enter Your Tracking Number" includes the instruction "Paste or type your Cargo Express PH tracking number above to get live shipment updates." Two green callout boxes provide additional information: "Tracking numbers follow the format CE-YYYYMMDD-XXXX" and "Contact Cargo Express PH staff if you need help locating it". At the bottom, there is a link to "Sign In" for users with accounts and a copyright notice "© 2026 Cargo Express PH".

Preview 5: Invalid Tracking Number

INPUT: No tracking number entered

Entered invalid format tracking number

PROCESS: System searches the database for the entered tracking number.

OUTPUT: Enter a tracking number to search.

Test Case 6: Creating New Trip

CARGO EXPRESS PH - Create Trip — Cargo Express PH

+ Create New Trip
Define route, schedule, capacity, and pricing for a cargo run.

Route

Bohol → Manila
Bohol → Manila

Manila → Bohol
Manila → Bohol

Schedule

Departure Date & Time: 08/21/2026, 11:36 AM
Estimated Arrival Date & Time: 08/24/2026, 11:36 AM

Departure date cannot be in the past.

Capacity & Pricing

Capacity (kg): 1000
Amount per Kilo (P): P 80

Maximum total cargo weight for this trip.
Cost per kilogram for bookings on this trip.

At P80.00/kg, a full trip of 1,000 kg = P80,000 max revenue

Notes (Optional)
Any special instructions, remarks, or conditions for this trip...

Martin Sarang
Administrator

Create Trip

Preview 6: Invalid Departure Date

INPUT: Set departure date and time that is in the past date and time

PROCESS: System validates the selected departure date and time against the current date and time.

OUTPUT: Departure date cannot be in the past.

Test Case 7: Payment

Pickup Processing

CE-20260820-3838
Cancel Future Sender → Cancel Future Receiver
E2E cancellation future m22gaw9k2gm

Actual Weight (kg) *
100
Estimated cost: ₱7,500.00

Who Pays? *
☒ Sender (pay now) ☐ Receiver (pay on delivery)
Booking says: Sender pays

Payment Type *
☒ Full Payment ☐ Pay Later

Amount Received (₱) *
7,499
Amount entered is less than the ₱7,500.00 total cost. Enter the exact full amount, or switch Payment Type to Pay Later to record a part-payment. ₱100 will still be owing after this.

Payment Method *
☒ Cash ☐ GCash

Pickup Proof Photos * (1-3)

Add Photo

Cancel

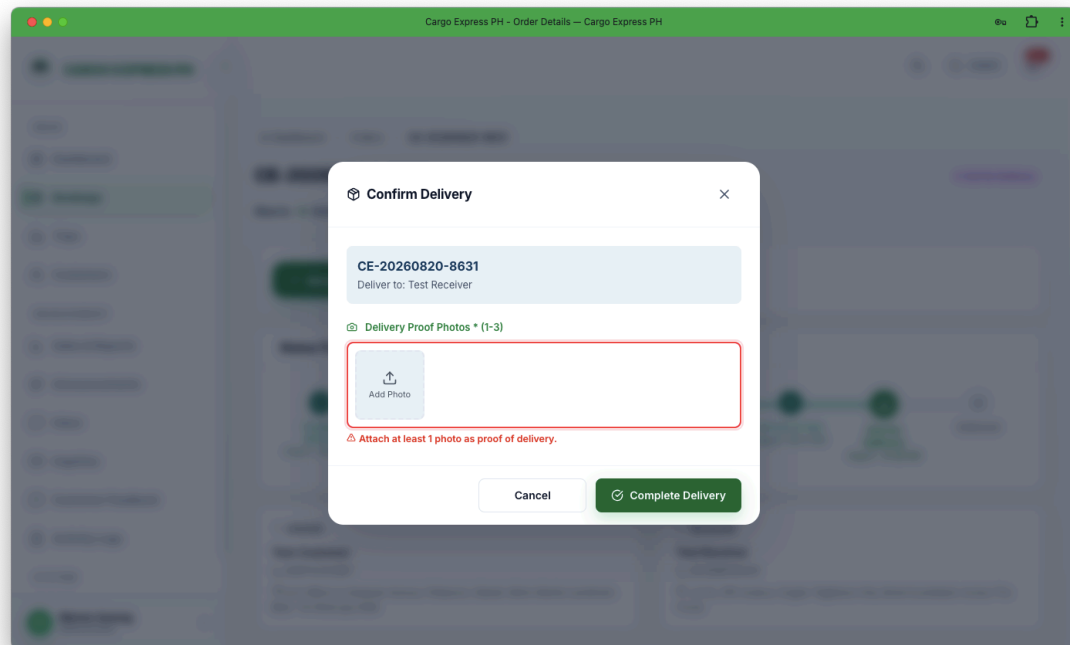
Preview 7: Insufficient Payment Amount

INPUT: Amount entered is less than the total cost for full payment

PROCESS: System verifies if the received amount matches the expected total cost for a full payment.

OUTPUT: Amount entered is less than the total cost. Enter the exact full amount, or switch Payment Type to Pay Later to record a part-payment.

Test Case 8: Proof of Delivery



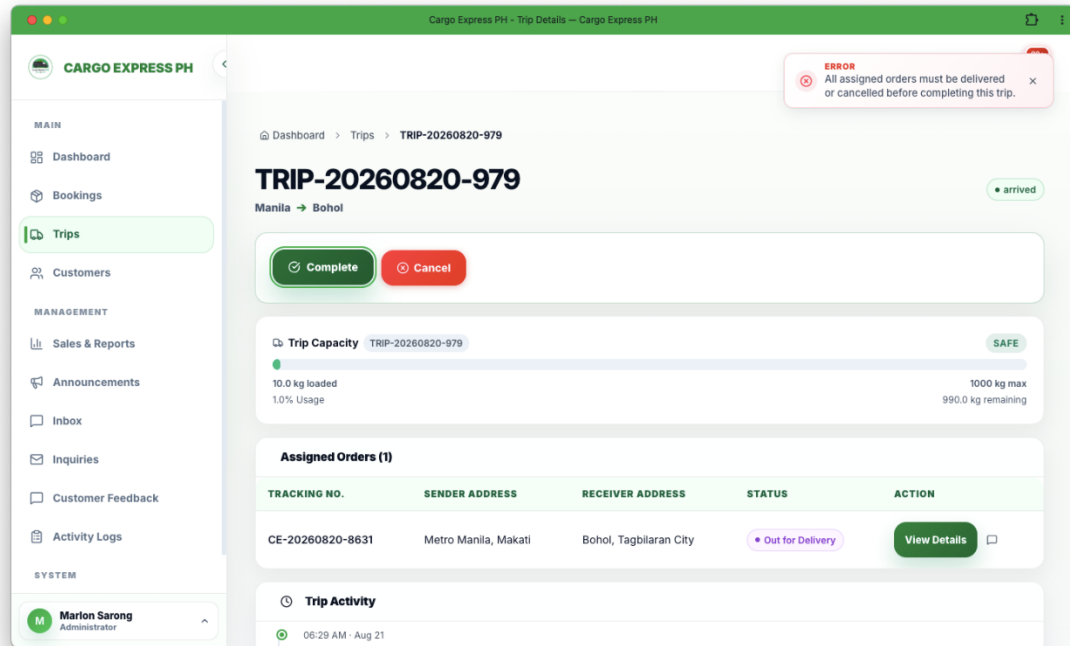
Preview 8: Missing Proof of Delivery

INPUT: No proof photos attached

PROCESS: System checks if at least one photo is uploaded for delivery confirmation.

OUTPUT: Attach at least 1 photo as proof of delivery.

Test Case 9: Trip Completion Validation



Preview 9: Trip Completion Validation

INPUT: Trip has assigned orders that are not delivered

PROCESS: System checks the status of all assigned orders for the trip before allowing completion.

OUTPUT: All assigned orders must be delivered or cancelled before completing this trip.

Test Case 10: Announcements

The screenshot shows the 'Announcements' form in the Cargo Express PH system. The form is titled 'Announcements' with a subtitle 'Publish operational updates customers can see in their dashboard.' and a '+ New' button. The form fields are:

- Category Tag:** A dropdown menu with 'Auto-Detect (Smart Category)' selected.
- Title * (Max 100 characters):** An empty text input field with a red border and a validation message below it: 'Please enter a title — this is what customers see first.'
- Content * (Max 1500 characters):** An empty text area with a red border and a validation message below it: 'Please write the announcement content.'

At the bottom right of the form are 'Publish' and 'Cancel' buttons. Below the form, there is a preview of the announcement: 'GENERAL UPDATE E2E Announcement mt23da2kp46c' with a trash icon.

Preview 10: Empty Announcements Fields

INPUT: No title entered No content entered

PROCESS: System validates that the announcement title and content fields are not empty.

OUTPUT: Please enter a title — this is what customers see first and Please write the announcement content.

Revision of the System

First Revision

- Change the application name
- Create the official system logo

Second Revision

- Integrate PayMongo for GCash Transactions
- Redesign the User Interface (UI)
- Add a Reports module

Third Revision

- Separate the full name fields into First Name and Last Name
- Implement a QR code system for tracking parcels
- Change the Sales Report to filter and data per month

Technical Requirements

Hardware, software, and peopleware are three essential requirements needed for the implementation of the web-based CargoExpress PH system. These components must work together effectively to ensure the system's stability, efficiency, and long-term usability for managing cargo delivery operations.

Hardware Specification

Hardware refers to the physical components of the computer system and mobile devices used to access and operate the CargoExpress PH system. Since CargoExpress PH is a purely web-based system designed for accessibility and convenience, devices such as computers, laptops, tablets, or mobile devices may be used to access system operations, including booking shipments, shipment status tracking, and report generation. The administrator and customer may use similar hardware specifications because the system's database and backend services are cloud-based, requiring users only to have a device capable of running a modern web browser. To ensure smooth and efficient performance, the following client specifications are recommended:

Random Access Memory (RAM)

- **Minimum requirement:** 4GB
- **Display:** 6-inch touchscreen display or higher (for mobile access) / 13-inch display with 1366 x 768 resolution (for desktop/laptop access)
- **Processor:** 1.5 GHz dual-core processor or higher
- **Storage:** at least 32GB of internal storage (for OS and browser operation)
- **Connectivity:** Minimum of 5 Mbps broadband or stable 4G/LTE mobile data connection.

Software Requirements

Software refers to the set of instructions that enables the computer to process, store, and retrieve data effectively. Without software, hardware components cannot function properly. The CargoExpress PH system requires both system software and application software to operate efficiently.

Operating System

The operating system serves as the platform that manages hardware resources and allows the application to run properly. Because CargoExpress PH is a web-based system, it is highly versatile and can run on any modern operating system equipped with a supported web browser. For desktop and laptop environments, it fully supports Windows, macOS, and Linux. For mobile access, the responsive web application is fully compatible with Android and iOS. By utilizing Progressive Web App (PWA) technologies such as web manifests and service workers, the system can be installed directly on supported devices without requiring a separate native mobile application.

Application Program: JavaScript, React, and Vite

The CargoExpress PH system uses JavaScript as its programming language and React as its primary application development framework, bundled using the Vite build tool. JavaScript is used to develop the core functions and interactions of the system, while React organizes the user interface into efficient, reusable components. These technologies allow the system to provide an

organized and responsive interface for online booking, shipment status tracking, trip management, and communication. Additionally, the system integrates the PayMongo API for online GCash payment processing and Firebase Cloud Messaging for delivering real-time push notifications to users.

Database: Supabase with PostgreSQL

The CargoExpress PH system uses Supabase with PostgreSQL as its cloud-based database and backend platform. It is responsible for storing and managing the system's data, including user accounts, shipment bookings, trip schedules, and payment records. The database provides a centralized repository that enables efficient data retrieval and updating. Supabase was selected because it offers a secure and reliable cloud environment that supports authentication, file storage for payment receipts and delivery photos, and real-time data updates across the system. These features enable the system to provide accurate cargo status updates, controlled access to system data based on user roles, and efficient management of cargo delivery operations.

Peopleware

Peopleware refers to the individuals who use and manage the CargoExpress PH system, and its effectiveness depends on their proper utilization of the application. The main users include the Administrator, Customer, and Public User. The Administrator manages cargo delivery operations by scheduling trips, updating shipment statuses, verifying payments, maintaining customer records, and generating PDF reports. The Customer uses the system to book shipments

securely, monitor cargo status, view transactions, and receive automated push notifications. The Public User can track a shipment through the public tracking page by entering a valid tracking number without creating an account. The Researchers or Developers maintain, improve, and update the CargoExpress PH system to ensure efficient, secure, and reliable operation. Proper user training and familiarity with the system are essential to maximize its functionality, improve operational efficiency, and provide reliable cargo delivery services.

Cost-Benefit Analysis

The cost-benefit analysis compares the existing cargo booking and tracking process with the proposed CargoExpress PH system to determine whether the development is financially advantageous. Since the existing process does not require significant direct expenses for manual record-keeping, the analysis focuses on the operational costs required to maintain the proposed web-based system and the benefits it provides to the business.

At present, cargo booking and shipment management rely mainly on direct communication and manual coordination between customers and administrators. Customers provide their booking and shipment details through available communication channels, while administrators coordinate bookings, trip assignments, payment monitoring, and shipment updates. Although this process does not require significant operating expenses, it requires additional time and effort due to the absence of a centralized system for managing cargo transactions.

To address these limitations and provide a more organized, efficient, and accessible way of managing cargo deliveries, the development of the CargoExpress PH system is proposed. The web-based system requires operational resources such as domain registration and web hosting to deploy the cloud-based application. The estimated annual operating costs of the proposed system are presented in the following table.

Table 16. Proposed System Annual Operating Cost

ITEM	DURATION	PRICE	AMOUNT
A. Domain			
Domain Name	1 Year	₱100.00	₱100.00
B. Web Hosting			
Vercel Hosting	12 Months	₱149.00	₱1,788.00
Total Proposed Annual Operating Cost			₱2,480.32

Benefits

The proposed CargoExpress PH system provides a more efficient and organized approach to managing cargo delivery operations by replacing manual recording with a centralized web-based platform. The system allows administrators to manage bookings, shipment schedules, customer records, and payment transactions in one place, while enabling customers to book shipments and monitor delivery status online. This improves the accuracy of records, speeds up transactions, and supports better monitoring of cargo operations.

Tangible Benefits

The CargoExpress PH system provides tangible operational benefits by reducing manual work and improving operational efficiency. Shipment records, customer information, and payment transactions are stored digitally in the PostgreSQL database, allowing administrators to quickly retrieve information and generate automated reports without manually searching through paper records. The system also reduces paperwork and office supply expenses while improving the accuracy of cargo records. Furthermore, the system enables faster booking, updated shipment tracking information, and centralized payment monitoring, helping administrators manage operations efficiently. These improvements can save time, reduce operational costs, and support productivity.

Intangible Benefits

The CargoExpress PH system also provides intangible benefits to administrators and customers by improving service quality through an organized, reliable, and user-friendly platform. Customers can conveniently book shipments, monitor delivery progress through synchronized updates, and receive notifications, providing greater transparency and a more convenient customer experience. The system also enhances the business's professional image through modern progressive web-based technology. Centralized data management and secure Supabase user authentication help protect customer information while supporting better decision-making through timely and accurate cargo records.

Interface Design

The screen layout of CargoExpress PH is designed to be simple, clear, and user-friendly, allowing customers and administrators to easily navigate and manage cargo services.

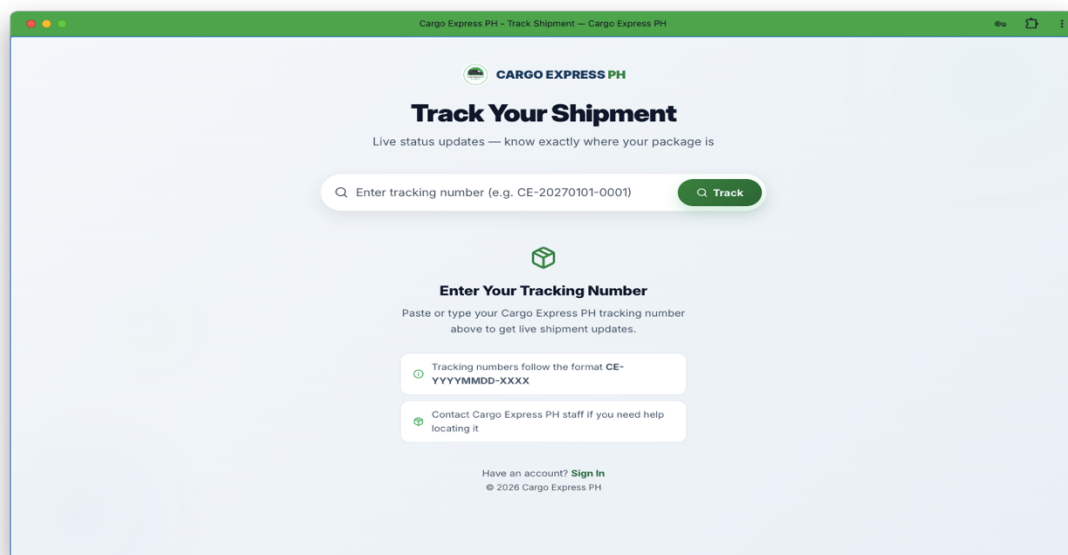
Objective ID: Public Tracking Page

Object Description: Publicly accessible tool for tracking shipments using a unique tracking number.

A. GENERAL DESCRIPTION

This allows individuals to view the current shipment status, route information, and tracking details using a unique tracking number without requiring an account.

B. SCREEN LAYOUT



Preview 11: Public Tracking Page

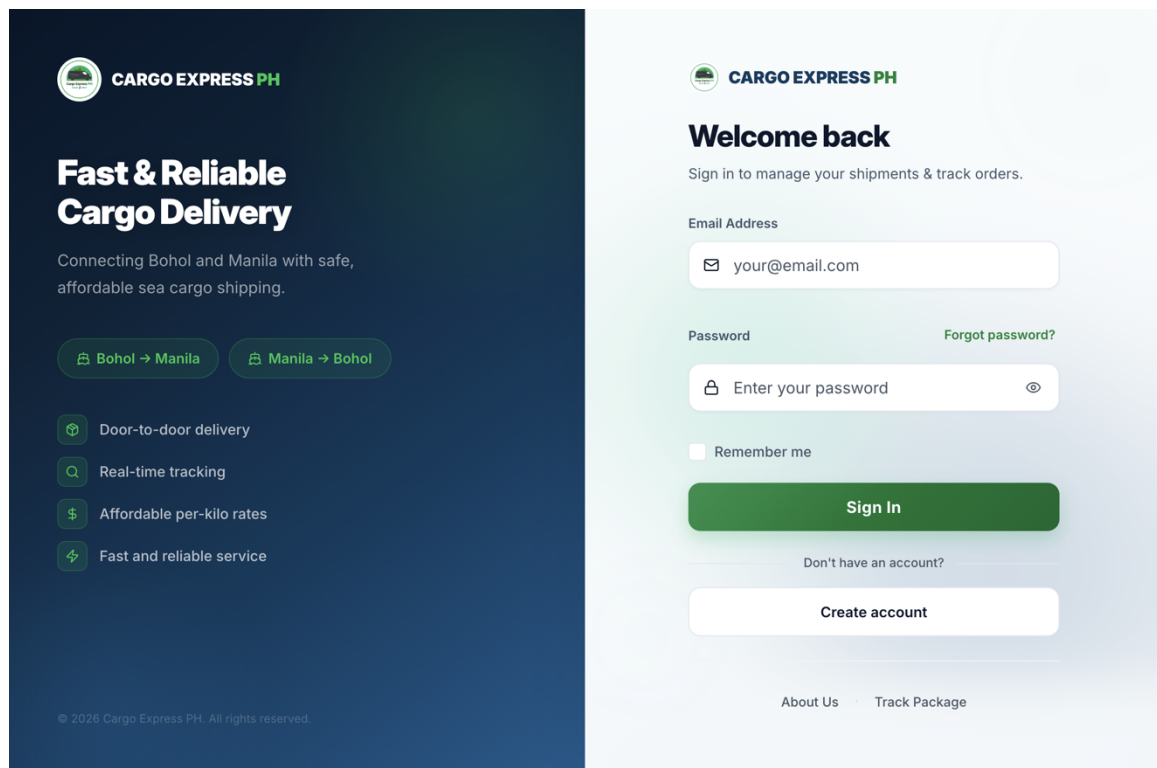
Object ID: Login Page / User Authentication Page

Object Description: Main entry screen for user authentication and account access.

A. GENERAL DESCRIPTION

This page serves as the login screen where registered users enter their email address and password to access the CargoExpress PH system. It also provides options for password recovery, account creation, and accessing the About Us and Track Package pages.

B. SCREEN LAYOUT



Preview 12: Authentication

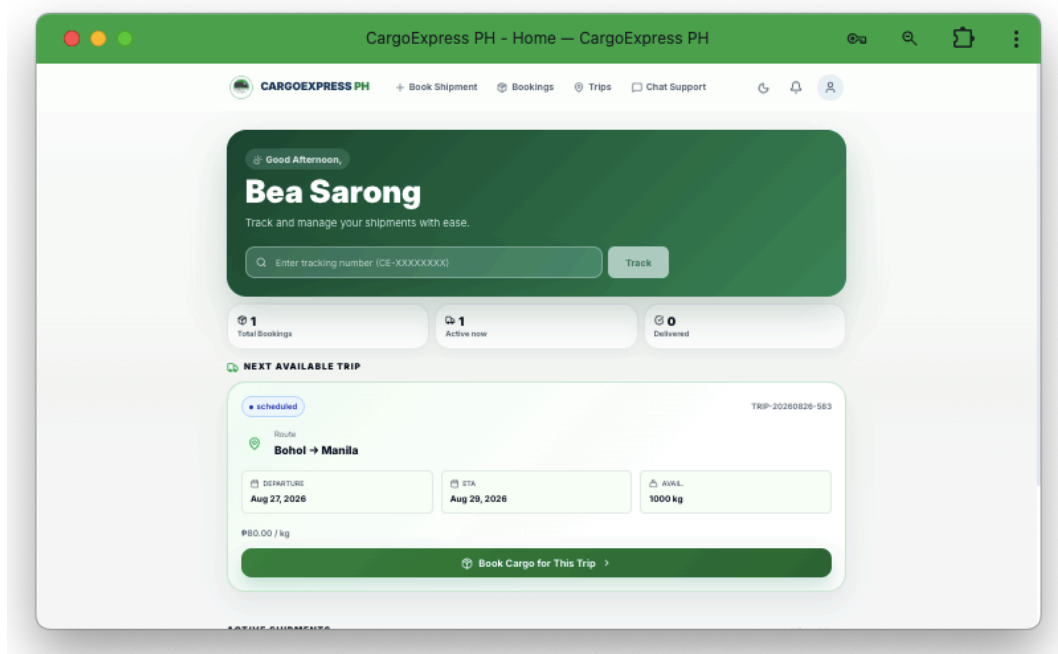
Object ID: Customer Dashboard

Object Description: Main dashboard displaying active shipments, upcoming trip schedules, and recent announcements.

A. GENERAL DESCRIPTION

This provides customers with an immediate overview of their active shipments, upcoming trip schedules, and recent announcements.

B. SCREEN LAYOUT



Preview 13: Customer Dashboard

Object ID: Book Shipment

Object Description: Form for submitting new cargo delivery requests and defining route details.

A. GENERAL DESCRIPTION

This enables customers to submit new cargo delivery requests by defining route details and providing comprehensive sender and receiver information.

B. SCREEN LAYOUT

The screenshot displays the 'Book Shipment' interface of the CARGO EXPRESS PH application. At the top, the app's header includes the logo, a '+ Place Order' button, and navigation links for Orders, Trips, and Chat Support. A user profile icon with a notification badge is also present. The main content area features a 'Book Shipment' title and a progress indicator with five steps: 1. Route (active), 2. Sender, 3. Receiver, 4. Package, and 5. Review. Below the progress bar, the 'Select Route' section provides a coverage area notice and two selectable route options: 'Bohol → Manila' and 'Manila → Bohol'. A green 'Continue' button is positioned at the bottom of the form.

Preview 14: Book Shipment Page

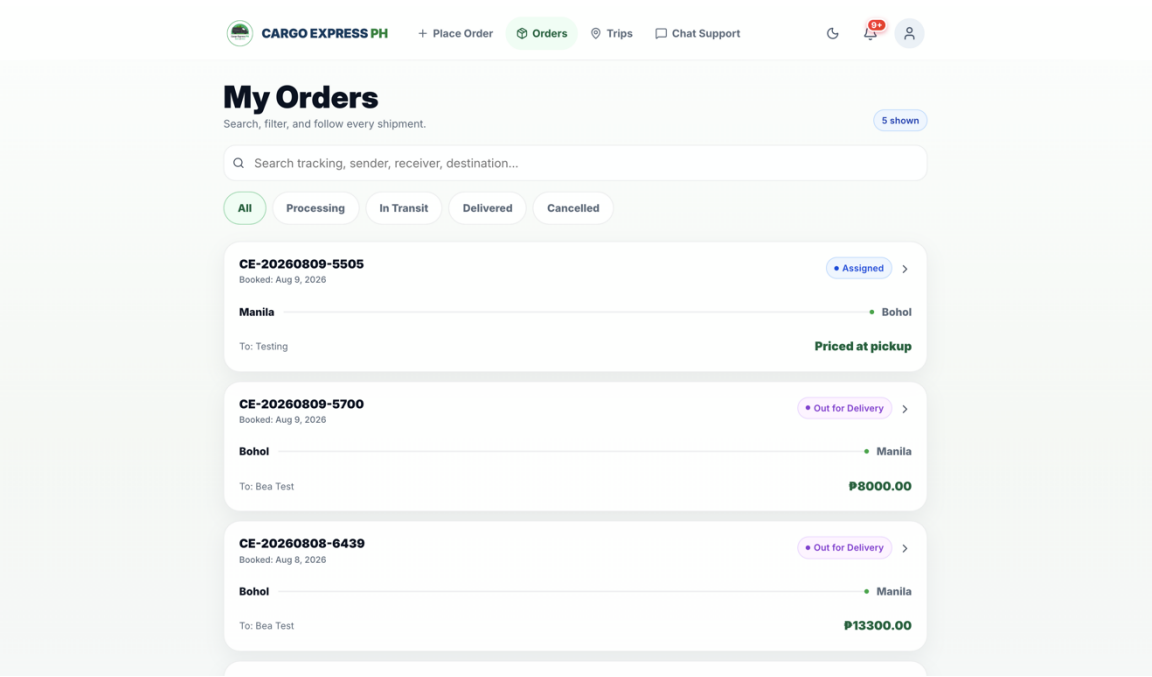
Object ID: Orders Page

Object Description: Comprehensive list of the customer's active and historical cargo shipments.

A. GENERAL DESCRIPTION

This displays a comprehensive and filterable list of the customer's active and historical cargo shipments.

B. SCREEN LAYOUT



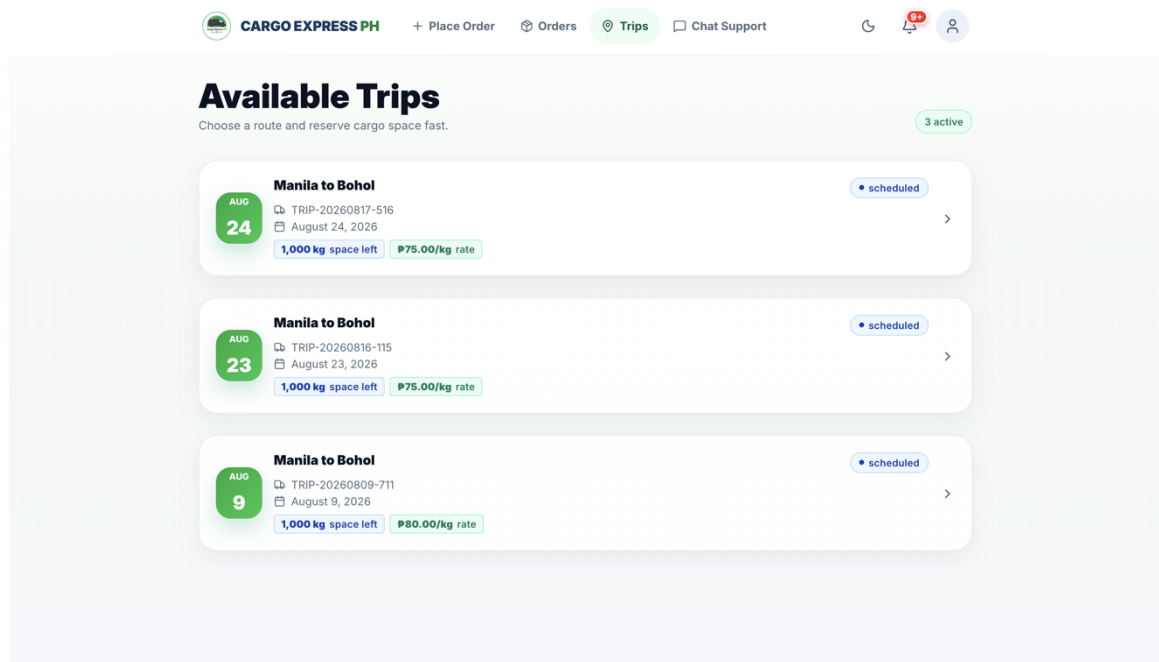
Preview 15: Orders Page

Object ID: Trips Schedules Page

Object Description: View of upcoming scheduled cargo trips and available capacities.

A. GENERAL DESCRIPTION

This presents the complete schedule of upcoming cargo trips along with their respective departure dates and available capacities.

B. SCREEN LAYOUT

Preview 16: Trip Schedules Page

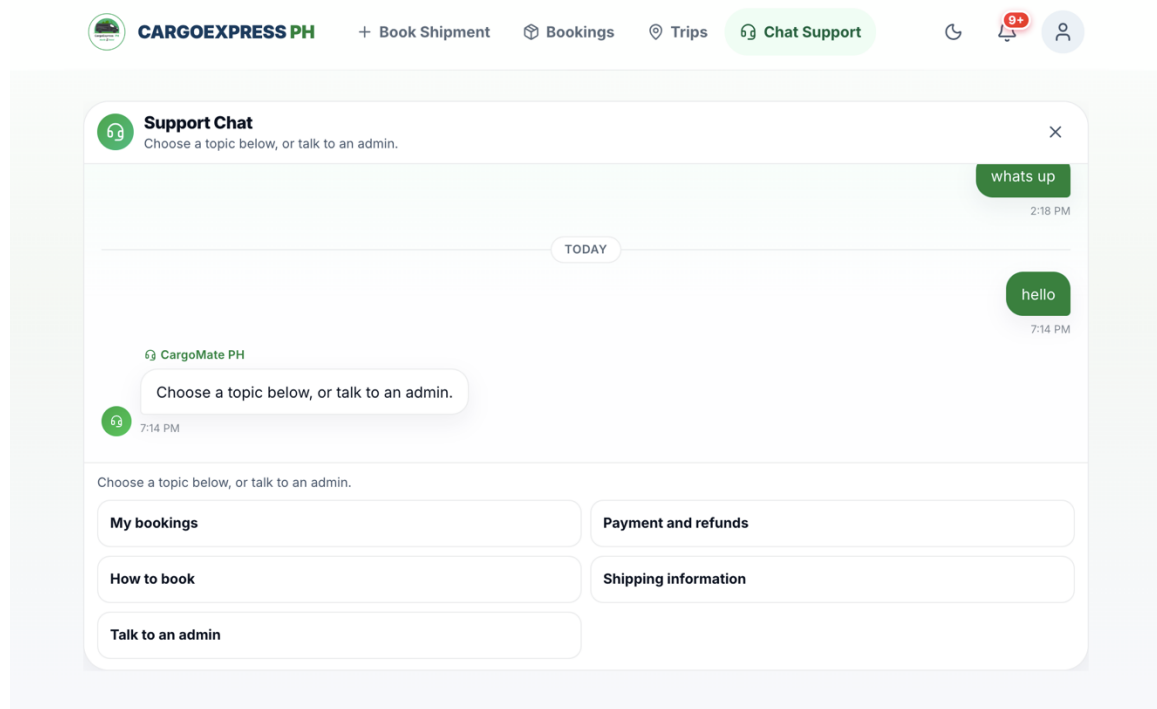
Object ID: Customer Support Chat

Object Description: An automated, menu-driven support feature that provides instant information and optional communication with system administrators.

A. GENERAL DESCRIPTION

This is an automated, menu-driven support system that retrieves real-time database information to instantly answer customer inquiries. It also allows users to escalate complex issues directly to the administrative support team.

B. SCREEN LAYOUT



Preview 17: Customer Support Chat

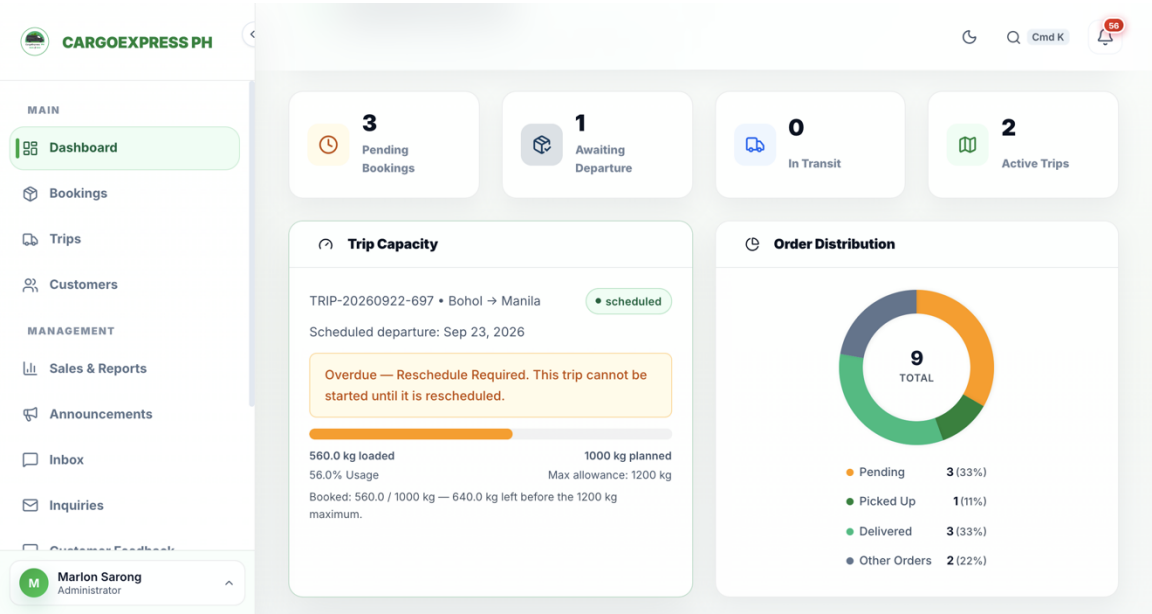
Object ID: Administrator Dashboard

Object Description: Centralized overview displaying key performance metrics and system alerts.

A. GENERAL DESCRIPTION

This aggregates vital operational data to present analytical summaries of recent bookings, active trips, and revenue metrics.

B. SCREEN LAYOUT



Preview 19: Administrator Dashboard

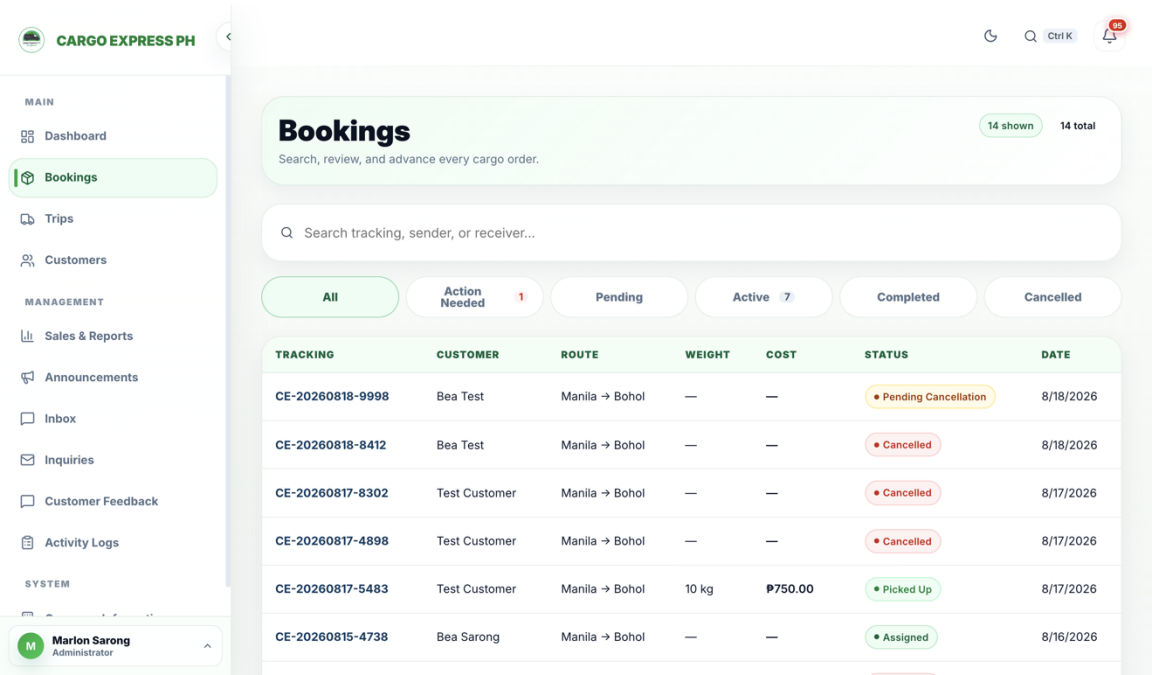
Object ID: Booking Management Page

Object Description: Administrative tool for processing, updating, and monitoring customer cargo shipments.

A. GENERAL DESCRIPTION

This equips administrators with comprehensive controls to oversee, update, and manage all customer cargo shipments.

B. SCREEN LAYOUT



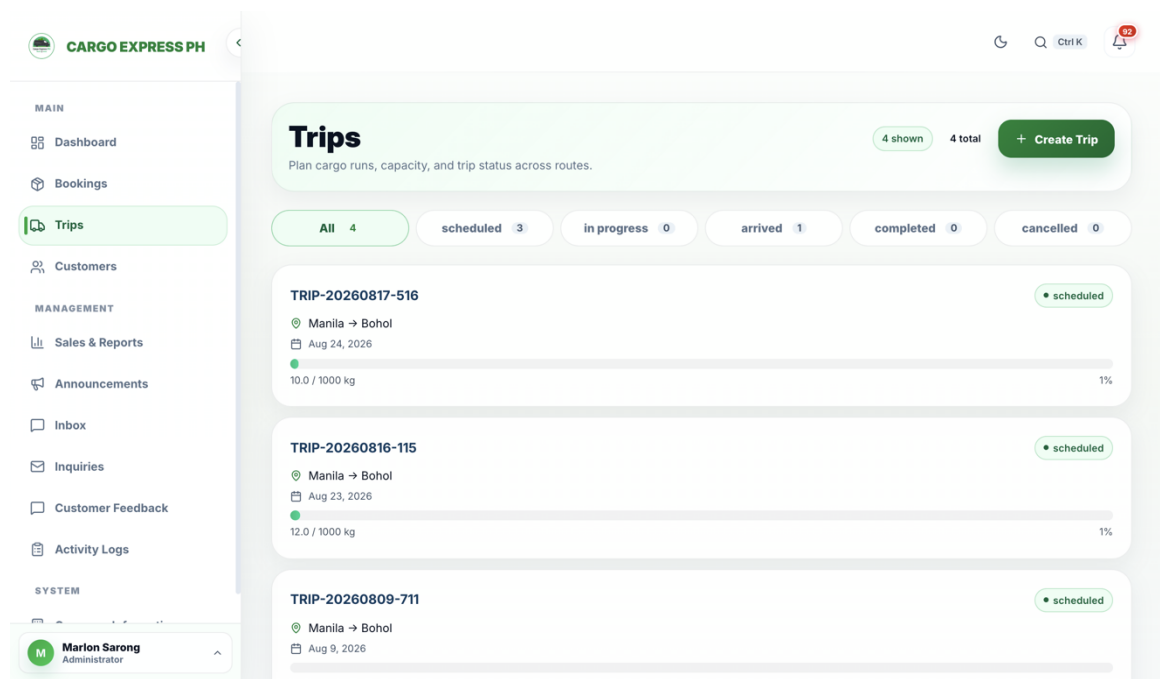
Preview 20: Booking Management Page

Object ID: Trip Management Page

Object Description: Controls for creating, scheduling, and updating cargo transport trips.

A. GENERAL DESCRIPTION

This provides administrators with the capabilities needed to schedule new cargo transport trips and manage vessel load capacities.

B. SCREEN LAYOUT

Preview 21: Trip Management Page

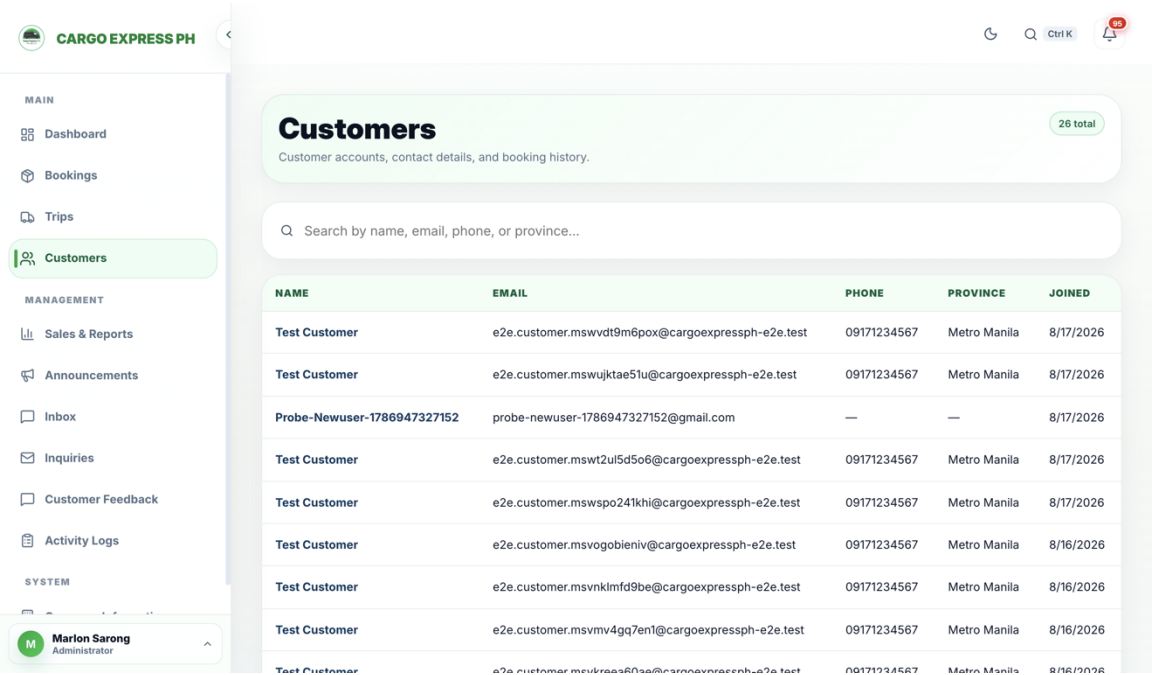
Object ID: Customer Management Page

Object Description: Directory for viewing registered customers and managing their account details.

A. GENERAL DESCRIPTION

This serves as a comprehensive directory for administrators to review registered user profiles and investigate transaction histories.

B. SCREEN LAYOUT



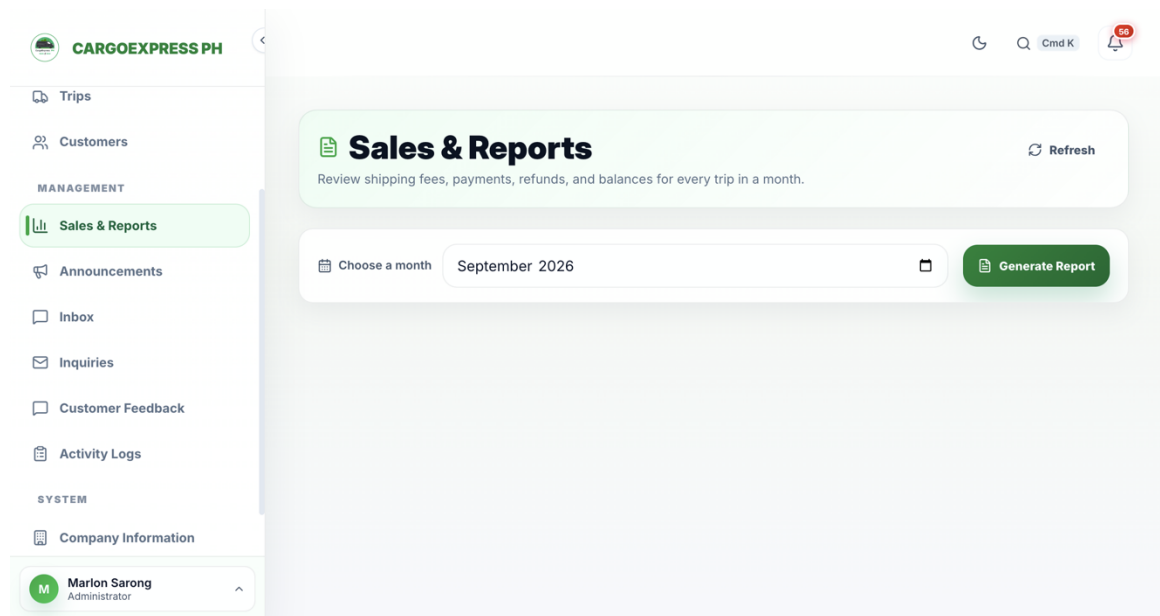
Preview 22: Customer Management Page

Object ID: Sales and Reports Page

Object Description: Analytics tool for generating revenue reports and tracking shipment volumes.

A. GENERAL DESCRIPTION

This generates analytical revenue reports and visualizes shipment volumes across different routes to support data-driven decision-making.

B. SCREEN LAYOUT

Preview 23: Sales and Reports Page

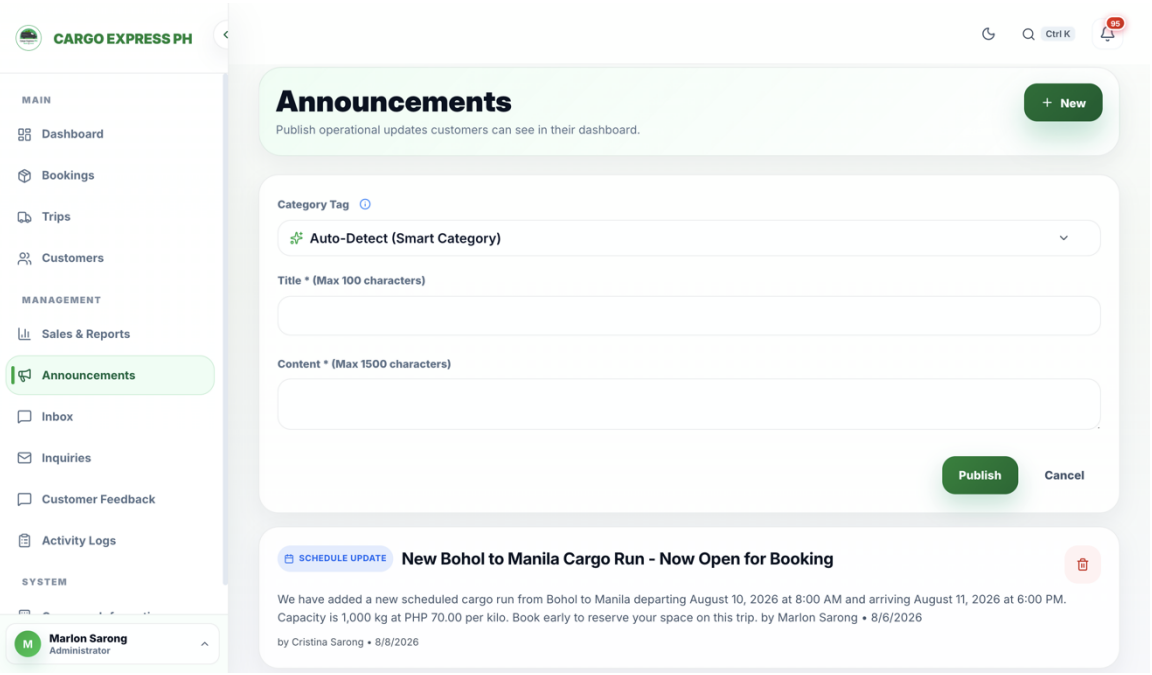
Object ID: Announcement Management

Object Description: Feature for composing and publishing system-wide advisories to customers.

A. GENERAL DESCRIPTION

This grants the administrative team the capability to compose and broadcast system-wide advisories directly to customers.

B. SCREEN LAYOUT



Preview 24: Sales and Reports Page

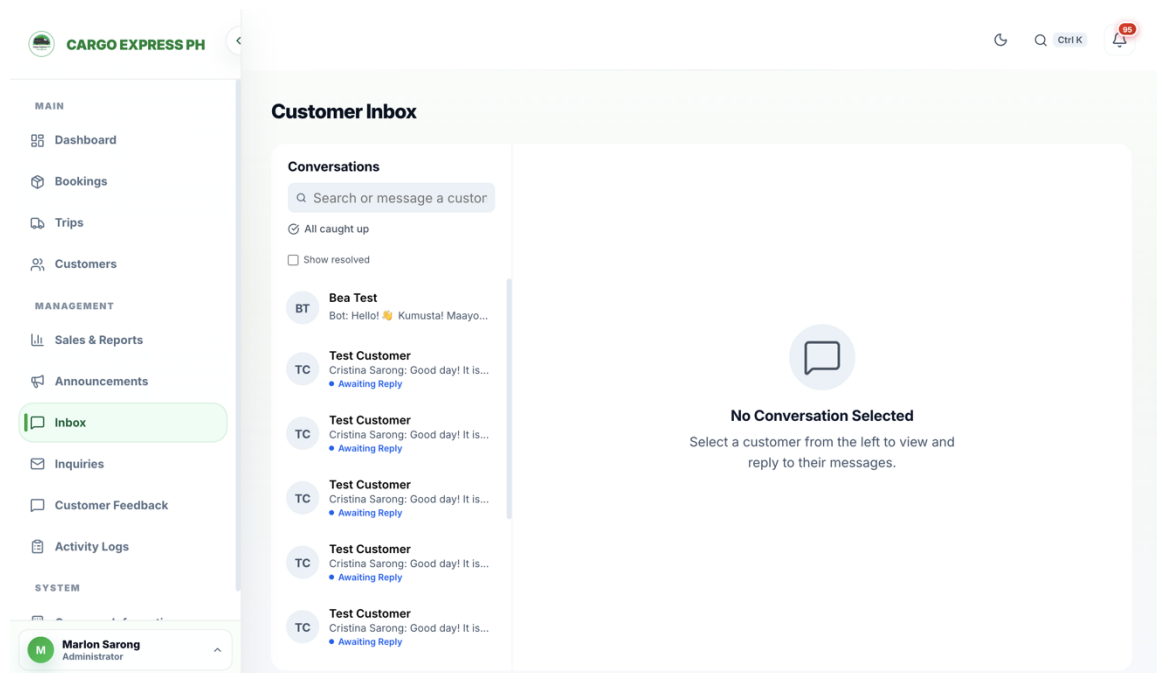
Object ID: Customer Support Inbox

Object Description: Centralized communication hub for responding to customer support chats.

A. GENERAL DESCRIPTION

This acts as a centralized communication hub for administrators to monitor and respond to support inquiries initiated by customers.

B. SCREEN LAYOUT



Preview 25: Customer Support Inbox

Chapter 3

SUMMARY, CONCLUSION AND RECOMMENDATIONS

A. Summary

This study aimed to design and develop the CargoExpress PH system, a Progressive Web App, to improve the efficiency, organization, accessibility, and reliability of cargo delivery operations. The system was designed to address the limitations associated with manual cargo booking, shipment monitoring, payment recording, trip scheduling, and customer communication. It sought to provide a centralized platform that integrated online cargo booking, shipment status tracking, trip scheduling and cargo assignment, payment processing and monitoring, public shipment tracking, customer support, and report generation. Through these functionalities, the system sought to reduce dependence on manual coordination, improve shipment visibility, organize cargo information, and provide a more convenient cargo delivery experience for administrators and customers.

The traditional cargo booking and tracking process relies heavily on Facebook Messenger, phone calls, written records, and direct communication between customers and administrators. This process creates delays in booking coordination and shipment updates, increases the possibility of incomplete or misplaced records, and makes the monitoring of payments, cargo information, and trip schedules more difficult as transactions increase. Customers also depend on administrators for shipment status updates, while administrators spend considerable time responding to repetitive inquiries and organizing cargo records.

Through the developed CargoExpress PH system, cargo transactions are centralized within a web-based platform where customers can submit bookings, monitor shipment progress, access tracking information, manage payment-related transactions, and communicate with administrators. Administrators benefit from organized booking management, trip scheduling, shipment status monitoring, payment recording, customer management, and report generation, resulting in a more efficient and organized cargo delivery workflow.

B. Conclusion

Based on the findings of the study, the CargoExpress PH system supports more efficient cargo delivery operations by providing a centralized and accessible platform for booking, shipment tracking, trip management, payment monitoring, customer communication, and operational reporting. The system improves the accuracy and organization of cargo records, reduces dependence on manual documentation and repetitive coordination, and enables administrators to retrieve shipment and payment information more efficiently. Its shipment status tracking feature enhances shipment visibility, while the public tracking page allows receivers and other users with a valid tracking number to monitor cargo progress without requiring an account.

Administrators can centrally manage bookings, delivery schedules, shipment status updates, customer records, payments, and reports, while customers can access online booking, shipment information, status updates, and support services through the system. The centralized management of information

also supports more accurate monitoring and better operational decision-making. Overall, the CargoExpress PH system supports more organized cargo operations, improved shipment visibility, easier access to records, and more convenient communication between administrators and customers.

C. Recommendations

Based on the study, the following recommendations are:

1. The CargoExpress PH system should be implemented to improve the accuracy, efficiency, and reliability of cargo delivery operations.
2. Administrators should undergo proper training covering booking management, trip scheduling, payment verification, and report generation to ensure correct and consistent use of the system.
3. Regular system maintenance and evaluation should be conducted to maintain system security, performance, and compatibility and to address future operational needs.
4. A regular and secure data backup and recovery schedule should be established to protect important system records and support business continuity.
5. Future development may include additional reporting and route analysis features to support the growing operational needs of the business.

TECHNOLOGY ACTION PLAN

CargoExpress PH: A Progressive Web-Based Cargo Booking and Tracking System

Rationale

The existing cargo delivery process relies on Facebook Messenger, phone calls, and manual record-keeping, which can lead to delays in booking coordination, incomplete or disorganized records, repetitive shipment inquiries, and difficulties in monitoring payments and delivery schedules. The implementation of the CargoExpress PH system provides a more centralized and organized approach to managing cargo delivery operations. Through online booking, shipment status tracking, trip scheduling, payment monitoring, customer communication, and centralized record management, the system reduces dependence on fragmented manual coordination and supports more accessible and organized services for administrators and customers.

Objectives

1. To implement the CargoExpress PH system as a centralized platform for cargo booking, shipment monitoring, trip management, and payment monitoring.
2. To provide proper training and orientation for administrators to ensure the correct and effective use of the system's operational functions.

3. To support the sustainable operation of the system through regular monitoring, maintenance, evaluation, and secure data backup and recovery procedures.

Schedule of Implementation

The CargoExpress PH system will be implemented after the completion of the necessary testing, final system review, and approval by the management. Administrator orientation and training will be conducted before the system is introduced for initial operational use. During the initial implementation period, system performance and usage will be monitored to identify possible issues and necessary adjustments. After the system has demonstrated proper functionality and usability, it may be adopted as the primary platform for managing CargoExpress PH operations.

Mechanics of Implementation

The researchers will coordinate with the CargoExpress PH management to present the system and demonstrate its major functions. Training sessions will be conducted for administrators to familiarize them with booking management, guest booking management, trip scheduling, cargo assignment, shipment status updates, payment verification and monitoring, customer support, and report generation. After the orientation and approval of management, the system will be introduced for operational use. Technical support, system monitoring, regular maintenance, and appropriate data backup and recovery procedures will also be

provided to support continuous and reliable system operation. Necessary updates may be applied based on future operational requirements.

Benefits

The implementation of the CargoExpress PH system provides several operational benefits by reducing dependence on manual record-keeping and centralizing cargo information within a single platform. It enables easier access to booking, shipment, trip, and payment records while providing customers with updated shipment status information. The system also supports organized trip scheduling, payment monitoring, customer communication, and report generation. These functions help administrators manage cargo operations more systematically while providing customers with more convenient access to booking and shipment information.

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APPENDICES

APPENDIX A

LETTER OF INTENT



Republic of the Philippines
BOHOL ISLAND STATE UNIVERSITY
 Magsija, Balilihan 6342, Bohol, Philippines
 Office of the College of Computing and Information Sciences



Management
 System
 ISO 9001:2015
 www.tuv.com
 ID: 612655238

Balance | Integrity | Stewardship | Uprightness

Feb. 16, 2026

MR. MARLON A. SARONG
 Owner
 Cargo Delivery Business
 Batuan, Bohol

Sir:

We, the Bachelor of Science in Information Technology junior students of Bohol Island State University–Balilihan Campus, are currently undertaking a system proposal as a major academic requirement of our course. In line with this requirement, our group has identified your cargo delivery business as the proposed locale of our study entitled "A Mobile Application for Cargo Delivery Booking and Tracking for Bohol–Manila Routes".

In this regard, we respectfully seek permission from your esteemed office to allow us to conduct interviews, administer surveys, perform site observations, and review pertinent documents deemed necessary for data gathering. These activities will greatly contribute to the successful completion of our academic study.

We assure you that all information gathered will be treated with utmost confidentiality and used solely for academic purposes. We sincerely hope for your favorable consideration and kind support.

Thank you very much for your time and consideration. May God bless you.

Very respectfully yours,

Quin Ann P. Amigo
 Bea A. Sarong
 Joven M. Tibuc

Noted:

(SGD.) SHELLA C. OLAGUIR, PhD
 Dean, CCIS

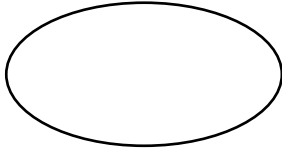
Approved:

MR. MARLON A. SARONG
 Owner

APPENDIX B

MODELING TOOLS

TERMINAL SYMBOL



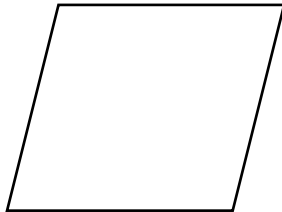
- Indicates the origin of data and its predetermined end
- Describes external entities referred to as a person or a group of departments with which the system communicates.

PROCESS SYMBOL



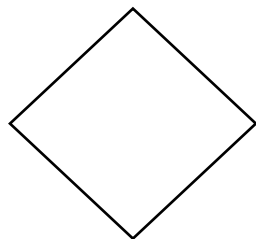
- Symbol used for processing of data into useful information
- Describes a part of the system that transforms data inputs and outputs in the same way

DATA SYMBOL



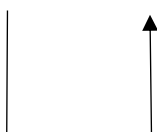
- Indicates where data is stored.
- An opened-end rectangle used to present storage
- A symbol used to represent input or output choices in a process.

DECISION SYMBOL

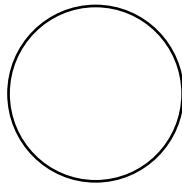


- A decision or branching point. Lines representing different decisions emerge from different points of the diamond

DATA FLOW SYMBOL



Represented by an arrow which



OUTPUT/DISPLAY SYMBOL

- Symbol used for displaying an output.
- It represents operations like displaying or printing information.

INTERVIEW GUIDE / QUESTIONS

1. Can you describe how you currently manage your daily cargo delivery operations, including customer inquiries, bookings, and shipment records?
2. What challenges do you encounter in monitoring cargo status and providing shipment updates to customers?
3. How do you currently record payments and remaining balances, and what difficulties do you encounter in following up on customers' promised payment dates?
4. What issues do you experience when organizing trip schedules, assigning cargo to trips, and coordinating pickups and deliveries?
5. How do you currently prepare reports, such as shipment summaries and payment records, and what problems do you encounter in doing so?
6. What specific features or improvements would you like to see in the proposed CargoExpress PH: A Progressive Web-Based Cargo Delivery Booking and Tracking System?

APPENDIX D

User's Guide

This guide provides detailed steps for accessing, installing, and using CargoExpress PH, a Progressive Web-Based Cargo Delivery Booking and Tracking System.

1. Prerequisites

Before using CargoExpress PH, ensure that the following are available:

- Device: Desktop computer, laptop, tablet, or smartphone
- Internet Connection: Stable Wi-Fi, broadband, or mobile data connection
- Web Browser: A modern web browser such as Google Chrome, Microsoft Edge, Safari, or another supported browser
- CargoExpress PH Website: Access to the official CargoExpress PH web address
- Email Address: Required when creating a customer account and accessing account-related features

2. Application Requirements

2.1 Hardware Requirements

- RAM: Minimum of 4 GB
- Processor: 1.5 GHz dual-core processor or higher
- Storage: At least 32 GB internal storage for the operating system and browser
- Mobile Display: 6-inch touchscreen display or higher
- Desktop/Laptop Display: 13-inch display with at least 1366 × 768

resolution

- Connectivity: Minimum of 5 Mbps broadband or stable 4G/LTE mobile data connection

2.2 Software Requirements

- Desktop Operating Systems: Windows, macOS, or Linux
- Mobile Operating Systems: Android or iOS
- Browser: A modern browser that supports standard web and PWA features
- Internet: Required to access and use the system
- JavaScript: Must be enabled in the browser

3. Accessing and Installing the Application

3.1 Accessing CargoExpress PH Through a Web Browser

1. Connect the device to the Internet.
2. Open a supported web browser.
3. Enter the official CargoExpress PH website address.
4. Wait for the CargoExpress PH landing or login page to load.
5. The user may proceed to log in, create an account, track a package, or view public information.

3.2 Installing CargoExpress PH as a PWA

For Android / Google Chrome

1. Open CargoExpress PH using Google Chrome.
2. Open the browser menu.
3. Select Install App or Add to Home Screen.

4. Confirm the installation.
5. Wait for the application icon to appear on the Home Screen.
6. Tap the CargoExpress PH icon to open the system.

For iPhone / iPad using Safari

1. Open CargoExpress PH using Safari.
2. Tap the Share button.
3. Select Add to Home Screen.
4. Confirm the application name.
5. Tap Add.
6. Open CargoExpress PH using the icon added to the Home Screen.

For Desktop Browsers

1. Open CargoExpress PH in a supported browser.
2. If an installation option is available, click the Install icon or browser installation option.
3. Confirm the installation.
4. CargoExpress PH may then be opened from the desktop, Start menu, Applications folder, or other application launcher depending on the operating system.

4. Creating a Customer Account

4.1 Customer Registration

1. Open the CargoExpress PH website.
2. Click Create Account.

3. Enter the required personal information.
4. Enter the required contact information.
5. Provide the complete address information.
6. Enter a valid email address.
7. Create a secure password.
8. Review the information entered.
9. Agree to the applicable Terms and Conditions and Privacy Policy.
10. Click Register or Create Account to complete the registration.

5. Logging In

1. Open CargoExpress PH.
2. Enter the registered email address.
3. Enter the account password.
4. Select Remember Me if desired.
5. Click Sign In.
6. After successful authentication, the customer will be redirected to the Customer Dashboard.

6. Navigating the Application

6.1 Customer Functions

Registered customers may use CargoExpress PH to:

- View their dashboard
- Book a cargo shipment
- View current and previous bookings
- Check available trip schedules

- Monitor cargo shipment status
- View shipment and payment information
- Receive notifications
- Access customer support
- Communicate with an administrator when necessary
- Manage profile and account settings
- Submit feedback after a completed delivery

6.2 Administrator Functions

The administrator may use CargoExpress PH to:

- View the administrative dashboard
- Manage cargo bookings
- Create walk-in or guest bookings
- Create and manage trip schedules
- Assign bookings to trips
- Process cargo pickup and weighing
- Record and monitor payments
- Update shipment status
- Process delivery information
- Manage registered customers
- Generate reports
- Manage announcements
- Respond to customer inquiries and support conversations
- Review customer feedback

- Maintain company information
- Review system activity logs
- Monitor system-related information

7. Troubleshooting

7.1 Website or Application Does Not Open

- Check the Internet connection.
- Refresh the browser.
- Close and reopen the browser.
- Verify that the correct CargoExpress PH website address is being used.
- Try using another supported browser.
- Restart the device if necessary.

7.2 Login Problems

- Check the Internet connection.
- Make sure the registered email address is entered correctly.
- Make sure the password is correct.
- Use Forgot Password? if the password cannot be remembered.
- Follow the password recovery instructions sent to the registered email address.

7.3 PWA Cannot Be Installed

- Make sure the browser supports PWA installation.
- Refresh the CargoExpress PH website.
- Check the browser menu for Install App or Add to Home Screen.
- On iOS, use Safari and select Share → Add to Home Screen.

- The system may still be used directly through the browser even if it is not installed.

7.4 Photo Upload Fails

- Check the Internet connection.
- Ensure that the selected file is a supported image.
- Try uploading the image again.
- Check whether the browser has permission to access photos or files.

7.5 Payment Does Not Update Immediately

- Check the Internet connection.
- Do not repeat a payment immediately if the previous transaction is still processing.
- Refresh the booking or payment information after waiting briefly.
- If the payment still does not appear, contact the administrator and provide the applicable transaction information.

7.6 Shipment Status Does Not Change

- Shipment status is updated based on the actual delivery process. If no new status appears, the customer may:
- Refresh the tracking page.
- Check notifications.
- Use Customer Support for additional information.
- Select Talk to an Admin when further assistance is required.

8. Support

For assistance, customers may use the available CargoExpress PH support channels.

- Customer Support Chat: Available to registered customers
- Talk to an Admin: Available through Customer Support when further assistance is needed
- Public Contact Form: Available to visitors who are not logged in
- Official Contact Information: Available through the CargoExpress PH website

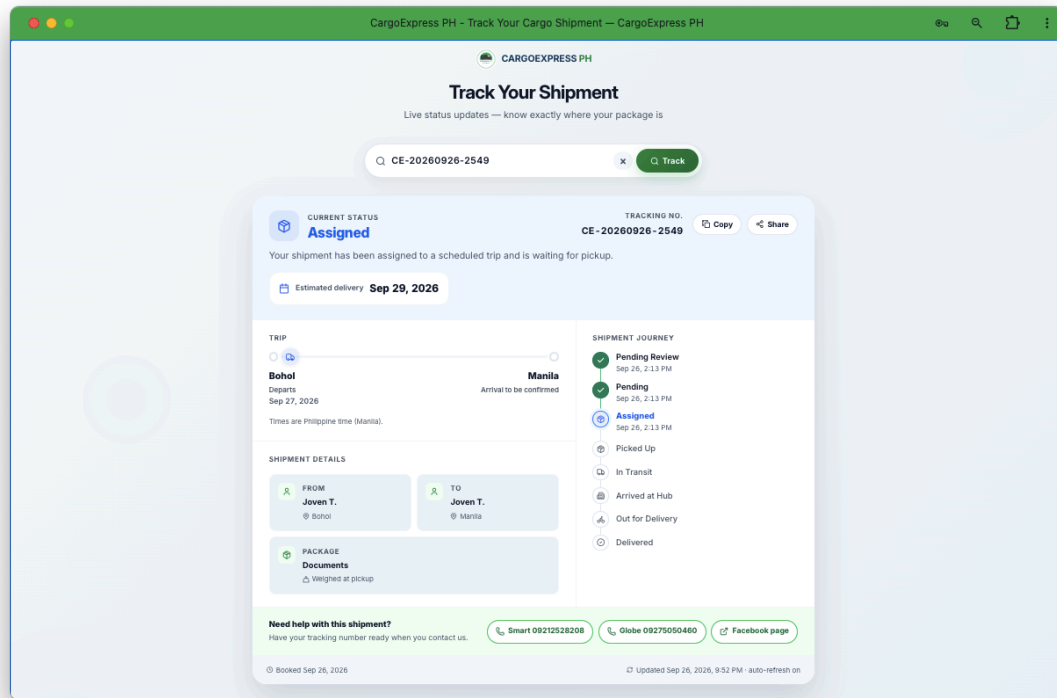
I would not invent an email address or phone number here. Use the official contact details already configured in your actual CargoExpress PH Company Information.

Running and Navigation of the System

This manual serves as a guide for users of CargoExpress PH by presenting step-by-step procedures for the major functions of the system. Visual illustrations should be included to support each procedure.

Process Name: Public Shipment Tracking

Description: Allows users to track a shipment using a valid tracking number.



Procedure:

1. Open the CargoExpress PH website.
2. Click Track Package.
3. Enter the tracking number.
4. Click Track.

Process Name: Customer Login

Description: Allows registered customers to access their account.

CargoExpress PH - Login - CargoExpress PH

CARGOEXPRESS PH

Fast & Reliable Cargo Delivery

Connecting Bohol and Manila with safe, affordable sea cargo shipping.

[Bohol → Manila](#) [Manila → Bohol](#)

- Door-to-door delivery
- Delivery status updates
- Affordable per-kilo rates
- Fast and reliable service

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CARGOEXPRESS PH

Welcome back

Sign in to manage your shipments & track orders.

Email Address

Password [Forgot password?](#)

☒ Remember me

[Sign In](#)

[Don't have an account?](#)

[Create account](#)

[About Us](#) [Track Package](#)

Procedure:

1. Open CargoExpress PH.
2. Enter email and password.
3. Click Sign In.

Process Name: Customer Registration

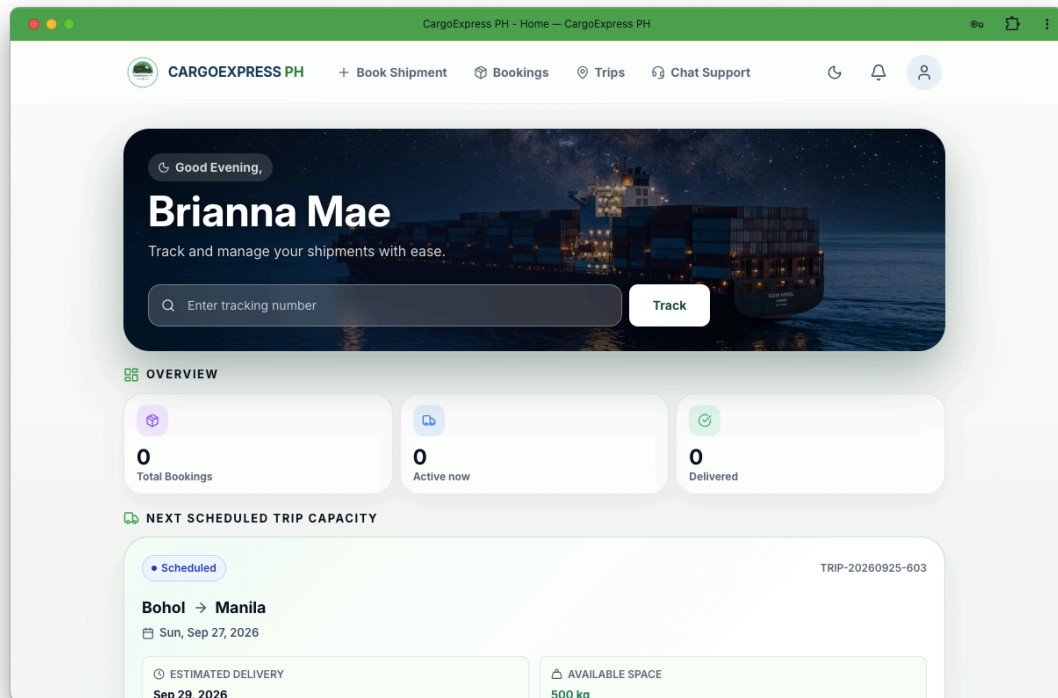
Description: Allows new customers to create an account.

Procedure:

1. Click Create Account.
2. Enter the required information.
3. Create a password.
4. Click Register.

Process Name: Customer Dashboard

Description: Displays the customer's shipment and account overview.



Procedure:

1. Log in to the customer account.
2. View the dashboard.
3. Select the needed menu.

Process Name: Book Shipment

Description: Allows customers to create new cargo booking.

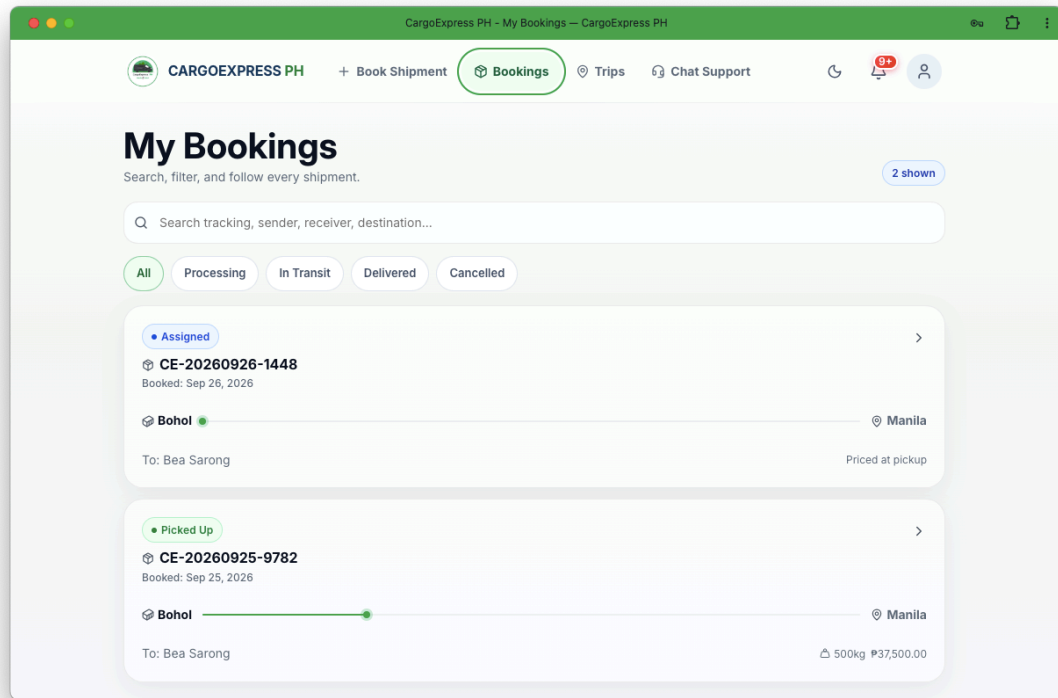
The screenshot shows a web application interface for 'CargoExpress PH'. The browser title bar reads 'CargoExpress PH - Book Shipment - CargoExpress PH'. The application header includes the logo, a '+ Book Shipment' button, and links for 'Bookings', 'Trips', and 'Chat Support'. A user profile icon is in the top right. The main content area has a 'Cancel' link and a 'Book Shipment' title. Below the title is a progress bar with five steps: 1. Route (active), 2. Sender, 3. Receiver, 4. Package, and 5. Review. The 'Select Route' section contains a note: 'Coverage Area: CargoExpress PH currently operates routes to and from Bohol only. Select a route below to view specific province rules.' There are two route selection buttons: 'Bohol → Manila' and 'Manila → Bohol'. A green 'Continue' button is at the bottom.

Procedure:

1. Click Book Shipment.
2. Enter the required booking details.
3. Review the information.
4. Submit the booking.

Process Name: View Customer Bookings

Description: Displays the customer's current and previous bookings.

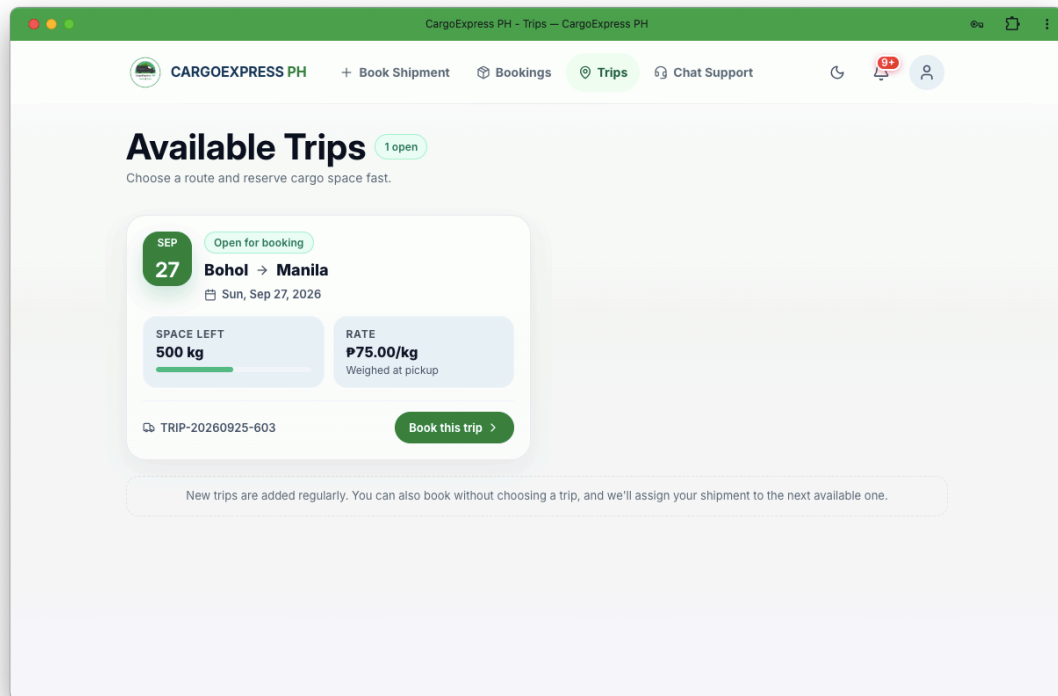


Procedure:

1. Click Bookings.
2. Select a booking.
3. View the booking details.

Process Name: View Available Trips

Description: Displays available and scheduled cargo trips.

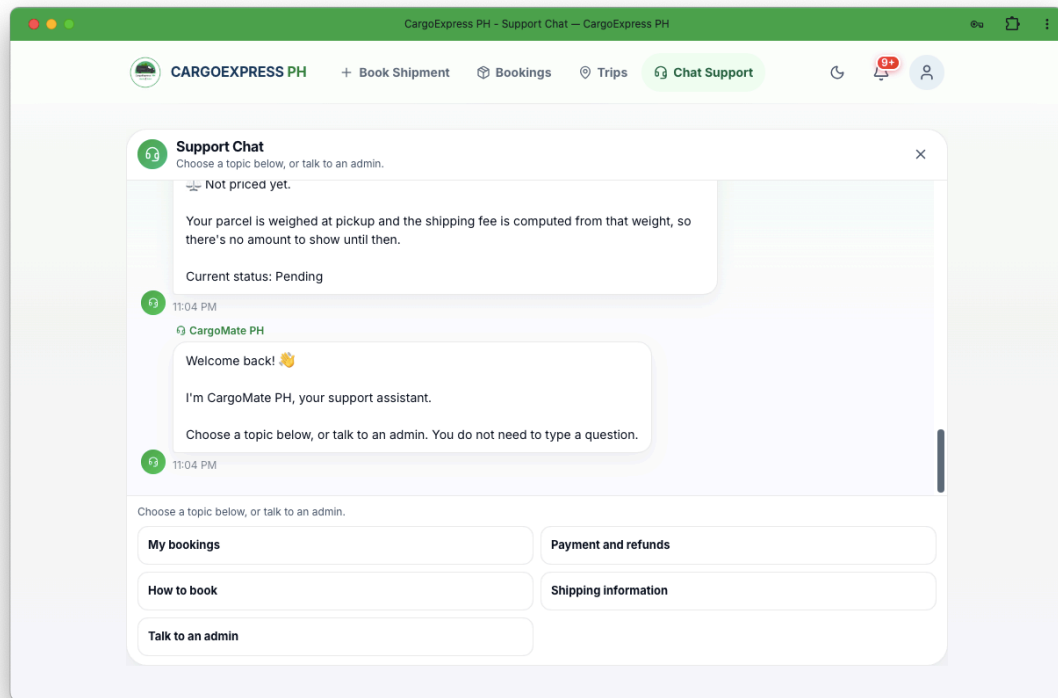


Procedure:

1. Click Trips.
2. View the available schedules.
3. Select a trip if wants to book

Process Name: Customer Support Chat

Description: Allows customers to ask for help and contact administrator.

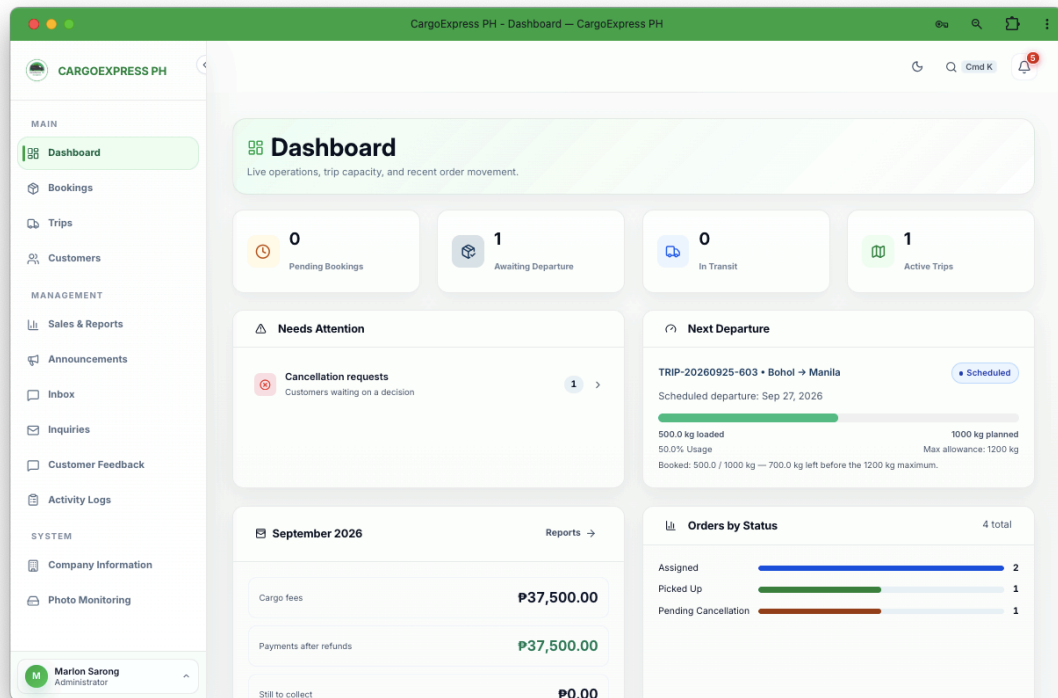


Procedure:

1. Click Chat Support.
2. Select a support option.
3. Send a message if needed

Process Name: Administrator Dashboard

Description: Displays an overview of cargo operations and system information.

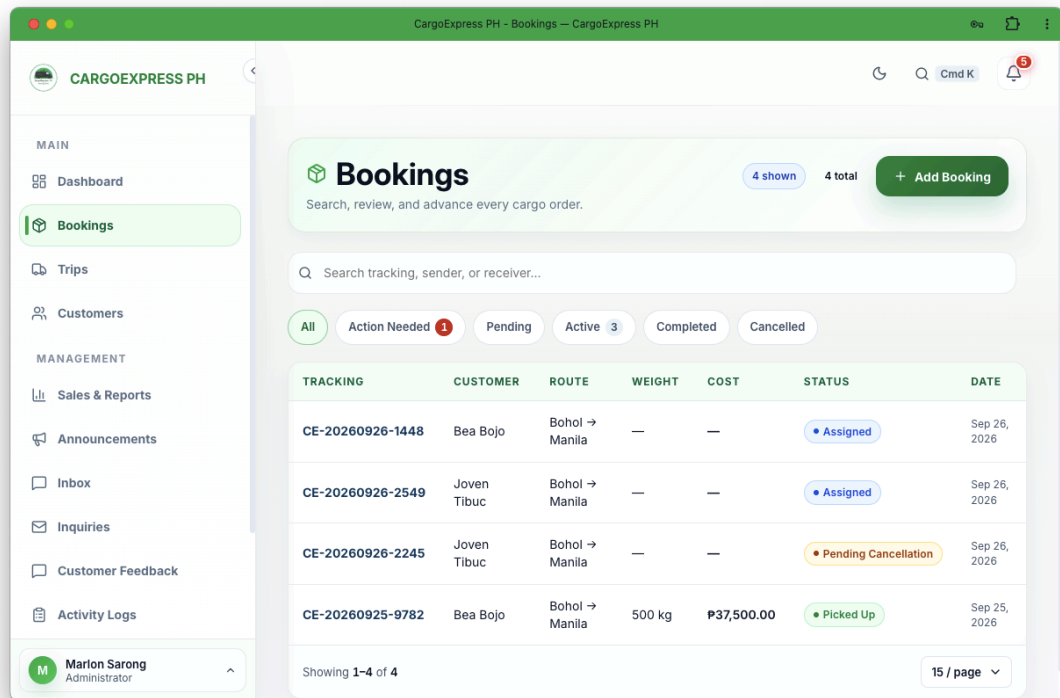


Procedure:

1. Log in as administrator.
2. View the dashboard.
3. Select the needed module.

Process Name: Booking Management

Description: Allows administrators to manage customer cargo bookings.



Procedure:

1. Click Bookings.
2. Select a booking.
3. Update or manage the booking.

Process Name: Trip Management

Description: Allows administrators to create and manage cargo trips.

CARGOEXPRESS PH

MAIN

- Dashboard
- Bookings
- Trips**
- Customers

MANAGEMENT

- Sales & Reports
- Announcements
- Inbox
- Inquiries
- Customer Feedback
- Activity Logs

SYSTEM

- Company Information
- Photo Monitoring

Marion Sarong
Administrator

Create New Trip
Define the route and schedule for a cargo run. Capacity and pricing come from your Company Information defaults.

Route

Bohol → Manila
Bohol → Manila

Manila → Bohol
Manila → Bohol

Schedule

Departure Date: mm/dd/yyyy

Estimated Shipment Delivery Date: mm/dd/yyyy

Capacity & Pricing

Capacity: 1,000 kg

Amount per Kilo: P75

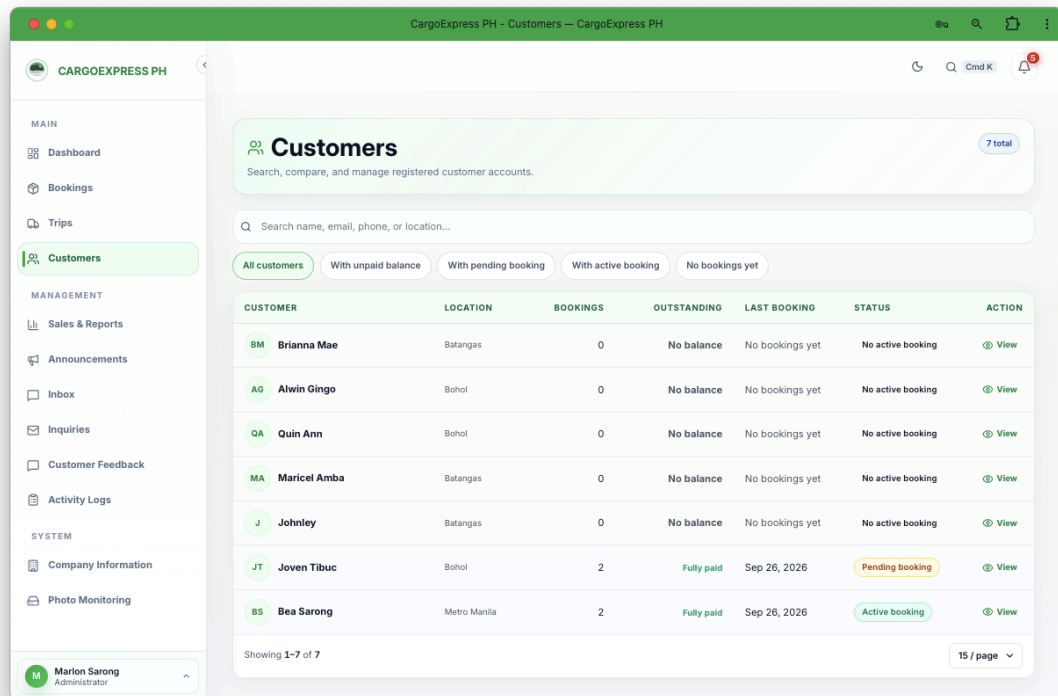
At P75.00/kg, a full trip of 1,000 kg = P75,000 max revenue

Procedure:

1. Click Trips.
2. Select or create a trip.
3. Save the trip details.

Process Name: Customer Management

Description: Allows administrators to view and manage customer records.

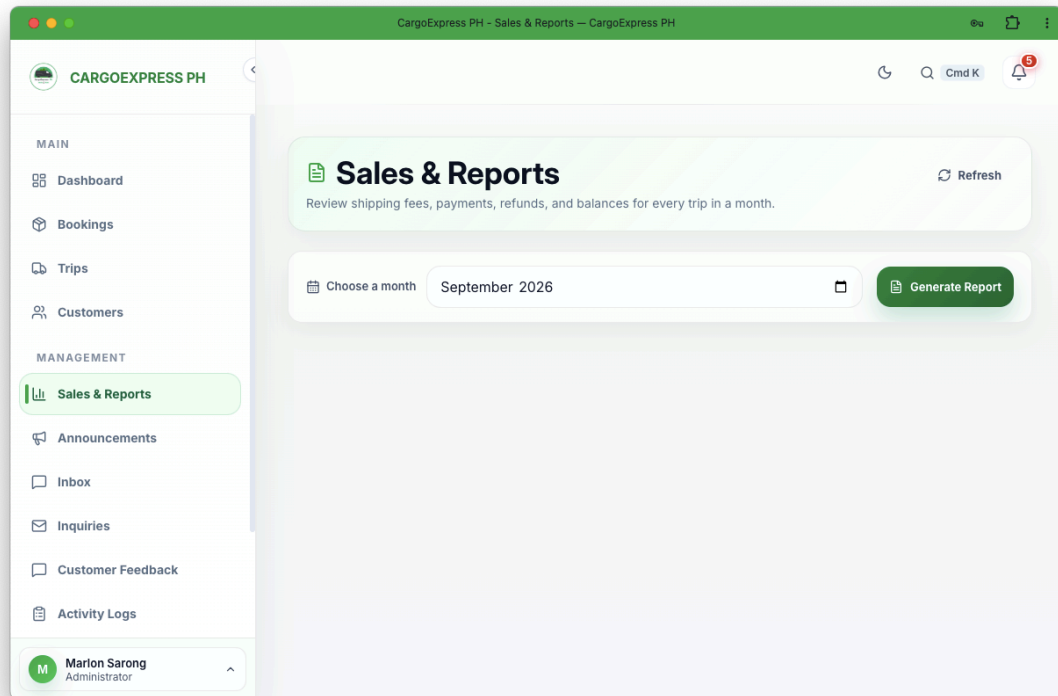


Procedure:

1. Click Customers.
2. Select a customer.
3. View the customer details.

Process Name: Sales and Report

Description: Allows administrators to view and generate system reports.

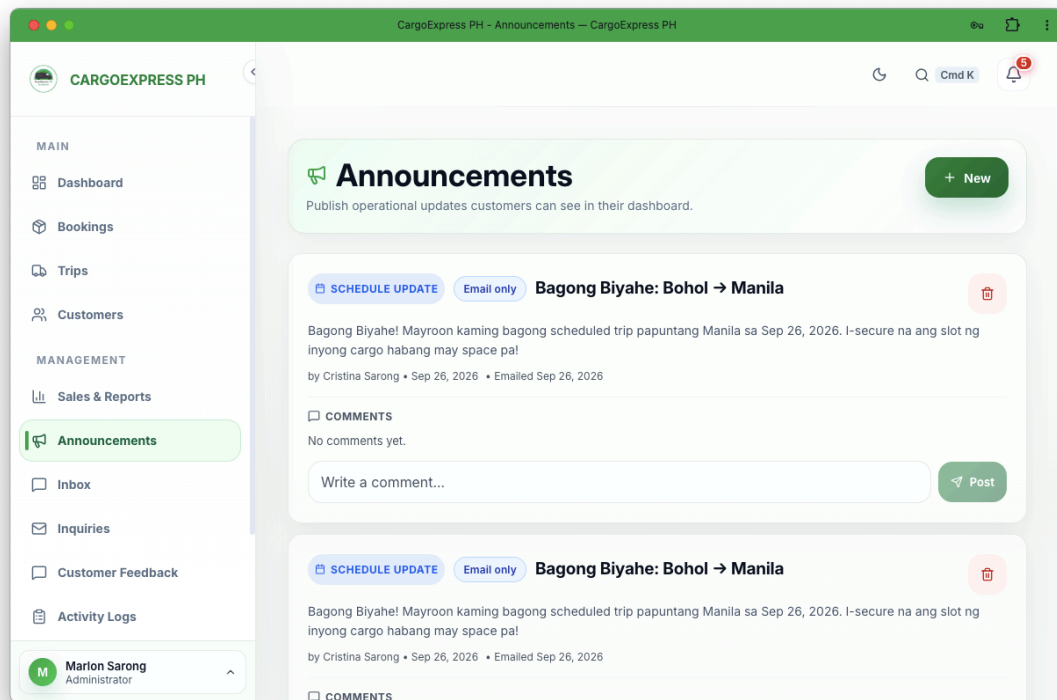


Procedure:

1. Click Sales & Reports.
2. Select the needed report.
3. View or generate the report.

Process Name: Announcement Management

Description: Allows administrators to create and publish announcements.



Procedure:

1. Click Announcements.
2. Create or select an announcement.
3. Enter the details.
4. Publish the announcement.

APPENDIX E

SOURCE CODE

1. User Authentication and Registration

```
const login = async (email, password) => {
  try {
    isAuthAction.current = true;
    const { data, error } = await
supabase.auth.signInWithPassword(
{ email, password });
    if (error) throw error;
    setLoading(true);
    const profileResult = await
fetchProfile(data.user.id);
    if (!profileResult.success) {
      await supabase.auth.signOut();
      throw new Error(`Failed to
retrieve user profile:
${profileResult.error.message ||
'Unknown database error'}`);
    }
  }
```

```
clearPasswordRecoveryPending();
  isAuthAction.current = false;
  return { success: true, user:
data.user, profile:
profileResult.profile };
} catch (error) {
  isAuthAction.current = false;
  setLoading(false);
  let msg = error.message || 'An
unexpected error occurred.';
  if
(msg.toLowerCase().includes('invalid
login credentials') ||
```

```
msg.toLowerCase().includes('invalid
login')) {
    msg = 'Incorrect password or
email.';
  } else if
(msg.toLowerCase().includes('email
not confirmed')) {
```

```
    msg = 'Your email is not
confirmed. Please check your
inbox.';
  } else if
(msg.toLowerCase().includes('rate
limit') ||
```

```
msg.toLowerCase().includes('too
many')) {
    msg = 'Too many failed
attempts. Please wait a few minutes
and try again.';
  }
  return { success: false, error:
msg };
}
```

```
const register = async (email, password, profileData) => {
  try {
```

```
setAuthTransition(AUTH_TRANSITI
ONS.REGISTERING);
    setLoading(true);
    const { legal_consent:
legalConsent, ...profileFields } =
profileData || {};
    if (
      legalConsent?.termsAccepted
!== true ||
      legalConsent?.privacyAccepted
!== true ||
      legalConsent?.termsVersion !==
LEGAL_DOCUMENTS.terms.version
||
      legalConsent?.privacyVersion
!==
LEGAL_DOCUMENTS.privacy.versi
on
    ) {
      throw new Error('You must
agree to the current Terms of
```

```

Service and Privacy Policy to create
an account.');
```

`}
 isAuthAction.current = true;
 const { data, error } = await
supabase.auth.signUp({
 email,
 password,
 options: {
 data: {
 name: profileFields.name,
 legal_terms_accepted: true,
 legal_privacy_accepted: true,
 legal_terms_version:
legalConsent.termsVersion,
 legal_privacy_version:
legalConsent.privacyVersion,
 },
 },
});
if (error) throw error;
const normalizedAddress =
normalizeProfileAddressFields(profil
eFields);
const profilePayload = {
 id: data.user.id,
 email,
 name: profileFields.name,
 facebook_name:
profileFields.facebook_name || null,
 phone: profileFields.phone ||
null,
 role: 'customer',
 address_lot_block:
normalizedAddress.address_lot_bloc
k || null,
 address_street:
normalizedAddress.address_street ||
null,
 address_barangay:
normalizedAddress.address_barang
ay || null,
 address_city:
normalizedAddress.address_city ||
null,
 address_province:`

```

normalizedAddress.address_provinc
e || null,
  address_landmark:
normalizedAddress.address_landma
rk || null,
  wants_announcements:
profileFields.wants_announcements
=== true,
};
let profileSaved = true;
try {
  await
createProfile(profilePayload);
} catch (profileError) {
  await new Promise(resolve =>
setTimeout(resolve, 800));
  try {
    await
createProfile(profilePayload);
  } catch (retryError) {
    profileSaved = false;
  }
}
const fetchResult = await
fetchProfile(data.user.id);
isAuthAction.current = false;
if (!fetchResult.success) {
  throw new Error(
    profileSaved
    ? `Account created, but we
could not load your profile:
${fetchResult.error?.message ||
'Unknown error'}. Please try logging
in.`
    : 'Your account was created,
but we could not save your address
and phone number. Please log in
and complete your profile from the
Profile page.'
  );
}

clearPasswordRecoveryPending();
return { success: true, user:
data.user, profileIncomplete:
!profileSaved };

```

```

    } catch (error) {
      isAuthAction.current = false;
      setAuthTransition(null);
      setLoading(false);
      let msg = error.message ||
'Registration failed. Please try
again.';
      if (msg.includes('already
registered')) {
        msg = 'This email is already
registered. Please sign in instead.';
      }
      return { success: false, error:
msg };
    }
  };
};

```

2. Role-Based Access Control and Security

```

const ProtectedRoute = ({ children,
requiredRole }) => {
  const { user, userProfile, loading } =
useAuth();
  const location = useLocation();
  const hasUsableProfile = !(user &&
userProfile && (!requiredRole ||
userProfile.role === requiredRole));
  if (loading && !hasUsableProfile)
return <LoadingScreen />;
  if (!user) return <Navigate
to="/login" state={{ from: location }}
replace />;
  if (!userProfile) return <Navigate
to="/login" state={{ from: location }}
replace />;
  if (requiredRole && userProfile.role
!== requiredRole) {
    if (!userProfile.role) return
<Navigate to="/login" replace />;
    return <Navigate
to={userProfile.role === 'admin' ?
'/admin' : '/customer'} replace />;
  }
  return children;
};

```

CREATE OR REPLACE FUNCTION

```

public.is_admin()
RETURNS BOOLEAN
LANGUAGE sql
SECURITY DEFINER
SET search_path = public
AS $$
  SELECT COALESCE(
    (SELECT role = 'admin' FROM
public.profiles WHERE id =
auth.uid()),
    FALSE
  );
$$;

```

```

DROP POLICY IF EXISTS "Users
can create own orders" ON
public.orders;
CREATE POLICY "Users can create
own orders" ON public.orders
FOR INSERT
WITH CHECK (
  user_id = auth.uid()
  AND status IN ('Pending',
'Assigned')
  AND actual_weight IS NULL
  AND payment_method IS NULL
  AND payment_status = 'unpaid'
  AND amount_paid = 0
  AND pickup_photos = '[]'::jsonb
  AND delivery_photos = '[]'::jsonb
  AND service_area_status IN
('standard', 'for_review')
  AND service_area_remarks IS
NULL
);

```

3. Customer Cargo Booking

```

const handleSubmit = async () => {
  if (submittingRef.current) return;
  submittingRef.current = true;
  setLoading(true);
  let focusingInvalidField = false;
  try {
    if (!selectedRoute) throw new
Error('Please select a route.');
```

const sErrs = validateSender(); if
(Object.keys(sErrs).length) {

```

setFieldErrors(sErrs);
skipStepScrollRef.current = true;
setStep(2); focusingInvalidField =
true; focusFirstInvalid(); throw new
Error('Please fix sender details.');
```

```

const rErrs = validateReceiver();
if (Object.keys(rErrs).length) {
setFieldErrors(rErrs);
skipStepScrollRef.current = true;
setStep(3); focusingInvalidField =
true; focusFirstInvalid(); throw new
Error('Please fix receiver details.');
```

```

const validation =
validateRouteProvinces(form.sender
_province, form.receiver_province,
selectedRoute);
if (!validation.valid) throw new
Error(validation.error);
if (form.sender_province ===
'Other Area' &&
selectedRoute.destination !==
'Bohol') {
throw new Error('CargoExpress
PH currently delivers to Bohol
destinations only.');
```

```

}
const payload = {
  user_id: user.id,
  origin: selectedRoute.origin,
  destination:
selectedRoute.destination, trip_id:
selectedTrip ? form.trip_id : null,
  sender_first_name:
normalizeName(form.sender_first_na
me), sender_last_name:
normalizeName(form.sender_last_na
me), sender_phone:
form.sender_phone,
  sender_facebook:
normalizeName(form.sender_facebo
ok), sender_city: form.sender_city,
sender_province:
form.sender_province === 'Other
Area' ? form.sender_other_province :
form.sender_province,
  sender_barangay:
```

```

form.sender_barangay,
sender_street: form.sender_street,
sender_lot_block:
form.sender_lot_block,
sender_landmark:
form.sender_landmark,
  receiver_first_name:
normalizeName(form.receiver_first_n
ame), receiver_last_name:
normalizeName(form.receiver_last_n
ame), receiver_phone:
form.receiver_phone,
  receiver_facebook:
normalizeName(form.receiver_faceb
ook), receiver_city:
form.receiver_city,
receiver_province:
form.receiver_province,
  receiver_barangay:
form.receiver_barangay,
receiver_street: form.receiver_street,
receiver_lot_block:
form.receiver_lot_block,
receiver_landmark:
form.receiver_landmark,
  package_description:
form.package_description,
  payer_type: form.payer_type,
payment_preference:
form.payment_preference, notes:
form.notes,
};
if (form.sender_province ===
'Other Area') {
  payload.service_area_status =
'for_review';
  payload.status = 'Pending
Review';
}
const data = await
createOrder(payload);
if (payload.service_area_status
=== 'for_review') {
  await logOrder('Out-of-
Coverage Booking Submitted',
data.id, data.tracking_number, {
```

```

details: `Special pickup request
submitted for
${orderPartyAddress(data, 'sender')}`
});
    } else {
      await logOrder('Booking
Created', data.id,
data.tracking_number, { details:
'Standard booking created via
Customer Portal.' });
    }
    setSuccess(data);

clearBookingDraftStorage(user.id);
  refreshRecentContacts();
} catch (err) {
  toast.error(err.message || 'An
unexpected error occurred while
saving the booking. ');
  if (!focusingInvalidField)
window.scrollTo({ top: 0, behavior:
'smooth' });
  setLoading(false);
} finally {
  submittingRef.current = false;
}
};

```

4. Shipment Status Workflow and Tracking

```

export const ORDER_STATUS = {
  PENDING_REVIEW: 'Pending
Review',
  PENDING: 'Pending',
  ASSIGNED: 'Assigned',
  PICKED_UP: 'Picked Up',
  IN_TRANSIT: 'In Transit',
  ARRIVED_HUB: 'Arrived at Hub',
  OUT_FOR_DELIVERY: 'Out for
Delivery',
  DELIVERED: 'Delivered',
  PENDING_CANCELLATION:
'Pending Cancellation',
  CANCELLED: 'Cancelled',
};
export const VALID_STATUSES =

```

```

Object.values(ORDER_STATUS);
export const STATUS_FLOW = {

[ORDER_STATUS.PENDING_REVI
EW]: ORDER_STATUS.PENDING,
[ORDER_STATUS.PENDING]:
ORDER_STATUS.ASSIGNED,
[ORDER_STATUS.ASSIGNED]:
ORDER_STATUS.PICKED_UP,
[ORDER_STATUS.PICKED_UP]:
ORDER_STATUS.IN_TRANSIT,
[ORDER_STATUS.IN_TRANSIT]:
ORDER_STATUS.ARRIVED_HUB,

[ORDER_STATUS.ARRIVED_HUB]:
ORDER_STATUS.OUT_FOR_DELI
VERY,

[ORDER_STATUS.OUT_FOR_DELI
VERY]:
ORDER_STATUS.DELIVERED,
};

```

5. Trip Management

```

export const reassignTrip = async
(orderId, newTripId, reason) => {
  if (newTripId) {
    const [{ data: trip }, { data: order }]
= await Promise.all([

supabase.from('trips').select('id').eq('i
d', newTripId).single(),

supabase.from('orders').select('actua
l_weight').eq('id', orderId).single(),
]);
    if (trip) {
      await
attachCompanyTripDefaults(trip);
      await assertTripCapacity(trip,
parseFloat(order?.actual_weight || 0)
|| 0, orderId);
    }
  }
  const { data, error } = await
supabase.rpc('reassign_trip', {

```

```

    p_order_id: orderId,
    p_new_trip_id: newTripId,
    p_reason: reason
  });
  if (error) throw error;
  return data;
};

export const
bulkUpdateOrdersStatusByTrip =
async (tripId, currentStatuses,
newStatus, actionDescription) => {
  const { data: orders, error: fetchErr }
= await supabase
  .from('orders')
  .select('id, tracking_number,
status, user_id')
  .eq('trip_id', tripId)
  .in('status', currentStatuses);
  if (fetchErr) throw fetchErr;
  if (!orders || orders.length === 0)
return 0;
  await Promise.all(orders.map(async
(order) => {
    const { error: updateErr } = await
supabase
    .from('orders')
    .update({ status: newStatus,
updated_at: new
Date().toISOString() })
    .eq('id', order.id);
    if (updateErr) throw updateErr;
    logOrder(`Status Changed to
${newStatus}`, order.id,
order.tracking_number, {
      previousValue: { status:
order.status },
      newValue: { status: newStatus },
      details: actionDescription || `Bulk
update via Trip ${tripId}`
    });
  }));
  return orders.length;
};

```

6. PayMongo GCash Payment Integration

```

const capturePayment = async
(sourceld: string, amount: number,
description: string | null) => {
  const response = await
providerFetch('https://api.paymongo.
com/v1/payments', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
      'Authorization':
paymongoAuthHeader(),
    },
    body: JSON.stringify({
      data: {
        attributes: {
          amount: Math.round(amount *
100),
          source: { id: sourceld, type:
'source' },
          currency: 'PHP',
          description: description ||
'CargoExpress PH Shipping
Payment',
        },
      },
    })
  const result = await response.json()
  if (!response.ok) {
    throw new
Error(result.errors?.[0]?.detail ||
'Failed to process payment')
  }
  return {
    paymentId: result.data.id,
    status:
result.data.attributes.status,
    amount:
result.data.attributes.amount / 100,
  }
}

const timingSafeEqual = (a: string, b:
string) => {
  if (a.length !== b.length) return false
  let result = 0
  for (let i = 0; i < a.length; i++) {

```

```

    result |= a.charCodeAt(i) ^
b.charCodeAt(i)
  }
  return result === 0
}
const parseSignature = (header:
string) => {
  return
header.split(',').reduce<Record<string,
string>>((acc, part) => {
    const [key, value] = part.split('=')
    if (key && value !== undefined)
acc[key.trim()] = value.trim()
    return acc
  }, {})
}
const verifySignature = async
(rawBody: string, signatureHeader:
string | null) => {
  const secret =
Deno.env.get('PAYMONGO_WEBHOOK_SECRET')
  if (!secret) throw new
Error('PAYMONGO_WEBHOOK_SECRET is not configured')
  if (!signatureHeader) return false
  const parts =
parseSignature(signatureHeader)
  if (!parts.t) return false
  const currentTimestamp =
Math.floor(Date.now() / 1000)
  if (currentTimestamp -
Number(parts.t) > 300) {
    console.error('[paymongo-
webhook] Signature timestamp is too
old')
    return false
  }
  const signedPayload =
`${parts.t}.${rawBody}`
  const key = await
crypto.subtle.importKey(
    'raw',
    new
TextEncoder().encode(secret),
    { name: 'HMAC', hash: 'SHA-256'

```

```

},
  false,
  ['sign'],
)
const expected = hex(await
crypto.subtle.sign('HMAC', key, new
TextEncoder().encode(signedPayload)))
return [parts.te, parts.li].some(value
=> value &&
timingSafeEqual(expected, value))
}

```

7. Partial Payment and Remaining Balance

```

export const isOrderPriced = (order)
=> {
  if (!order) return false;
  return
(parseFloat(order.actual_weight || 0)
|| 0) > 0;
};
export const finalShippingFee =
(order) => {
  if (!order) return 0;
  const cost =
parseFloat(order.shipping_cost || 0)
|| 0;
  const discount =
parseFloat(order.discount_amount ||
0) || 0;
  return Math.max(0,
Math.round((cost - discount) * 100) /
100);
};
export const hasDiscount = (order)
=>
(parseFloat(order?.discount_amount
|| 0) || 0) > 0;
export const outstandingBalance =
(order) => {
  if (!order) return 0;
  const paid =
parseFloat(order.amount_paid || 0) ||
0;
  return Math.max(0,

```

```

Math.round((finalShippingFee(order)
- paid) * 100) / 100);
};

export const
recordAdditionalPayment = async
(orderId, amount, method, ref, notes,
paymentDate = null, receiptUrl = null,
idempotencyKey = null,
adminVerifiedReceipt = false) => {
  const { data, error } = await
supabase.rpc('record_additional_pay
ment', {
    p_order_id: orderId,
    p_amount: amount,
    p_payment_method: method,
    p_reference: ref || null,
    p_notes: notes || null,
    p_payment_date: paymentDate ||
null,
    p_receipt_url: receiptUrl || null,
    p_idempotency_key:
idempotencyKey || null,
    p_admin_verified_receipt:
adminVerifiedReceipt || false,
  });
  if (error) throw error;
  return {
    newAmountPaid:
data.amount_paid,
    newBalance:
data.remaining_balance,
    newStatus: data.payment_status,
  };
};

```

8. Notifications

```

export const getNotifications = async
(userId, { limit = 50, offset = 0 } = {})
=> {
  const { data, error } = await
supabase
  .from('notifications')
  .select('*')
  .eq('user_id', userId)
  .order('created_at', { ascending:
false })

```

```

  .order('id', { ascending: false })
  .range(offset, offset + limit - 1);
  if (error) throw error;
  return data || [];
};

export const markNotificationRead =
async (id) => {
  const { error } = await supabase
    .from('notifications')
    .update({ is_read: true })
    .eq('id', id);
  if (error) throw error;
  emitNotificationsChanged();
};

export const
markAllNotificationsRead = async
(userId) => {
  const { error } = await supabase
    .from('notifications')
    .update({ is_read: true })
    .eq('user_id', userId)
    .eq('is_read', false);
  if (error) throw error;
  emitNotificationsChanged();
};

```

9. Customer Support Chat

```

export const
getOrCreateConversation = async
(customerId) => {
  let { data: convRows, error } = await
supabase
  .from('conversations')
  .select('*')
  .eq('customer_id', customerId)
  .order('created_at', { ascending:
true })
  .limit(1);
  if (error) throw error;
  let conv = convRows &&
convRows.length > 0 ? convRows[0]
: null;
  if (!conv) {
    const { data: newConv, error:
createError } = await supabase
      .from('conversations')

```

```

      .insert({ customer_id:
customerId, status: 'bot_active' })
      .select()
      .single();
      if (createError) throw createError;
      conv = newConv;
    }
    return conv;
  };

export const sendMessage = async
(conversationId, senderId,
senderRole, text) => {
  const { data, error } = await
supabase
  .from('chat_messages')
  .insert({
    conversation_id: conversationId,
    sender_id: senderId,
    sender_role: senderRole,
    message: text
  })
  .select()
  .single();
  if (error) throw error;
  return data;
};

```

10. Admin Booking Management and Cancellation

```

export const
requestOrderCancellation = async
(orderId, reason) => {
  const trimmed = (reason || "").trim();
  if (trimmed.length < 5) {
    throw new Error('Please tell us
why you are cancelling (at least 5
characters).');
  }
  const { data, error } = await
supabase.rpc('request_order_cancell
ation', {
    p_order_id: orderId,
    p_reason: trimmed,
  });
  if (error) throw error;
  return data;
};

```

```

};
export const
reviewOrderCancellation = async
(orderId, approve, notes = null) => {
  const { data, error } = await
supabase.rpc('review_order_cancell
ation', {
    p_order_id: orderId,
    p_approve: approve,
    p_notes: notes || null,
  });
  if (error) throw error;
  return data;
};

```

11. Sales Reports

```

export const
buildPerTripSalesReport = ({
  trip,
  orders = [],
  activityByOrder = {},
  settlementsByOrder = {},
}) => {
  const rows = orders.map(order =>
buildRow(
  order,
  activityByOrder[order.id] || [],
  settlementsByOrder[order.id],
));
  const activeRows = rows.filter(row
=> row.status !==
ORDER_STATUS.CANCELLED);
  const cancelledRows =
rows.filter(row => row.status ===
ORDER_STATUS.CANCELLED);
  const reviewRows =
cancelledRows.filter(row =>
row.cancellation.needsReview);
  const dataInconsistent =
rows.some(row =>
row.cancellation?.dataInconsistent);
  const paymentsReceivedCents =
sumRows(rows,
'paymentsReceived');
  const moneyReturnedCents =
sumRows(rows, 'moneyReturned');
};

```

```

    const
    cancelledAwaitingDecisionCents =
    reviewRows.reduce((total, row) => {
      const amount =
    toCents(row.cancellation.netRetained);
      return amount > 0 ?
    addCents(total, amount) : total;
    }, 0);
    const shippingFeesCents =
    activeRows.reduce((total, row) => (
      row.shippingFee === null ? total :
    addCents(total,
    toCents(row.shippingFee))
    ), 0);
    const amountStillToCollectCents =
    activeRows.reduce((total, row) => (
      row.amountStillToPay === null ?
    total : addCents(total,
    toCents(row.amountStillToPay))
    ), 0);
    return {
      trip,
      rows,
      activeRows,
      cancelledRows,
      summary: {
        shippingFees:
    fromCents(shippingFeesCents),
        paymentsReceived:
    fromCents(paymentsReceivedCents)
      },
      moneyReturned:
    fromCents(moneyReturnedCents),
      paymentsAfterRefunds:
    fromCents(paymentsReceivedCents
    - moneyReturnedCents),
      amountStillToCollect:
    fromCents(amountStillToCollectCents),
      activeBookingCount:
    activeRows.length,
      completedBookingCount:
    activeRows.filter(row =>
    row.isCompleted).length,
      cancelledBookingCount:

```

```

    cancelledRows.length,
      unpricedActiveCount:
    activeRows.filter(row =>
    !row.isPriced).length,
      cancelledMoneyAwaitingDecision:
    fromCents(cancelledAwaitingDecisionCents),
      cancelledReviewCount:
    reviewRows.length,
      pendingRefundAmount:
    fromCents(sumRows(rows,
    'refundPending')),
      failedRefundAmount:
    fromCents(sumRows(rows,
    'refundFailed')),
      uncertainRefundAmount:
    fromCents(sumRows(rows,
    'refundUncertain')),
      dataInconsistent,
    },
    basis: 'Shows recorded payments
    and refunds for bookings assigned to
    this trip, regardless of payment
    date.',
    assignmentNote: 'This report uses
    each booking's current trip
    assignment. Reassigned bookings
    appear only under their current trip.
    A cancelled booking with no current
    assignment cannot be placed here
    reliably.',
  };
};

```

RESEARCHER BIODATA

PERSONAL DATA

Name : Bea A. Sarong

Nick Name : Bey

Date of Birth : April 22, 2002

Place of Birth : Mandaluyong City, Metro Manila

Home Address : Cantigdas, Batuan, Bohol

Email Address : itsbeasarong@gmail.com

Civil Status : Single

Citizenship : Filipino

Father's Name : Marlon A. Sarong

Mother's Name : Cristina A. Sarong

EDUCATIONAL BACKGROUND

Elementary : Andres Bonifacio Integrated School
Welfareville Cmpd, Brgy. Addition Hills, Mandaluyong City

Secondary : Andres Bonifacio Integrated School
Welfareville Cmpd, Brgy. Addition Hills, Mandaluyong City

Tertiary : Bohol Island State University – Balilihan Campus
Magsija, Balilihan, Bohol

Motto : "Orbit your own dreams, not someone else's planet."

RESEARCHER BIODATA

PERSONAL DATA

Name : Quin Ann P. Amigo
Nick Name : Kwen
Date of Birth : April 26, 2005
Place of Birth : Cabad, Balilihan, Bohol
Home Address : Cabad, Balilihan, Bohol
Email Address : quinann26@gmail.com
Civil Status : Single
Citizenship : Filipino
Father's Name : Rainero B. Amigo
Mother's Name : Annie P. Amigo

EDUCATIONAL BACKGROUND

Elementary : Cabad-Candasig Elementary School
Cabad, Balilihan, Bohol

Secondary : Carmel Academy
Del Carmen Norte, Balilihan, Bohol

Tertiary : Bohol Island State University – Balilihan Campus
Magsija, Balilihan, Bohol

Motto : "Live your life, you only live once, do what makes you."

RESEARCHER BIODATA

PERSONAL DATA

Name : Joven M. Tibuc
Nick Name : Joven
Date of Birth : October 16, 2004
Place of Birth : Cantonimbo, Balilihan, Bohol
Home Address : Cantonimbo, Balilihan, Bohol
Email Address : joventibuc12345@gmail.com
Civil Status : Single
Citizenship : Filipino
Father's Name : Jerry T. Tibuc
Mother's Name : Antonia M. Tibuc

EDUCATIONAL BACKGROUND

Elementary : Sal-ing Elementary School
Sal-ing, Balilihan, Bohol

Secondary : Congressman Pablo Malasarte National High School
Cabad, Balilihan, Bohol

Tertiary : Bohol Island State University – Balilihan Campus
Magsija, Balilihan, Bohol

Motto : "Done is better than perfect."